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MOHSIN IDNA HOSSAIN
AIUB, Device Note

Electronic
Devices:

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#Semicon-
Ductor
Dood
Choo

#Semi-Conductor Materials:-

⇒ Semiconductors are materials which have a Conductivity between Conductors and non-conductors, or insulators.

The most used Semi-conductor materials are:-

1. Silicon (Si)

2. Germanium (Ge)

3. Gallium Arsenide (GaAs)

#P-type Semiconductor:-

1. It is an Intrinsic Semiconductor doped with Boron and Indium.

2. The majority of carriers in a P-type Semiconductors are holes.

3. Electrons are minority Carriers in P-type Semiconductors.

4. In P-type Semi-Conductor the hole density is much greater than the electron density.

N-type Semi-Conductor:-

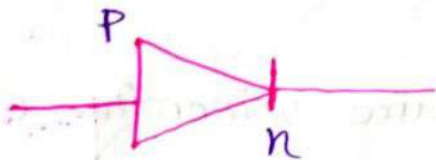
1. The majority of charge carriers in n-type Semi-Conductors are electrons.

2. Holes are minority carriers in n-type Semi-Conductor.

3. In n-type Semiconductor electron density is much greater than the hole density.

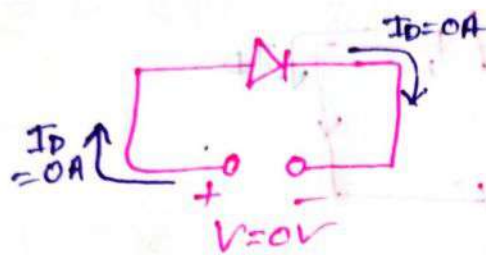
Semi-Conductor Diode:-

⇒ It is a p-n junction diode. It is a two-terminal Device that conduct current only in one direction.



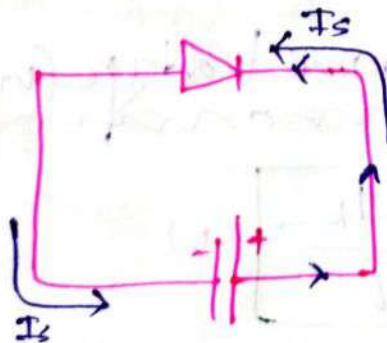
Diode in No Applied Bias ($V=0V$)

⇒ In the absence of an applied bias, across a Semiconductor diode, the net flow of charge in one direction is zero.



Reverse-Bias ($V_D < 0V$)

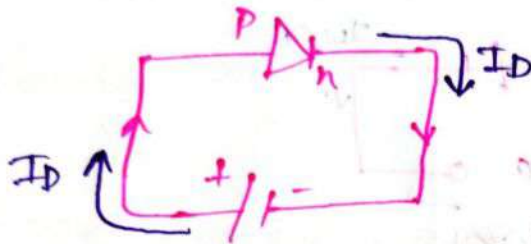
⇒ The Current that exists reverse-bias conditions is called the reverse saturation current and is denote by I_s .



I_s = Minority-Carrier flow
Majority $\approx 0A$

Forward-Bias ($V_D > 0V$)

⇒ A forward-bias or "on" condition is established by applying positive potential to the p-type and negative potential to the n-type.



$$I_D = I_S (e^{V_D/nV_T} - 1)$$

I_S = Reverse Saturation Current

V_D = Voltage across diode.

n = Range between 1 & 2

V_T = Thermal Voltage.

Thermal Voltage (V_T)

$$V_T = \frac{k T_K}{q} \quad (V)$$

k = Boltzman's Constant = $1.38 \times 10^{-23} \text{ J/K}$

T_K = Temperature in Kelvins = $273 + ^\circ C$

q = $1.6 \times 10^{-19} \text{ C}$

#problem:1:-
mon

⇒ At the temperature of 27°C , determine the voltage V_T

⇒ As given:-
mon

$$T_K = 27 + 273 = 300\text{K}$$

$$q = 1.6 \times 10^{-19}\text{C}$$

$$K = 1.38 \times 10^{-23}\text{ J/K}$$

we know:-
mon

$$V_T = \frac{KT}{q}$$

$$= \frac{1.38 \times 10^{-23} \times 300}{1.6 \times 10^{-19}}$$

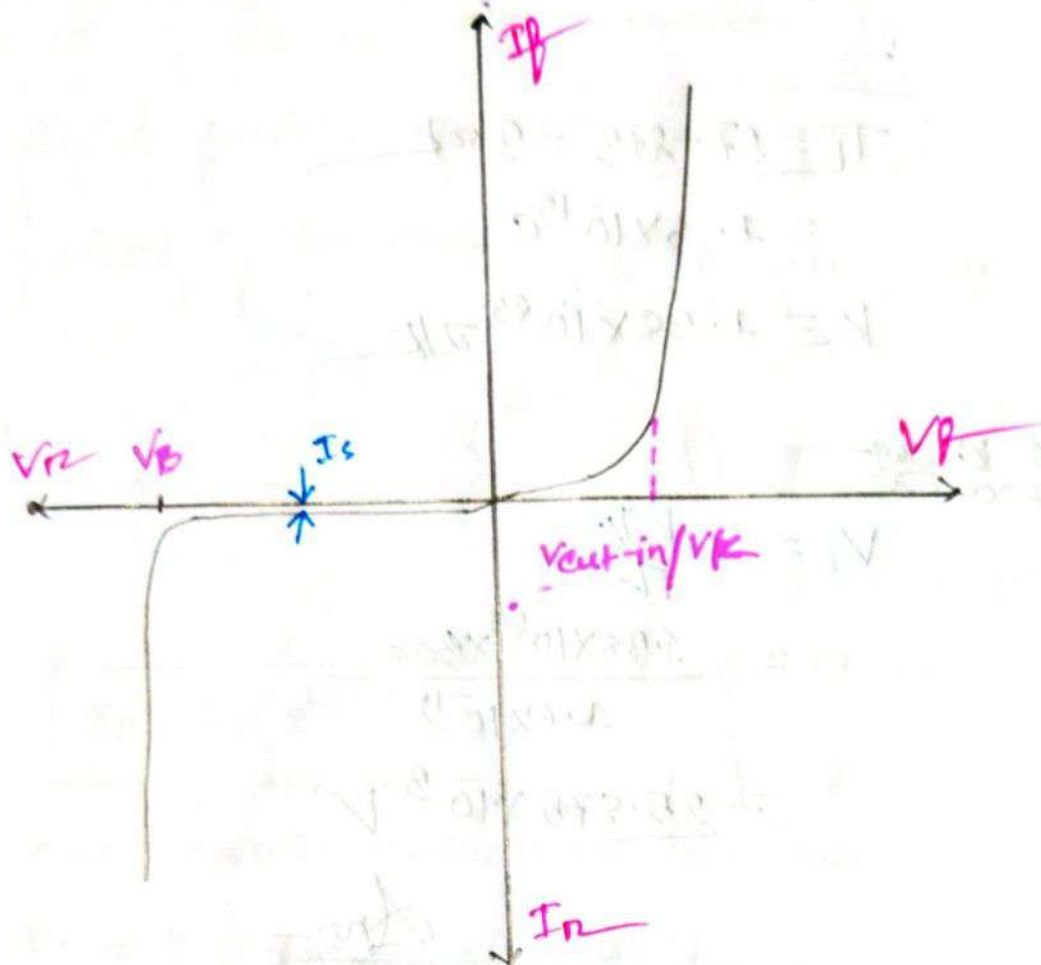
$$= 25.875 \times 10^{-3}\text{ V}$$

Ans:-

Breakdown Region:-
mon

⇒ The region where the reverse current rapidly increases by the breakdown phenomenon is called breakdown region.

#1 Diode Characteristics



V_f = Forward Voltage.

I_f = Forward Current

V_r = Reverse Voltage.

I_r = Reverse Current

V_{cut-in} = Cut-in Voltage

V_B = Breakdown Voltage.

I_s = Saturation Current.

V_K = Knee Voltage.

Knee Voltages of Ge, Si & GaAs

Semi-Conductors

Knee Voltage (V_K)

1. Ge $\rightarrow 0.3V$

2. Si $\rightarrow 0.7V$

3. GaAs $\rightarrow 1.2V$

Ideal Vs Practical!

$\Rightarrow$ Ideally, if the diode is to behave like a closed switch in the forward-bias region, the resistance of diode should be 0Ω

Reverse - Bias Region:-

$\Rightarrow$ Acts like an open switch with infinite resistance ($\infty\Omega$)

Practically:-

$\Rightarrow$ Threshold Voltage (V_f)! In forward region:-

$\Rightarrow$ A small V_f is required to start conducting.

Reverse-Bias Region:-

1. Leakage Current

2. Breakdown Voltage.

Resistance Levels:-

⇒ Semiconductors react differently to DC And AC Current.

There are three types of resistance

1. DC (Static) resistance.

2. AC (dynamic) resistance.

3. Average AC resistance.

DC/Static Resistance:-

⇒ For a specific applied DC voltage V_D , the diode has a specific current I_D , and a specific resistance R_D .

$$R_D = \frac{V_D}{I_D}$$

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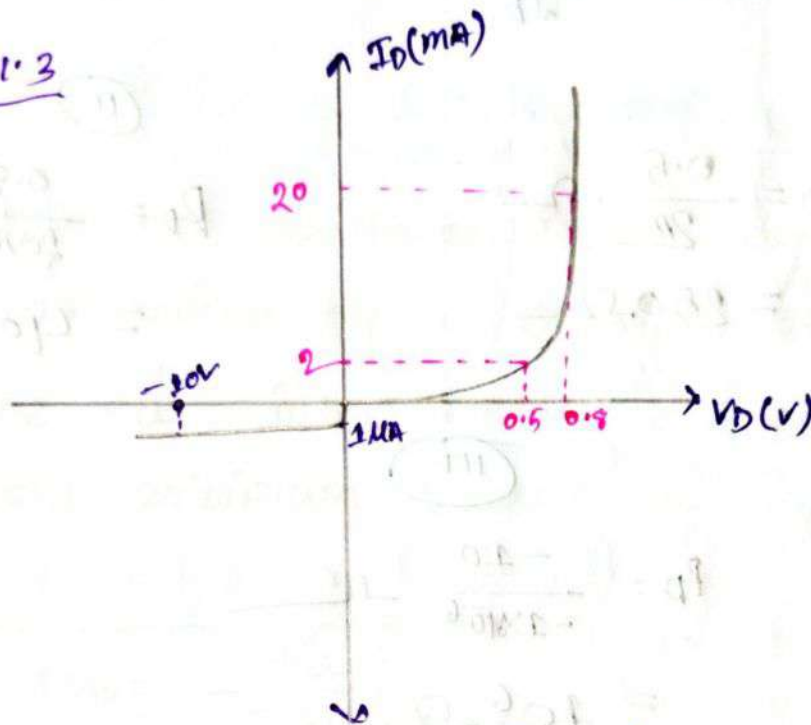
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Note:- The higher the current through a diode, the lower the dI_D/dV_D resistance level.

Problem: 1.3
mon



⇒ Determine the dI_D/dV_D resistance level for the diode at

(i) $I_D = 2\text{mA}$

(ii) $I_D = 20\text{mA}$

(iii) $I_D = -10\text{V}$

⇒ I_D given:-
mon

$I_D = 2\text{mA} ; V_D = 0.5\text{V}$

$I_D = 20\text{mA} ; V_D = 0.8\text{V}$

$I_D = -10\text{V} ; V_D = -10\text{V}$

We know:-

$$I_D = \frac{V_D}{R_D}$$

$$\therefore R_D = \frac{V_D}{I_D}$$

(i)

$$R_D = \frac{0.5}{2m} \Omega$$

$$= 250 \Omega$$

(ii)

$$R_D = \frac{0.8}{20m} \Omega$$

$$= 40 \Omega$$

(iii)

$$R_D = \frac{-10}{-1 \times 10^6} \Omega$$

$$= 10^6 \Omega$$

Ans:-

AC / Dynamic Resistance:-

⇒ In the forward bias:-

$$r'_d = \frac{26mV}{I_D} + r_{eD}$$

⇒ In the reverse bias:-

$$r'_d = \infty$$

Note:-

⇒ The lower the Q point of operation (low current or low voltage) the higher the AC resistance.

Average AC Resistance:-

⇒ The average ac resistance is, by definition, the resistance determined by a straight line drawn between the two intersection established by the maximum and minimum values of input voltage.

$$r_{av} = \frac{\Delta V_d}{\Delta I_d} \quad \text{Pt. to Pt.}$$

Load-Line Analysis:-

1. Load-Line plots combinations of diode current (I_D) and voltage (V_D) for a given circuit.

2. Maximum $I_D = \frac{E}{R}$

3. Maximum $V_D = E$

4. Intersection of load line & characteristic curve is the "Q" point.

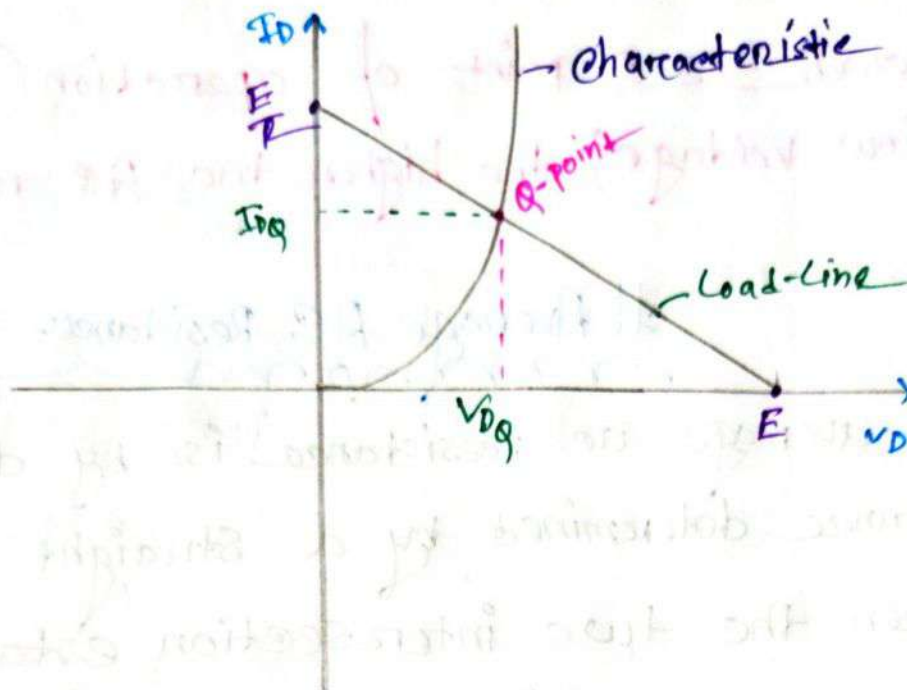
5. 'Q' point identifies I_D & V_D for the diode in circuit.

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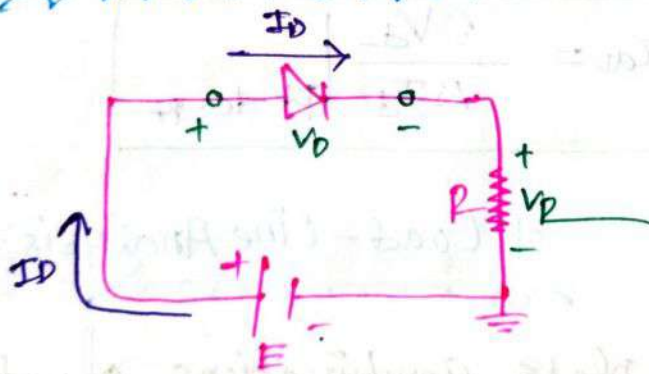
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How to Draw a Load line



According to KVL:-

$$+E - V_D - V_R = 0$$

$$\therefore \boxed{E = V_D + I_D R}$$

Since $V_R = I_D \times R$
 $I_D = I_R$

Now, let $V_D = 0$ [in y axis]

Then:- $E = I_D \times R$

$$\therefore \boxed{I_D = \frac{E}{R}}$$

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Again

Let, $I_D = 0$ [in x axis]

$$\therefore E = V_D$$

We need two points to draw a load-line of a diode:-
and those two points are:-

$$(V_D, I_D) = \left\{ (E, 0), \left(0, \frac{E}{R} \right) \right\}$$

Problem: 2.1:-

⇒ For the series diode configuration of Fig: a, employing the diode characteristics of Fig: b determine

(a) V_D and I_D

(b) V_D

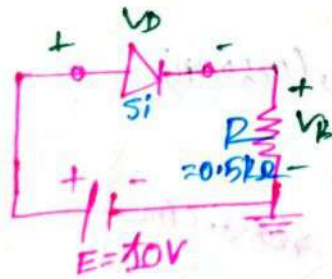


Fig: a

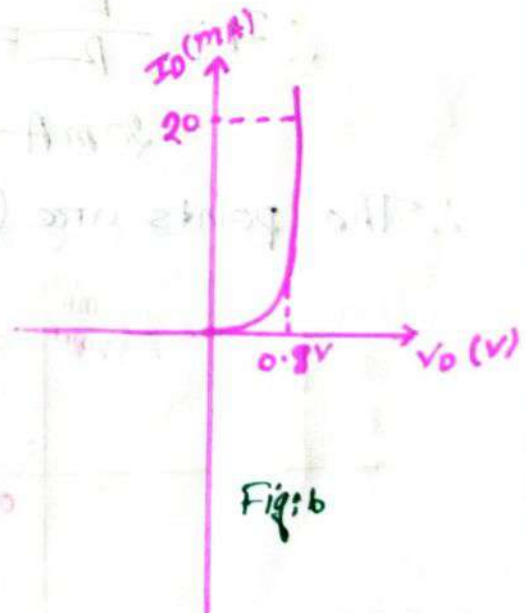


Fig: b

(a)

Is given:-

$$E = 10V$$

$$R = 0.5k\Omega$$

We know:-

$$E - V_D - I_D R = 0$$

$$\therefore E = V_D + I_D R$$

Now, for $I_D = 0$:-

$$E = V_D$$

$$\therefore V_D = 10V$$

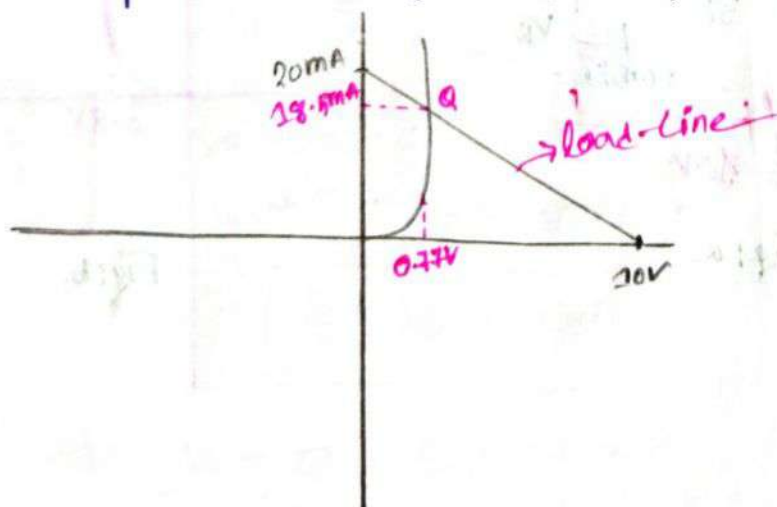
for $V_D = 0$:-

$$E = I_D R$$

$$\therefore I_D = \frac{E}{R} = \frac{10}{0.5k}$$

$$= 20mA$$

$\therefore$ The points are $(10, 0), (0, 20mA)$



$$\therefore V_{DQ} = 0.77V$$

$$I_{DQ} = 18.5mA$$

we have:-

$$I_{DQ} = 18.5 \text{ mA}$$

∴ we know:-

$$V_D = I_D \times R$$

$$= 18.5 \text{ mA} \times 0.6 \text{ k}$$

$$= 0.25 \text{ V}$$

Ans:-

OR

we have:-

$$V_D = 0.77 \text{ V}$$

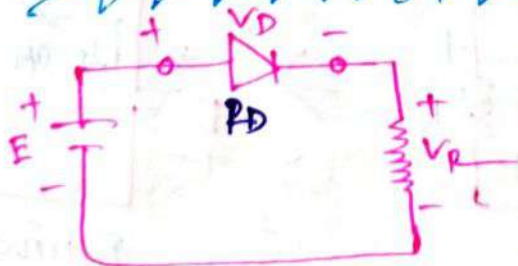
we know:-

$$V_R = E - V_D$$

$$= 10 - 0.77$$

$$= 9.23 \text{ V}$$

Calculate Diode Resistance:-



$$R_D = \frac{V_{DQ}}{I_{DQ}} \Omega$$

$$I_D = \frac{E}{R_D + R}$$

Then:-

And:-

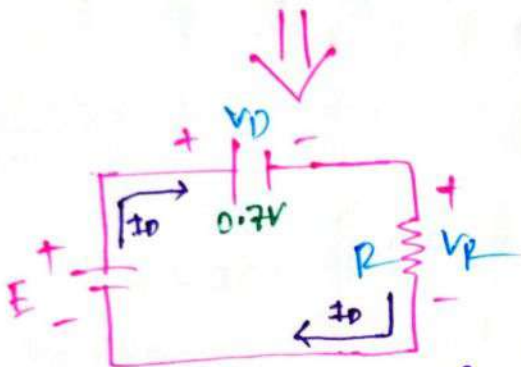
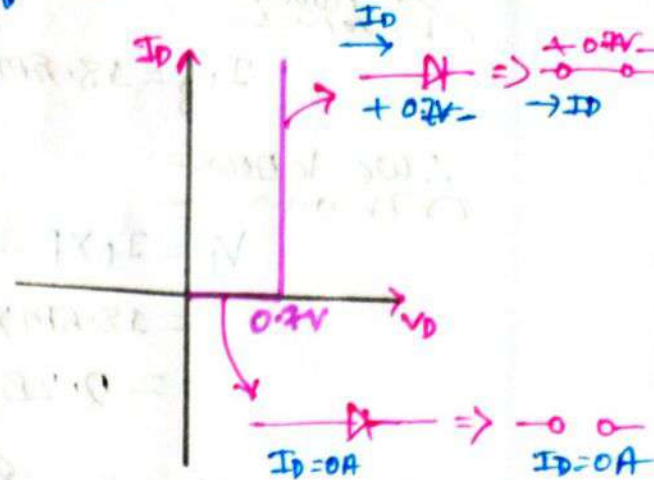
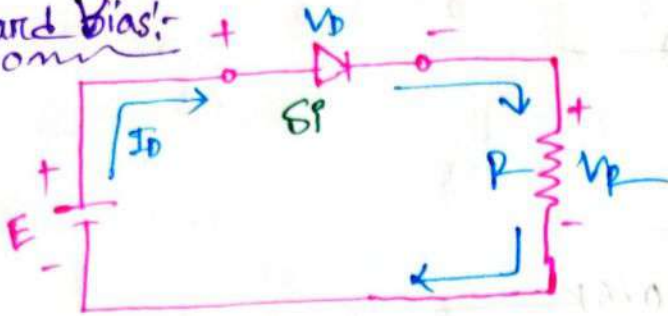
$$V_R = \frac{R}{R_T} \times E$$

Where

$$R_T = R_D + R$$

Series Diode Configurations:-

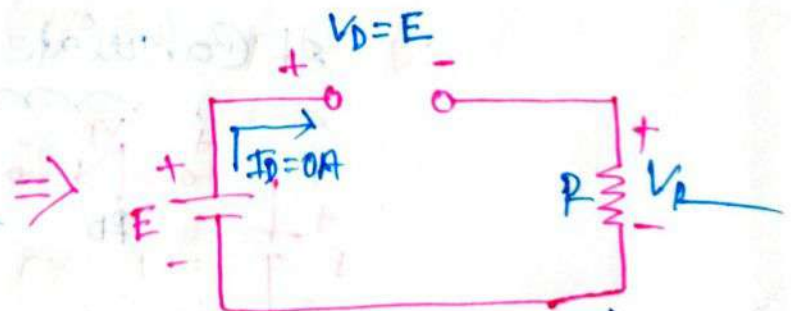
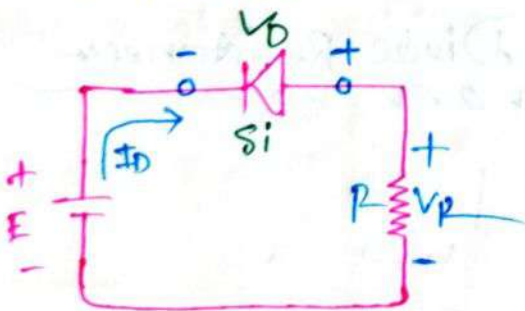
Forward Bias:-



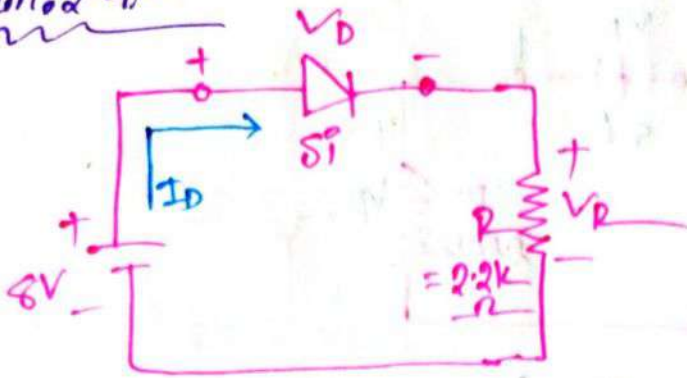
$$I_D = \frac{E - V_D}{R}$$

equivalent model for
"on" diode.

Reverse Bias:-



Equivalent model for
"off" diode.

#problem 2.4:-
mon

⇒ For the circuit determine V_D , V_R , I_D

⇒ In given:-
mon

$$E = 8V$$

$$V_D = 0.7V \text{ [Since Si diode]}$$

$$R = 2.2k\Omega$$

we know:-
mon

$$E - V_D - V_R = 0$$

$$\therefore V_R = E - V_D$$

$$= 8 - 0.7$$

$$V_R = 7.3V$$

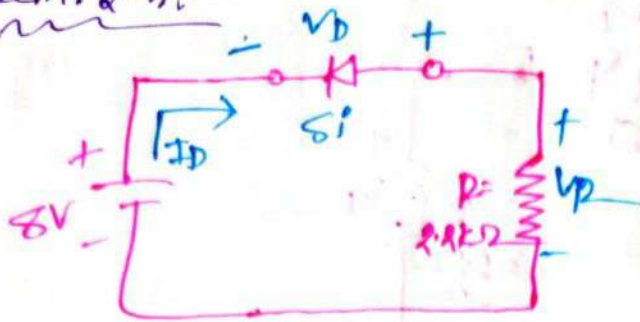
$$V_D = 0.7V$$

$$I_D = \frac{V_R}{R} = \frac{8 - 0.7}{2.2k}$$

$$I_D = 0.0032mA$$

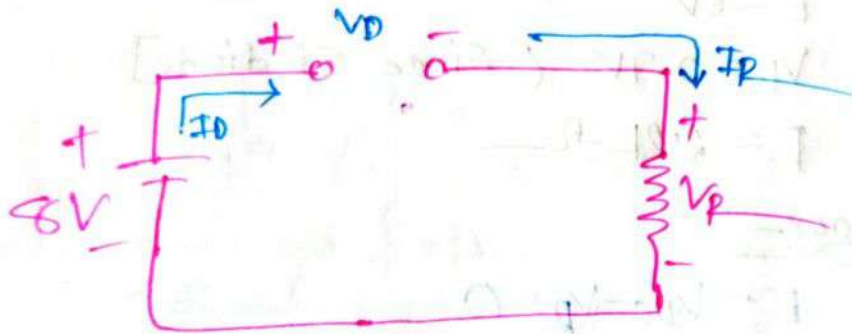
Ans:-

problem: 2.51-



⇒ Determine V_D , I_D , I_R , V_R

⇒ 1st Re-draw the circuit for Reverse Bias:-



We know-

$$E - V_D - V_R = 0$$

$$\therefore V_D = E - V_R = E - 0 = E = 8V \quad [\text{Since } I = 0A]$$

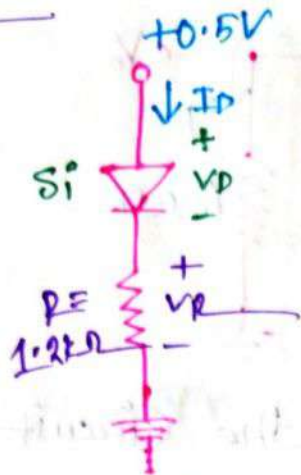
$$I_D = 0A$$

$$I_R = 0A$$

$$V_R = I_R \times R = 0 \times 2.2K = 0V$$

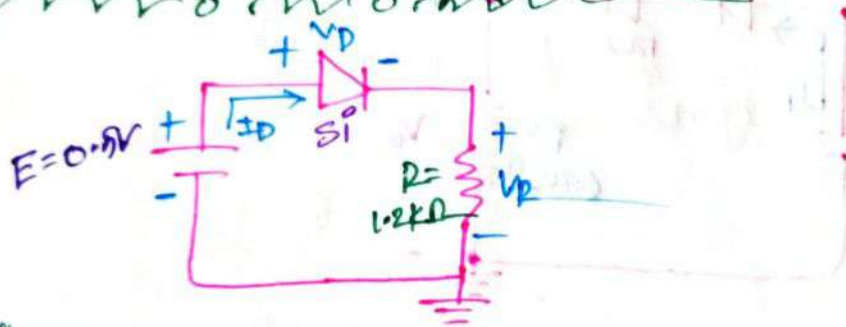
Ans:-

Problem 2.6:-



⇒ For the Series diode configuration determine V_D , V_R , I_D

⇒ 1st redraw the configuration:-



In given:-

$$E = 0.5V$$

$$R = 1.2k\Omega$$

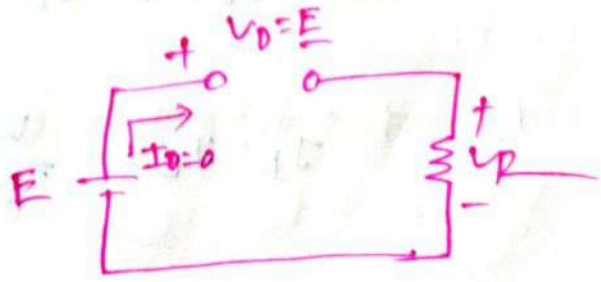
We know:-

To Active a Si diode, we need minimum 0.7V Voltage. here Supplied voltage 0.5V only. So the diode will not be active and act like an open circuit.

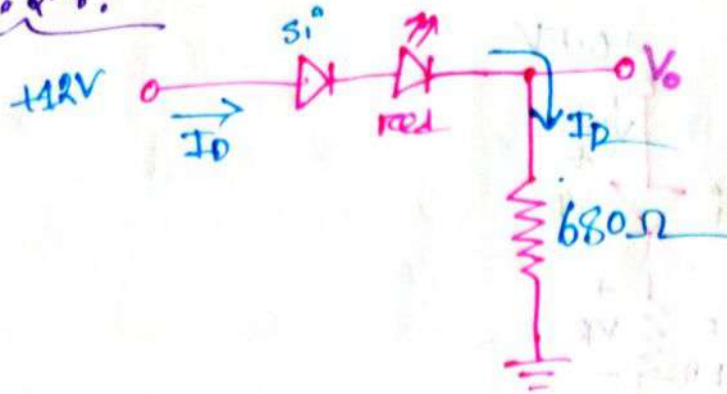
$$\therefore V_D = E = 0.5V$$

$$I_D = 0A$$

$$V_R = 0V$$

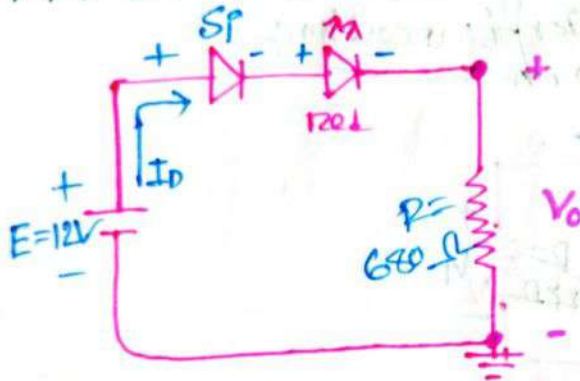


problems Q.7 :-
mon



⇒ Determine V_0 and I_D for the circuit

⇒ 1st redraw the circuit:-
mon



we have:-
mon

$$E = 12V$$

$$V_D = 0.7V \text{ [Since Si diode]}$$

$$V_{red} = 1.8V \text{ [Since Red LED diode]}$$

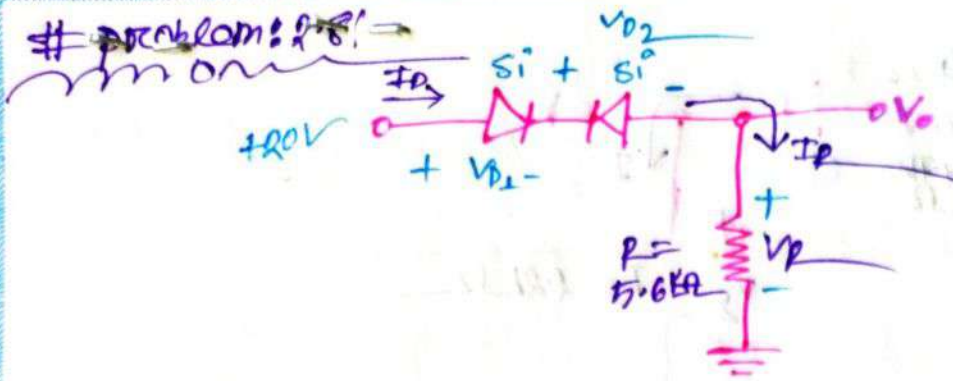
$$R = 680\Omega$$

we know:-
mon

$$E - V_D - V_{red} - V_0 = 0$$

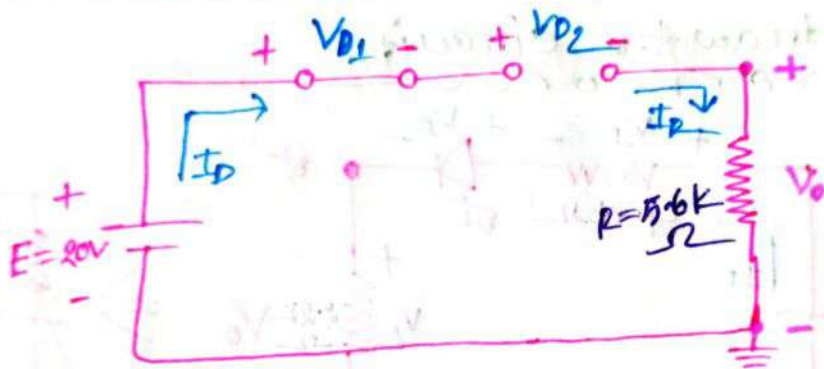
$$\therefore V_0 = E - V_D - V_{red} = 12 - 0.7 - 1.8 = 9.5V$$

$$I_D = I_R = \frac{V_0}{R} = \frac{9.5}{680} = 13.97mA$$



⇒ Determine I_D , V_{D2} , and V_O

⇒ 1st redraw the circuit:-



⇒ We have
now

$$E = 20V,$$

$$V_{D1} = 0.7V \text{ [Since there is no current]}$$

$$V_{D2} = 0$$

$$R = 5.6k\Omega$$

$$V_O = 0V$$

We know:-
now

$$E - V_{D1} - V_{D2} - V_O = 0$$

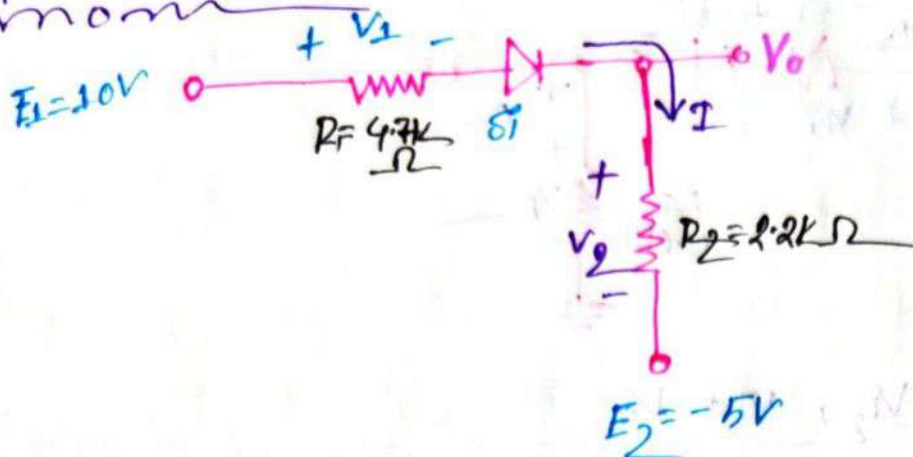
$$\therefore V_{D2} = E - V_{D1} - V_O = 20 - 0 - 0 = 20V$$

$$I_D = 0A$$

$$V_O = 0V$$

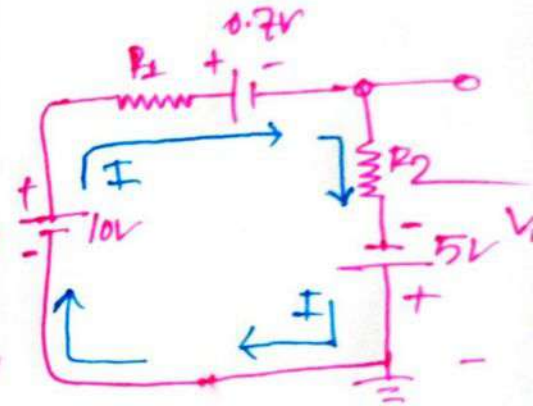
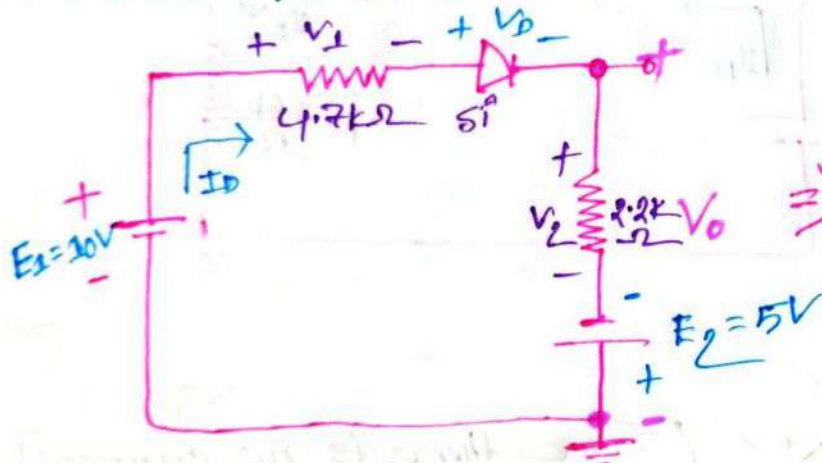
[Since the circuit is open]

problems 2.2! —



⇒ Determine I , V_1 , V_2 and V_0

⇒ Int redraw the circuit! —



In given! —

$$R_1 = 4.7k\Omega$$

$$R_2 = 2.2k\Omega$$

$$E_1 = 10V$$

$$E_2 = +5V$$

$$V_D = 0.7V$$

$$\Rightarrow I = \frac{E_1 + E_2 - V_D}{R_1 + R_2}$$

$$= \frac{10 + 5 - 0.7}{4.7k + 2.2k}$$

$$= 2.07mA$$

$$V_1 = IR_1$$

$$= 2.07m \times 4.7k$$

$$= 0.74V$$

$$V_2 = IR_2$$

$$= 2.07m \times 2.2k$$

$$= 4.55V$$

we know —

$$E_1 - V_1 - V_D - V_2 = 0$$

$$V_2 = E_1 - V_1 - V_D \dots \textcircled{1}$$

put all the values in eq (1) $\Rightarrow$

$$\therefore V_0 = E_1 - V_1 - V_D$$

$$= 10 - 0.74 - 0.7$$

$$= -0.44 \text{ V}$$

$$\therefore I = 2.07 \text{ mA}$$

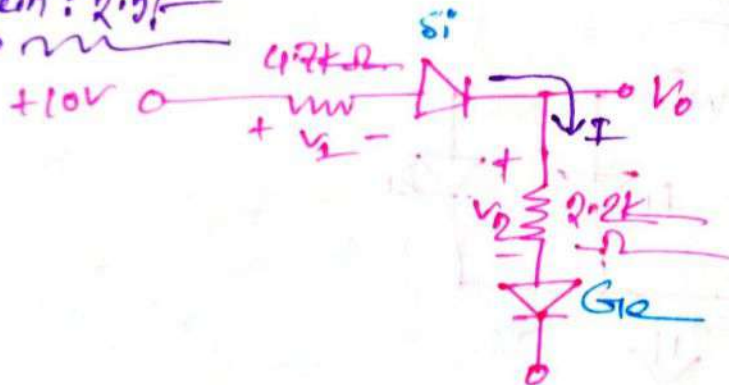
$$V_1 = 0.74 \text{ V}$$

$$V_2 = 4.55 \text{ V}$$

$$V_0 = -0.44 \text{ V}$$

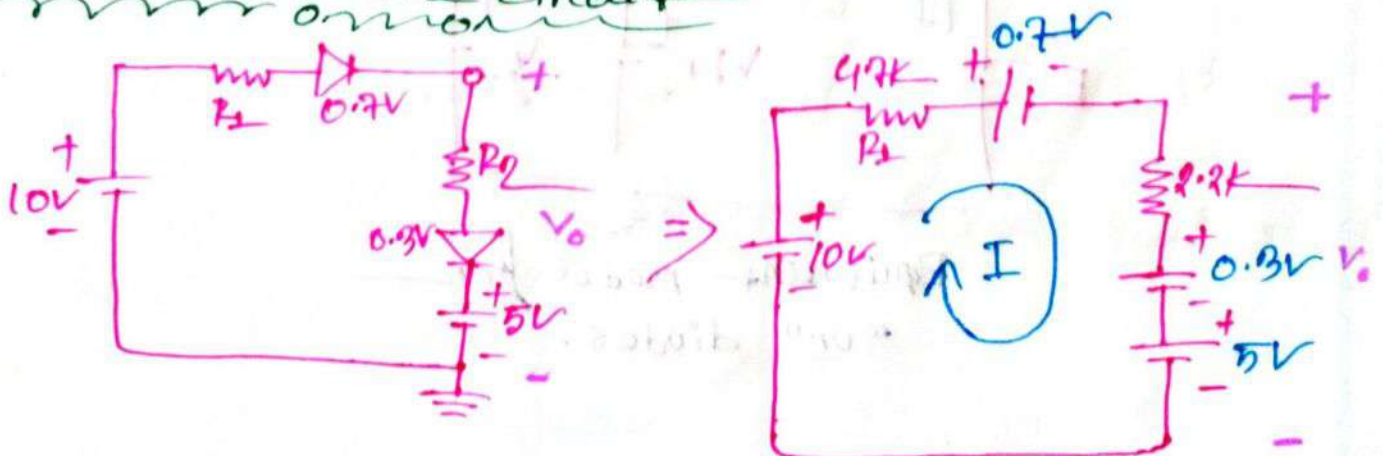
Ans:-

problem : 2.91 -



$\Rightarrow$ Determine the current I , V_0

$\Rightarrow$ Int redraw the circuit



We know:-

$$\Rightarrow I = \frac{E_1 - V_{D1} - V_{D2} - E_2}{R_1 + R_2} = \frac{10 - 0.7 - 0.3 - 5}{4.7k + 2.2k} = 570.71 \mu A$$

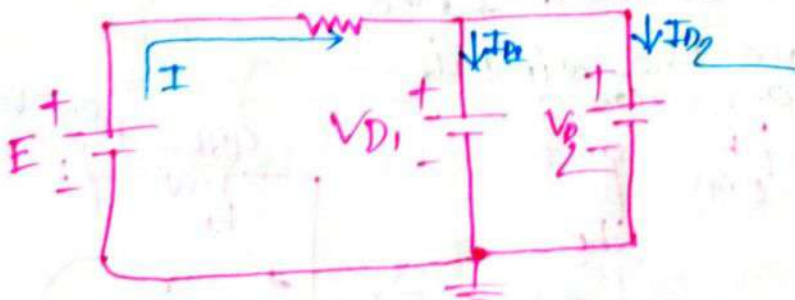
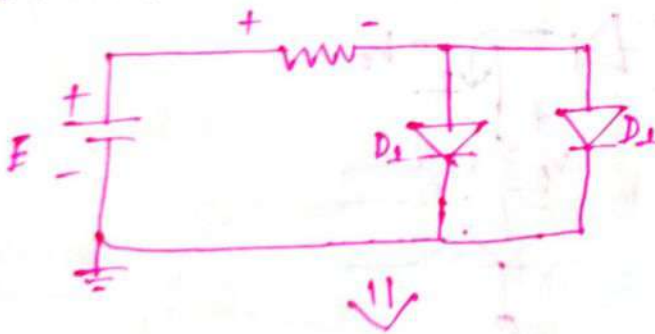
$$\Rightarrow V_1 = IR_1 = 570.71 \mu A \times 4.7k = 0.74V$$

$$\Rightarrow E_1 - V_1 - V_{D1} - V_0 = 0$$

$$\therefore V_0 = E - V_1 - V_{D1} = 10 - 0.74 - 0.7 = -0.44V$$

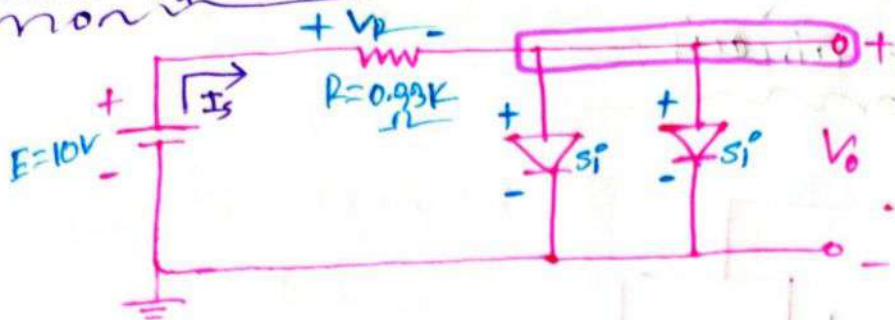
Series-Parallel Configurations:-

Forward Bias:-



Equivalent model for
"on" diodes.

problems 2.10 :-



⇒ Determine V_O , V_R , I_S , I_{D1} , I_{D2} for the configuration.

⇒ I_S given:-

$$E = 10V$$

$$R = 0.33k\Omega$$

We know:-

⇒ $V_O = 0.7V$ [Since V_O parallel to Si diode]

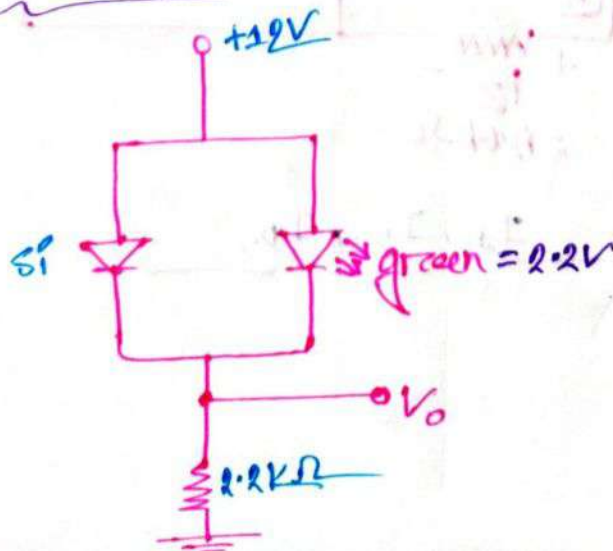
⇒ $V_R = E - V_O = 10 - 0.7 = 9.3V$

⇒ $I_S = \frac{V_R}{R} = \frac{9.3}{0.33k} = 28.18mA$

⇒ $I_{D1} = \frac{I_S}{2} = \frac{28.18mA}{2} = 14.09mA$

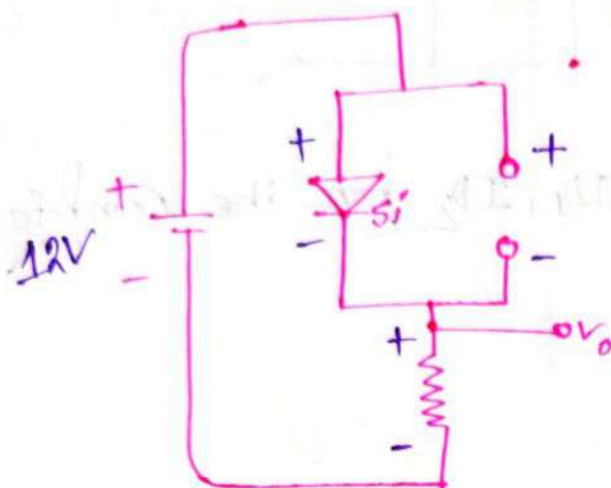
⇒ $I_{D1} = I_{D2}$ [Since Diodes are same]

problems 2.12 :-



⇒ Determine Voltage V_0

⇒ Redraw the circuit:-



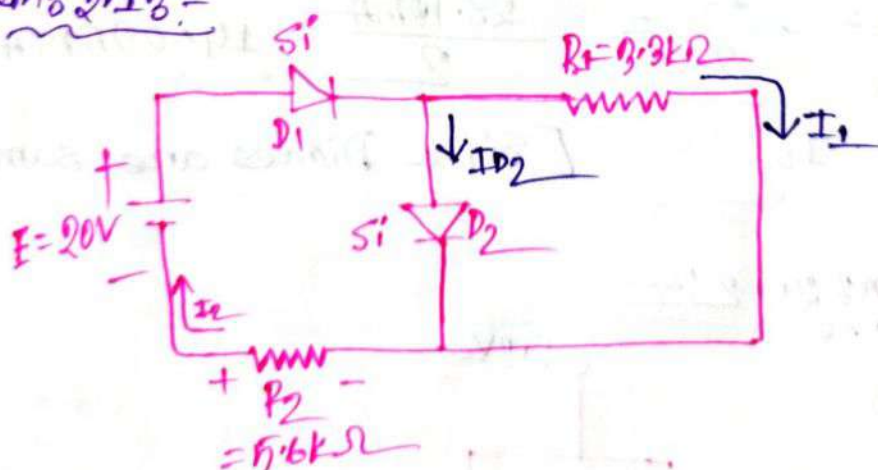
[Since current like to pass through the less resistance path]

we know:-

$$\Rightarrow E - V_D - V_0 = 0$$

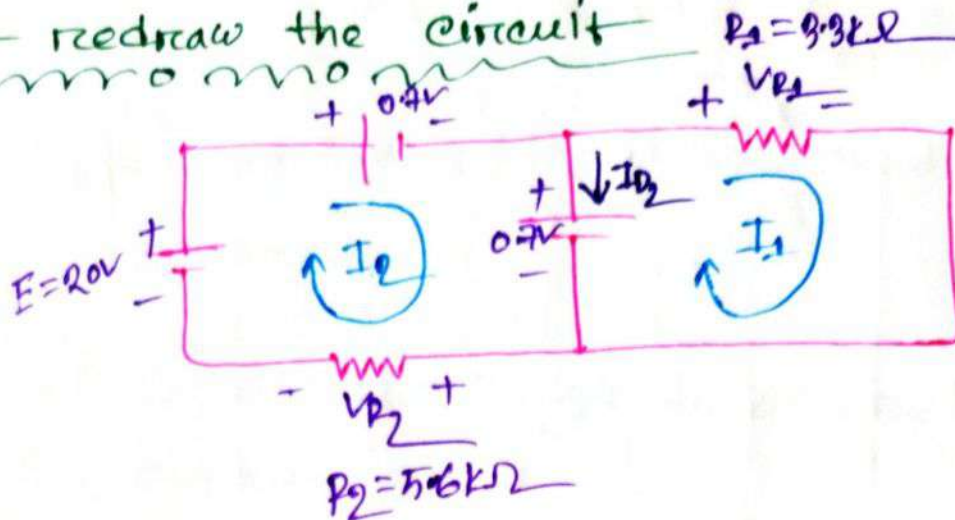
$$\therefore V_0 = E - V_D = 12 - 0.7 = 11.3 \text{ V}$$

#problem 2.18:-



⇒ Determine I_1, I_2, I_{02}

⇒ 1st redraw the circuit



we know:-

$$\Rightarrow E - V_{D1} - V_{D2} - V_{R2} = 0$$

$$\therefore V_{R2} = E - V_{D1} - V_{D2} = 20 - 0.7 - 0.7 = 18.6V$$

$$\Rightarrow V_{D2} - V_{R1} = 0$$

$$\therefore V_{R1} = V_{D2} = 0.7V$$

$$\Rightarrow I_1 = \frac{V_{R1}}{R_1} = \frac{0.7}{9.3k} = 0.212mA$$

$$\Rightarrow I_2 = \frac{V_{R2}}{R_2} = \frac{18.6}{5.6k} = 3.32mA$$

$$\Rightarrow I_{D2} = I_2 - I_1 = 3.32mA - 0.212mA$$

$$= 3.11mA$$

Ans:-

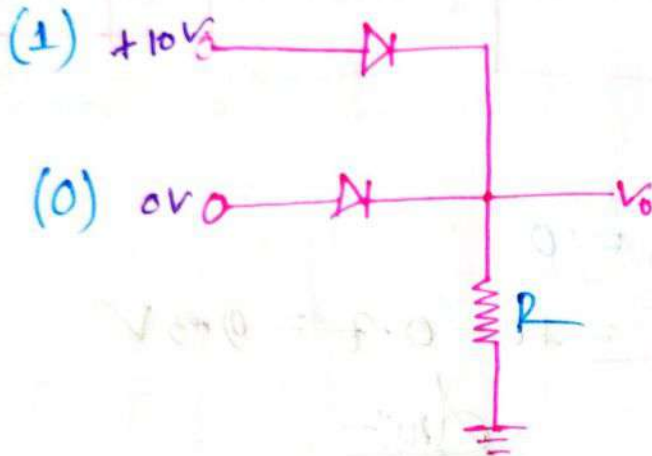
#Diodes Application

Ch 02

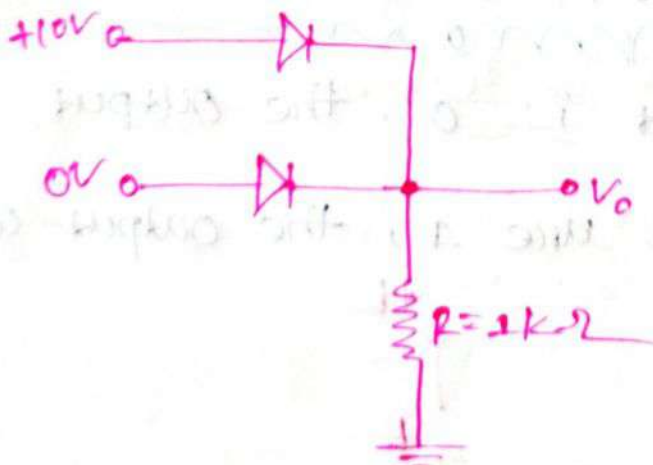
Logic OR gates:-

1. if any input is 1, its diode conducts, pulling the output 1.

2. if all inputs are 0, all diodes are reverse-biased, the output 0.

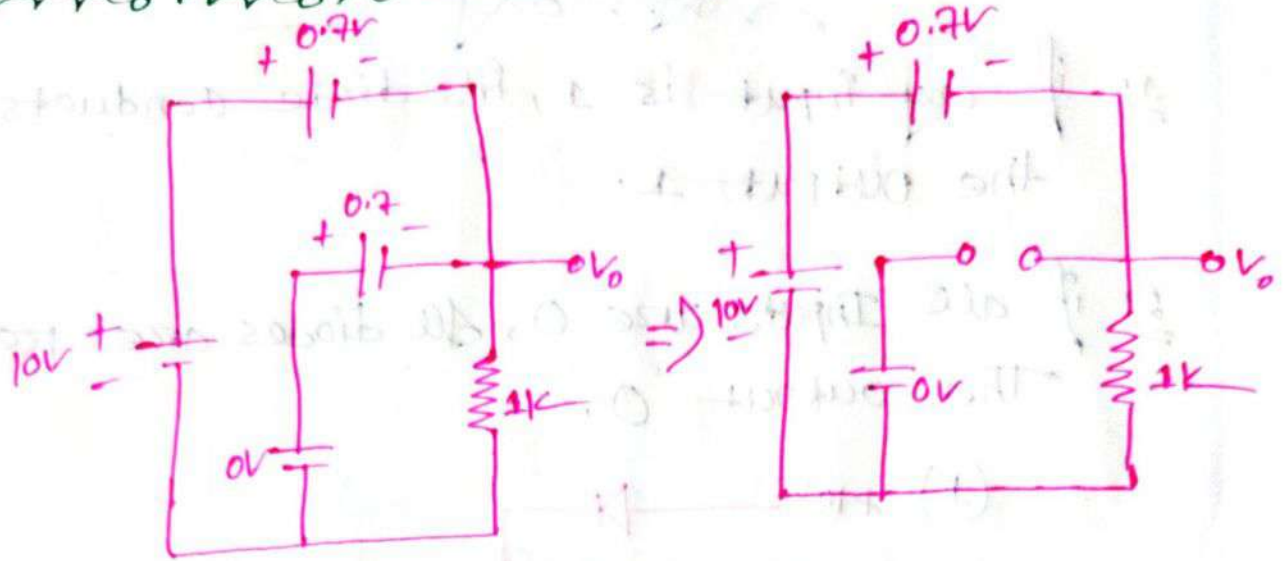


problem: 2.14:-



⇒ Determine V_O for the network.

⇒ not redraw the circuit:-



We know:-

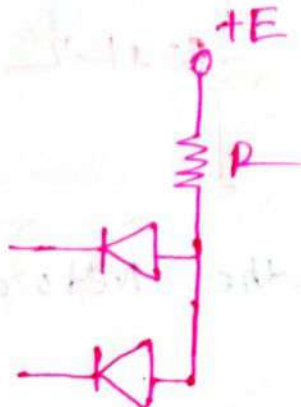
$$\Rightarrow E - V_{D1} - V_o = 0$$

$$\therefore V_o = E - V_{D1} = 10 - 0.7 = 9.3V$$

Ans:-

AND - Gates:-

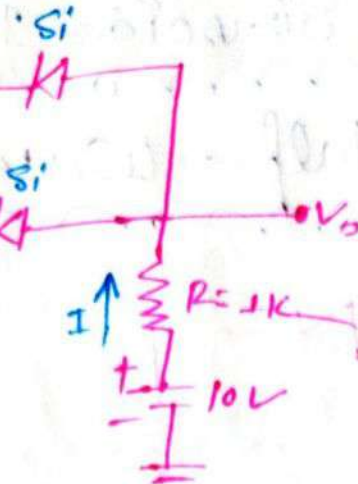
1. if any input is 0, the output will be zero(0)
2. if all inputs are 1, the output will be 1.



Problem Q.15:-

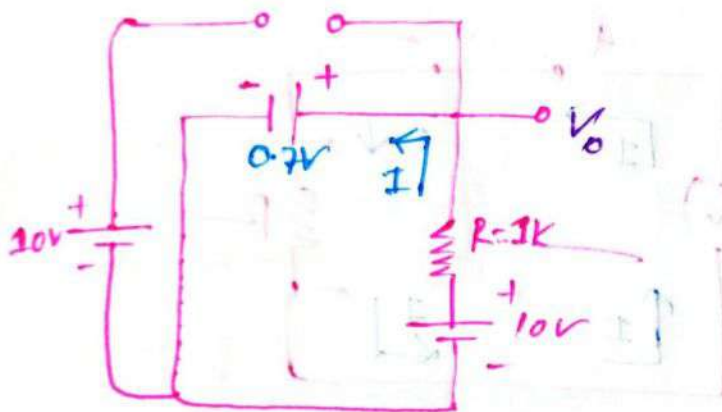
(1) $E_1 = 10V$

(2) $E_2 = 0V$



⇒ Determine the output level and I, V_0

⇒ Re draw the circuit:-



We know:-

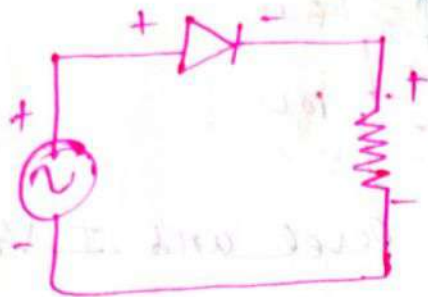
$$\Rightarrow I = \frac{E - V_D}{R} = \frac{10 - 0.7}{1k} = 9.3mA$$

$$\Rightarrow V_0 = E - V_D = 10 - 0.7 = 9.3V$$

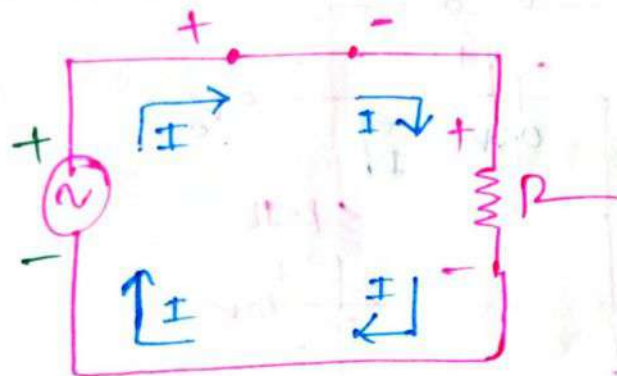
Sinusoidal Inputs:-

Half-Wave:-

☒

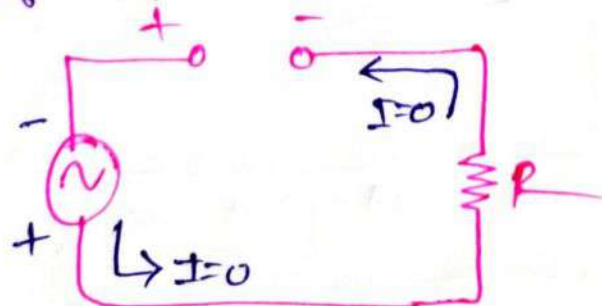


☒ For positive half-cycle:-



⇒ The Diode is Forward Bias. So that it acts like a short circuit.

☒ For Negative half-cycle:-



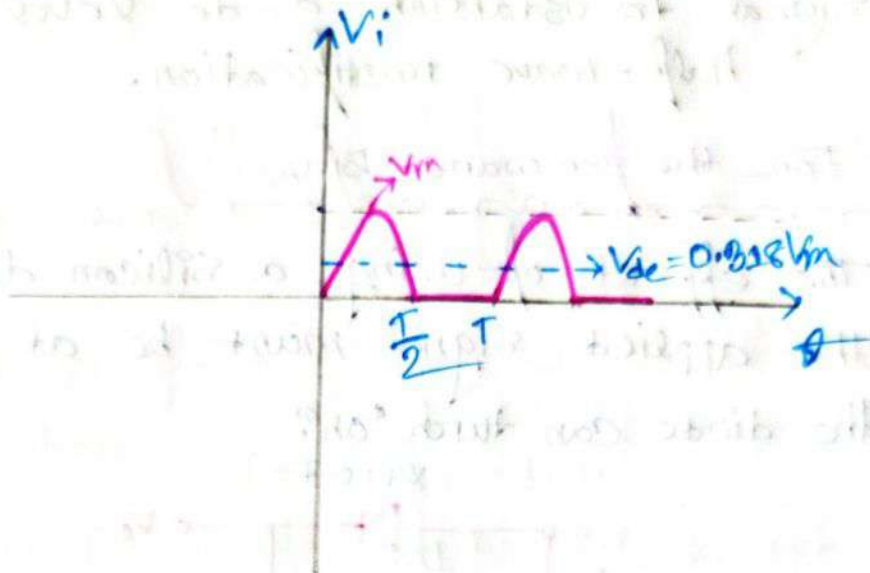
⇒ Now the diode is Reverse Bias. So that it acts like an open circuit.

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⇒ During this process the Sinusoidal wave look like this:-
mon



$$\therefore V_{dc} = 0.318 V_m$$

Rectification:-
mon

⇒ It is a process of converting Alternating Currents (AC) into Direct Currents (DC).

There are three types of diode rectification

1. Half-wave rectifier.

2. Full-wave rectifier

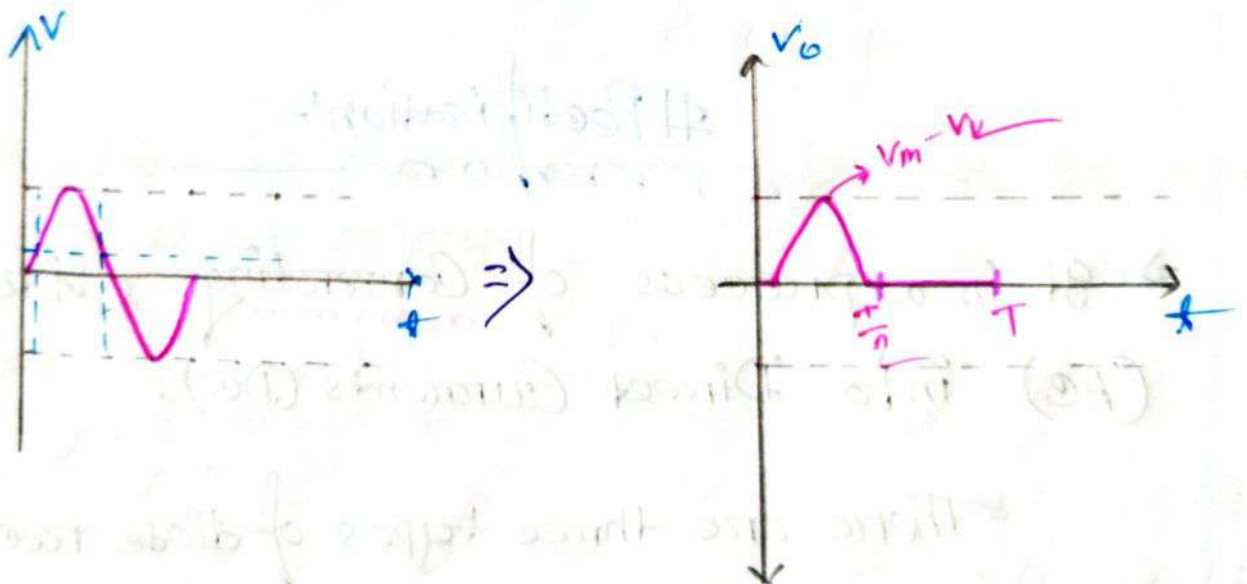
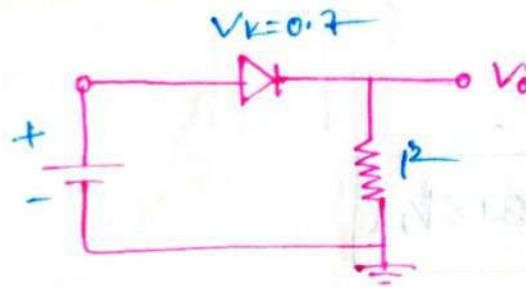
3. Center-tapped full-wave rectifier.

HALF-WAVE Rectification:-

⇒ The process of removing one-half the input signal to establish a dc level is called half-wave rectification.

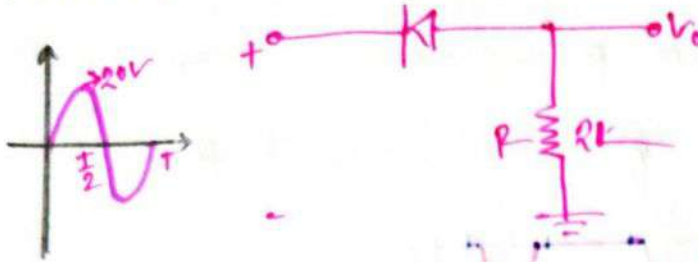
For the forward Bias:-

⇒ The effect of using a silicon diode with $V_K = 0.7V$ the applied signal must be at least $0.7V$ before the diode can turn "on".



$$\therefore V_{De} = 0.318(V_m - V_K)$$

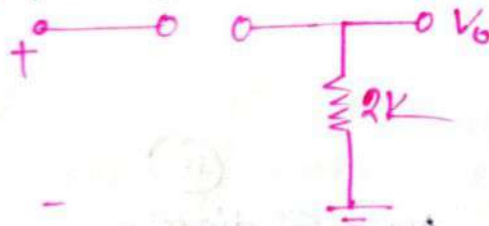
problems 2:16:-



- Sketch the output V_o & determine the dc level of the output.
- Repeat part (a) if the ideal diode is replaced by a Silicon diode.
- Repeat part (a), (b) if the V_m is increased to 200V, and compare the solutions.

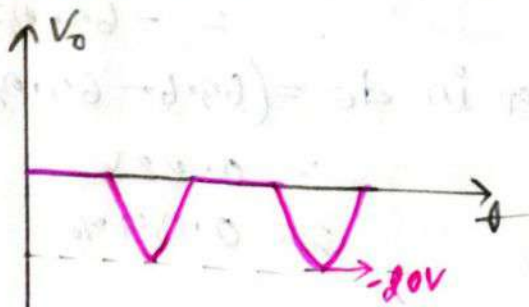
(a)

⇒ For positive half-cycle

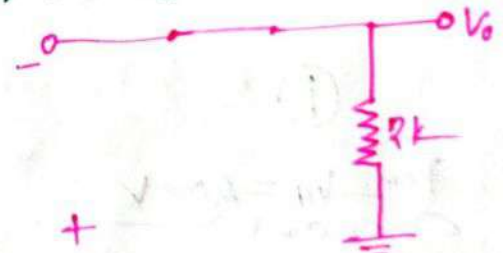


It's act like an open circuit

∴ the output:-



For Negative half-cycle:-



It's act like a short circuit

We know:

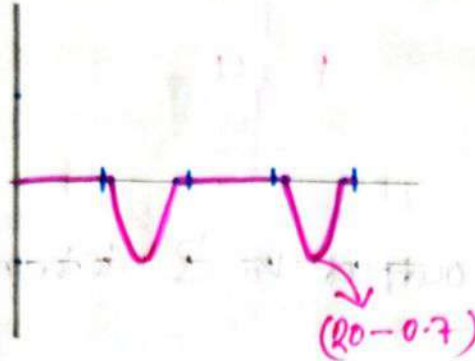
$$\begin{aligned} V_{oe} &= 0.318 \times V_m \\ &= 0.318 \times (-20) \\ &= -6.36V \end{aligned}$$

Ans:-

(b)

For Si diode:-
mon

⇒ The output
mon



We know:-
mon

$$\begin{aligned} V_{De} &= -0.318(20 - V_K) \\ &= -0.318(20 - 0.7) \\ &= -6.14V \end{aligned}$$

$$\begin{aligned} \text{The result drop in } D_e &= (6.36 - 6.14)V \\ &= 0.22V \\ &\approx 3.46\% \end{aligned}$$

(c)

(i)

For $V_m = 200V$
mon

We know:-
mon

$$\begin{aligned} V_{De} &= -(0.318 \times V_m) \\ &= -0.318 \times 200 \\ &= -63.6V \end{aligned}$$

$$\begin{aligned} \text{The result drop in } d_e &= (63.6 - 63.3774)V \\ &= 0.22V \\ &\approx 0.34\% \end{aligned}$$

(ii)

Now Si diode:-
mon

We know:-
mon

$$\begin{aligned} V_{De} &= -0.318 \times (V_m - V_K) \\ &= -0.318 \times (200 - 0.7) \\ &= -63.3774V \end{aligned}$$

$$\begin{aligned} &= 0.22V \\ &\approx 0.34\% \end{aligned}$$

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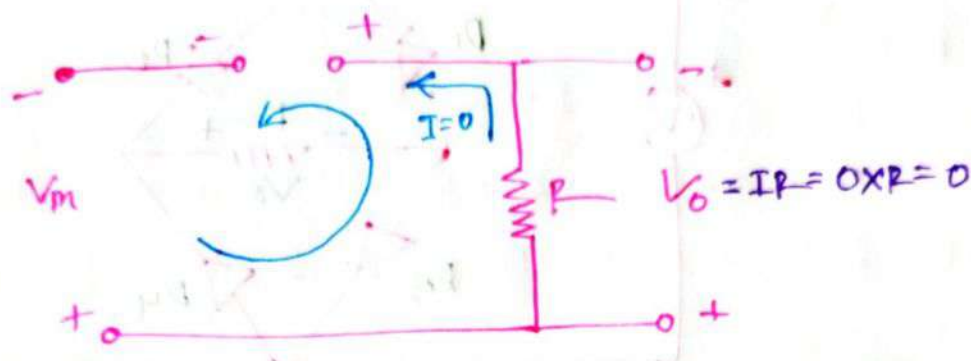
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Peak Inverse Voltage (PIV):-

⇒ The maximum voltage a diode can withstand in the reverse biased direction before breakdown.



⇒ Now Applying KVL:-

→ It is obvious that the PIV rating of the diode must equal or exceed the peak value of the applied voltage.

$$\therefore \text{PIV rating} \geq V_m$$

→ for half-wave rectifier & full-wave bridge rectifier

$$\text{PIV} \geq 2V_m$$

→ for CT transformer full-wave rectifier.

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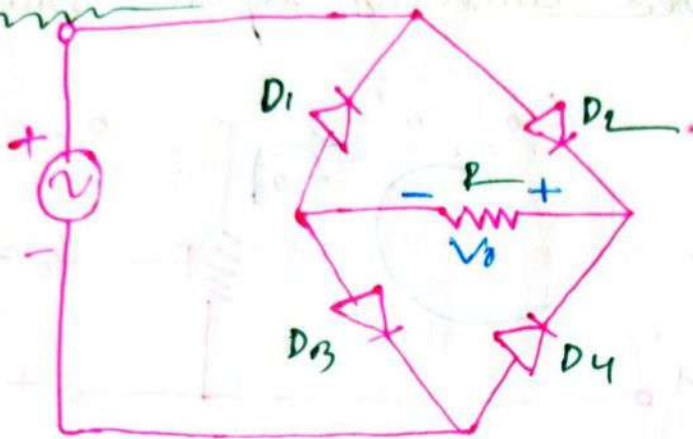
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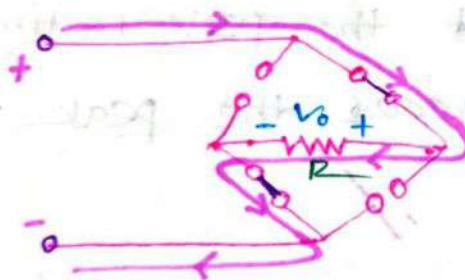
Full-wave Rectification:-

1. Bridge Network

For Ideal Diode:-

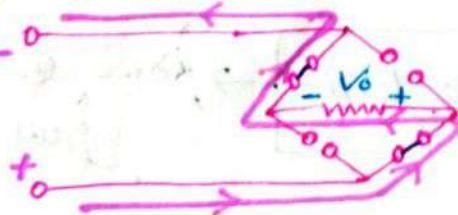


For positive half-cycle:-



$\Rightarrow$ The D_2 & D_3 diodes Acts like a short circuit, on the other hand D_1 & D_4 Acts like an open circuit.

For the negative half-cycle:-



$\Rightarrow$ The D_1 & D_4 diodes Acts like a short circuit, on the other hand D_2 & D_3 are Acts like an open circuit

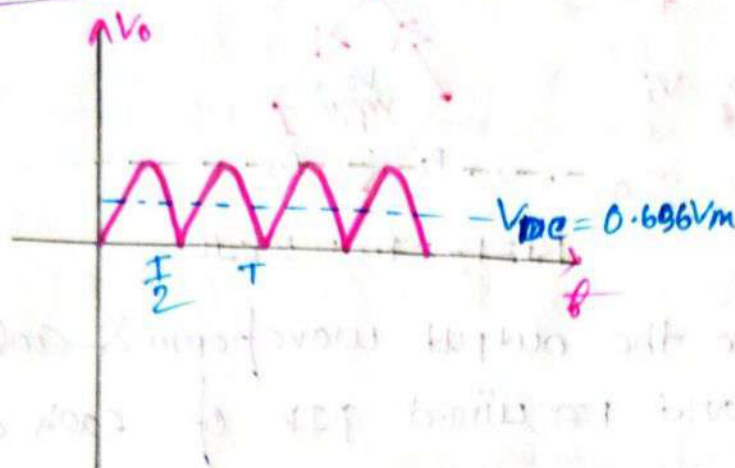
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⇒ During this process the sinusoidal wave look like this:-

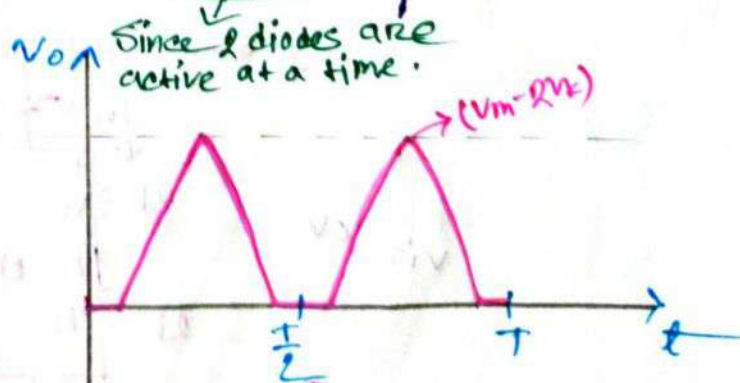
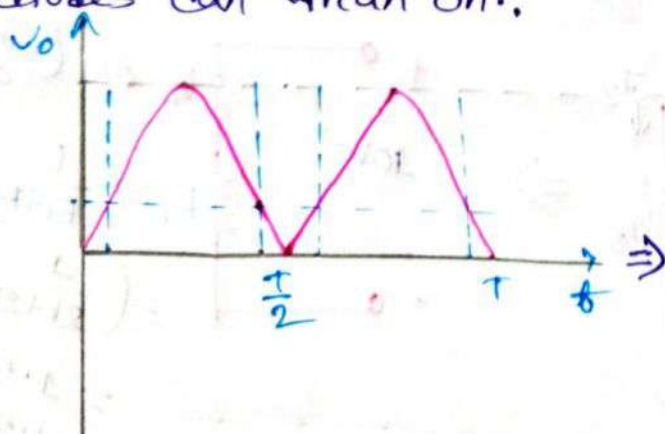


⇒ This happens because the direction of current through the load resistance is always the same.

$$\therefore V_{dc} = 0.636 V_m$$

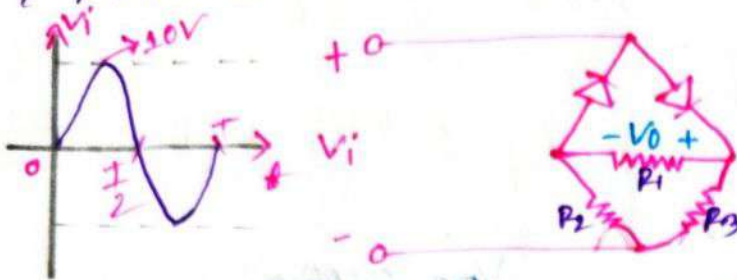
✗ For Silicon Diode:-

⇒ The effect of using silicon diodes with $V_k = 0.7V$, The Applied signal must be at least 0.7×2 before the diodes can turn "on".



$$\therefore V_{dc} = 0.636 (V_m - 2V_k)$$

problems 2.17:-



$$R_1 = R_2 = R_3 = 2k\Omega$$

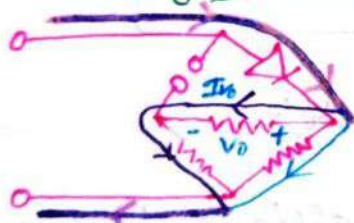
⇒ Determine the output waveform & calculate the output DC level and required PIV of each diode.

⇒ In given:-

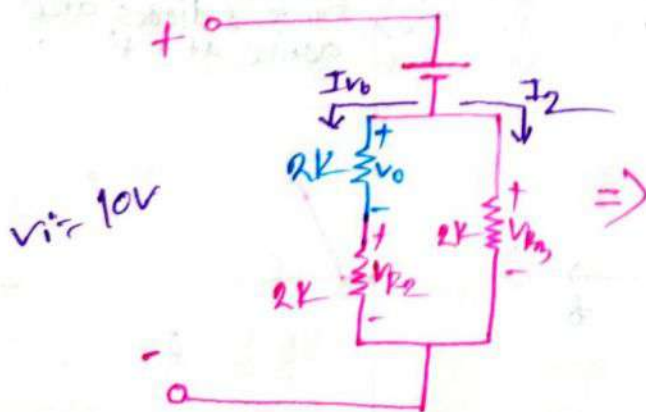
$$V_m = 10V$$

$$R_1 = R_2 = R_3 = 2k\Omega$$

Now:- For positive half cycle:-



Now redraw the circuit:-



$$V_0 = 0V \quad [\text{Since it was an ideal diode}]$$

$$R_T = \left(\frac{1}{R_1 + R_2} + \frac{1}{R_3} \right)^{-1}$$

$$= \left(\frac{1}{2k + 2k} + \frac{1}{2k} \right)^{-1}$$

$$= 1.33 \times 10^3 \Omega$$

$$= 1.33k\Omega$$

$$[V_{PIV} = 10V]$$

We know:-

$$I = \frac{V_i}{R_T} = \frac{10}{1.33k} = 7.52mA$$

∴ The current passing through the load voltage:-

using CDR:-

$$I_{V_0} = \frac{R_T}{R'} \times I$$

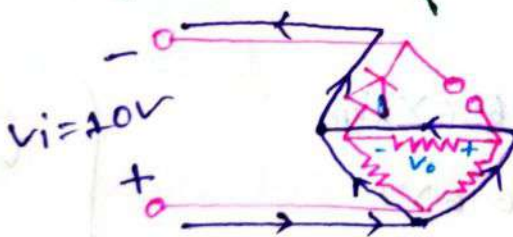
$$= \frac{1.33k}{4k} \times 7.52mA$$

$$= 2.5mA$$

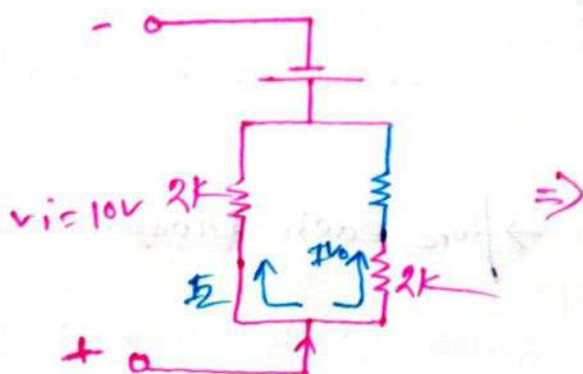
∴ The output:-

$$V_0 = I_{V_0} \times R = 2.5 \times 2 = 5V$$

Now:- For Negative half-cycle:-



Now redraw the circuit



$$V_0 = 0V$$

$$R_T = \left(\frac{1}{R_1 + R_2} + \frac{1}{R_3} \right)^{-1}$$

$$= \left(\frac{1}{4k} + \frac{1}{2k} \right)^{-1}$$

$$= 1.33k\Omega$$

We know:-

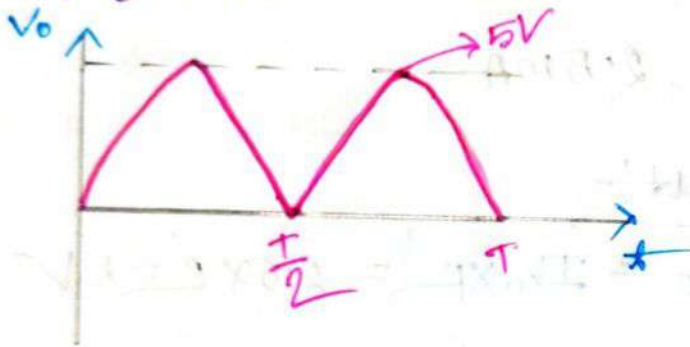
$$I = \frac{V_i}{R_T} = \frac{10}{1.33k} = 7.52mA$$

$$\therefore I_{v0} = \frac{R_T}{R'} \times I = \frac{2.33k}{4k} \times 7.52mA = 2.5mA$$

$\therefore$ The output:-

$$V_0 = I_{v0} \times R_L = 2.5mA \times 2k = 5V$$

$\therefore$ The output wave:-



For DC:-

$\Rightarrow$ We know:-

$$\begin{aligned} V_{dc} &= 0.636 \times V_m \\ &= 0.636 \times 5 \\ &= 3.18V \end{aligned}$$

$$\left. \begin{array}{l} \text{here,} \\ V_m = 5V \end{array} \right\}$$

For PIV required:-

$\Rightarrow$ We know:-

$$PIV \geq V_m$$

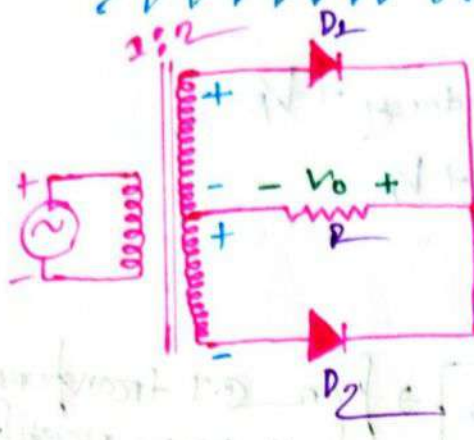
$$\therefore PIV \geq 5V \rightarrow \text{for each Diode.}$$

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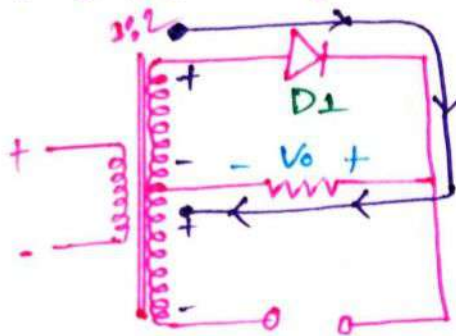
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2. Center-Tapped Transformer

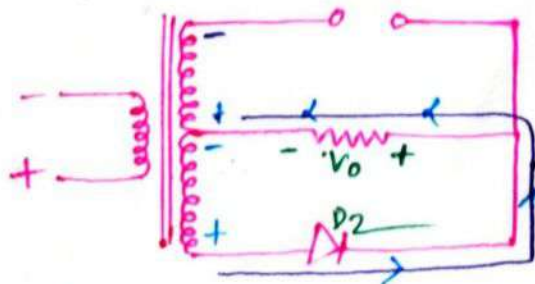


⇒ For positive half-cycle:-



⇒ The D_1 diodes act like a short circuit & the D_2 diode act like an open circuit [Since for positive half cycle the D_2 Diode is in reverse Bias.]

⇒ For Negative half-cycle:-



⇒ The D_2 diodes act like a short circuit & the D_1 diode act like an open circuit [Since for negative half cycle the D_1 diode is in reverse Bias & the D_2 diode is Forward Bias]

PIV Analysis:-

$$\begin{aligned}\therefore \text{PIV} &= V_{\text{secondary}} + V_r \\ &= V_m + V_m \\ \text{PIV} &= 2V_m\end{aligned}$$

$\therefore \boxed{\text{PIV} \geq 2V_m}$ $\rightarrow$ for C.T transformer,
full wave rectifier.



#CLIPPERS
of
LITLES

Clippers:-

⇒ A circuit which removes or cuts unwanted portion of a waveform.

- 1. It is used for Amplitude limiting
- 2. Also for noise cancelling

Classification of Clipper:-

⇒ There are two types.

1. Series Clipper

2. Shunt/parallel clipper.

Series Clipper:-

1. UnBiased Series Clipper/Basic $(-V_c, +V_c)$

2. Biased Series Clipper $(-V_c, +V_c)$

Parallel Clipper:-

1. UnBiased shunt Clipper $(-V_c, +V_c)$

2. Biased shunt clipper $(-V_c, +V_c)$

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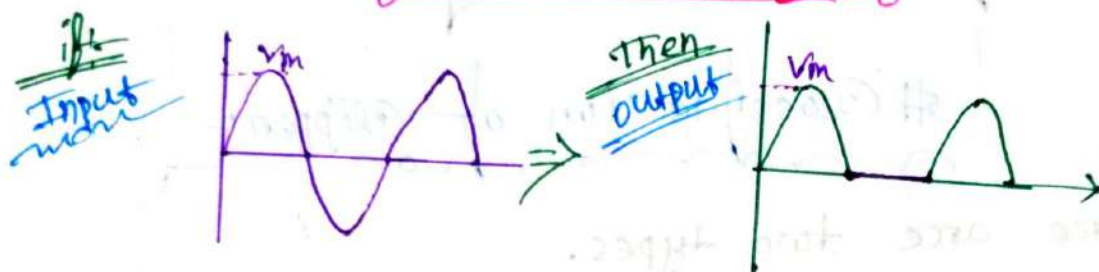
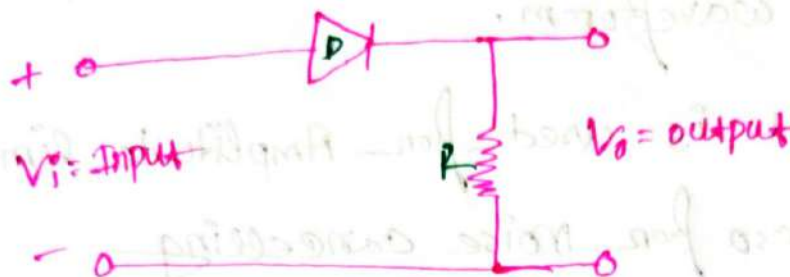
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UnBiased Series Clipper :-

Negative Series Clipper :-



Case: 01 :-

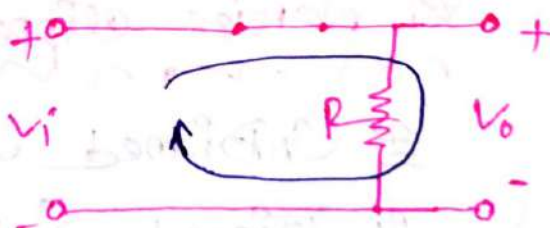
→ If $V_i > 0$ / For positive half cycle :-

⇒ Diode D is in Forward Bias & Act AS short circuit.

∴ Now Apply KVL :-

$$V_i - V_o = 0$$

$$\therefore V_o = V_i$$



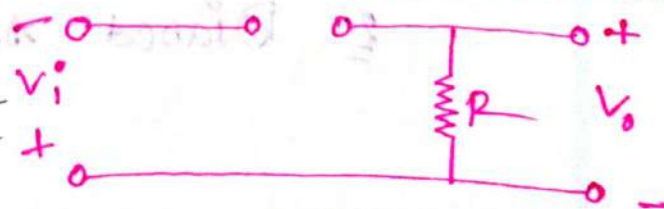
Case: 02 :-

→ If $V_i < 0$ / For Negative half cycle :-

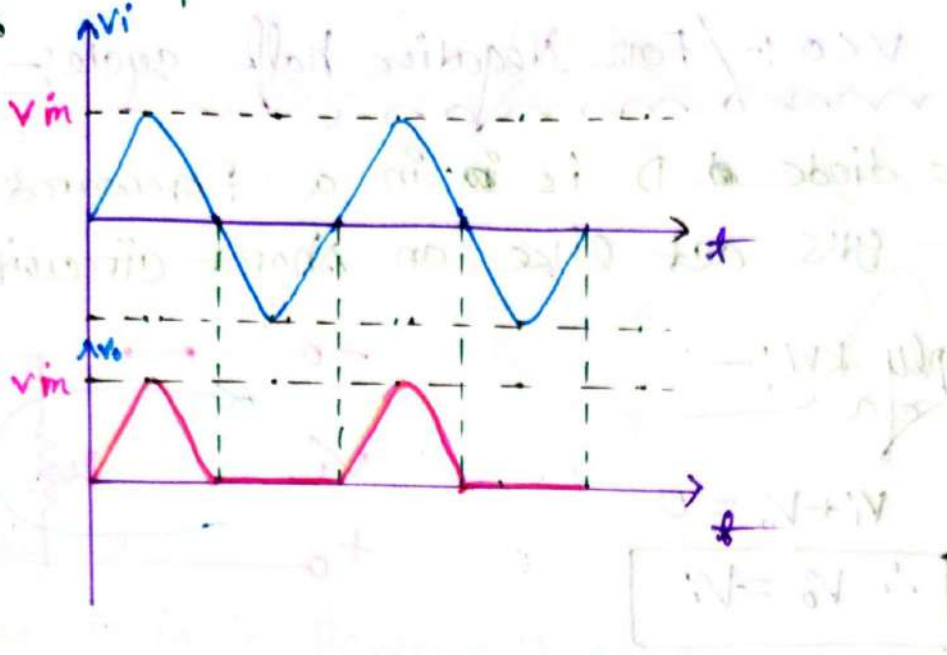
⇒ Diode D is in Reverse Bias & Act as open circuit

$$\therefore V_o = 0$$

[Since the circuit is not closed & there is no current supply]



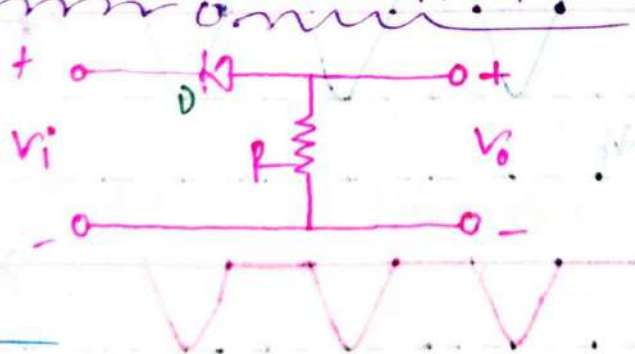
⇒ During this process the wave look like this:-



Note:-

"Since the negative portion is clipped, we called it a negative Series clipper."

Positive Series Clipper:-

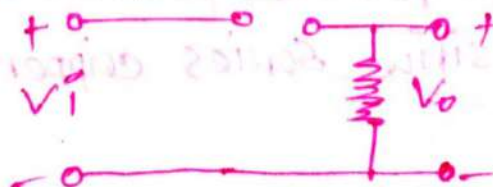


Case: 01:-

⇒ if $V_i > 0$:- / For positive half cycle:-

⇒ Now Diode D is in Reverse Bias, so that acts like an open circuit

$\therefore V_o = 0$



Case: 02:-

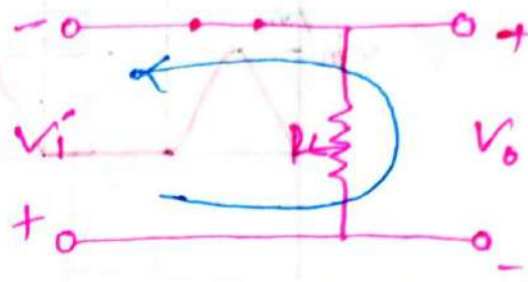
→ if $V < 0$:- / For Negative half cycle:-

⇒ Now diode D is in a Forward Bias, So that gts act like an short circuit.

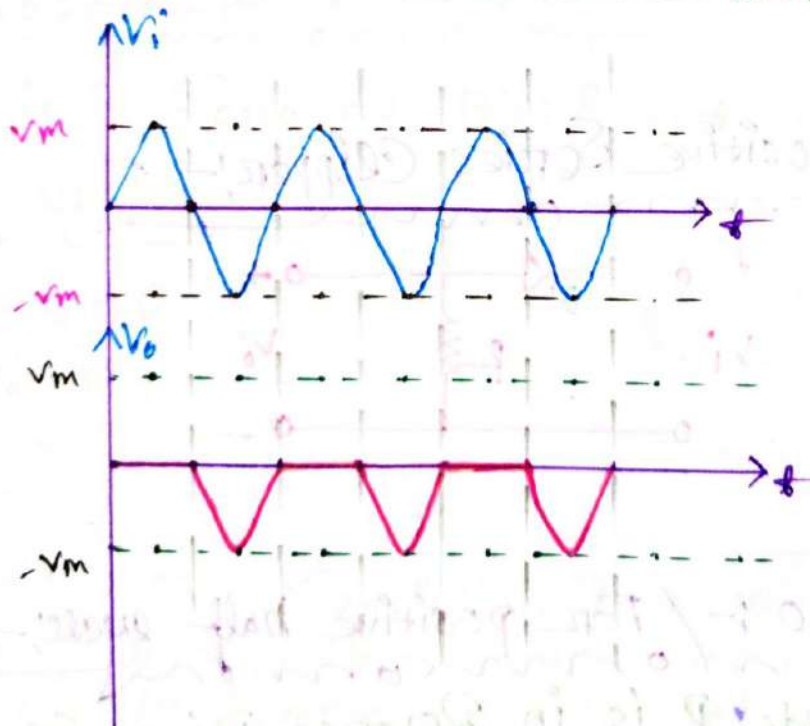
∴ Apply KVL:-

$$V_i + V_o = 0$$

$$\therefore V_o = -V_i$$



⇒ During this process the wave look like this:-



Note:-

"Since positive position is clipped, we called it in a positive series clipper."

TOPIC NAME : _____

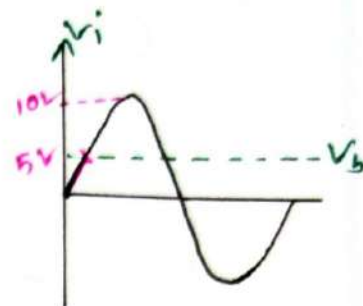
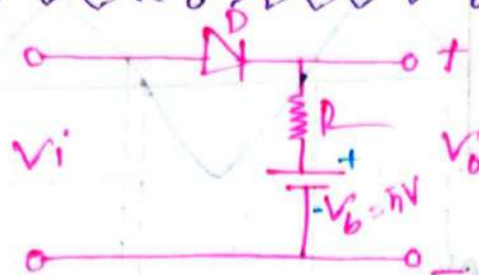
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#Biased Series Clipper-

Bianed Negative Series clipper:-



Case 1:-

if $V_i < V_b$:-

⇒ The Diode D is in Reverse Bias And Act as open circuit

∴ Apply KVL:-

$$V_b - V_o = 0$$

$$\therefore V_o = V_b$$

Case 2:-

if $V_i > V_b$:-

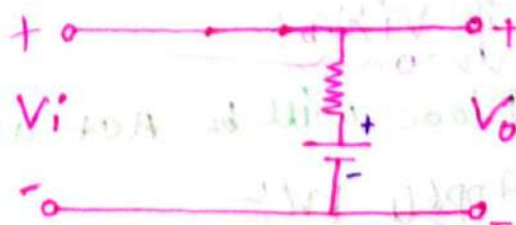
⇒ Diode D is in Forward Bias And Act as short circuit

∴ Apply KVL:-

$$V_o - V_i = 0$$

$$\therefore V_i = V_o$$

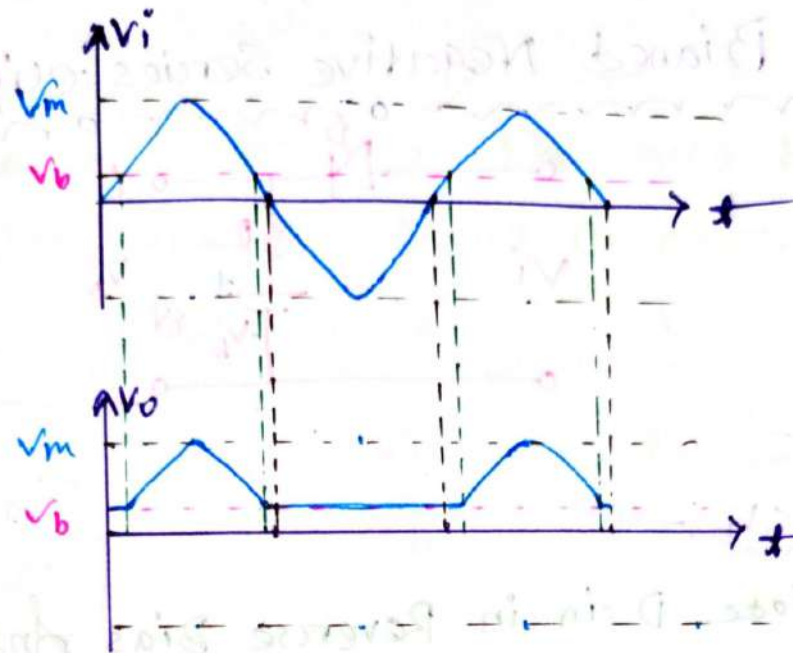
$$V_o = V_i$$



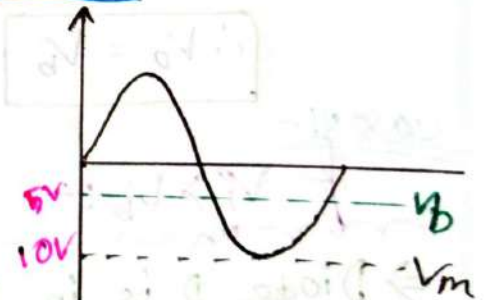
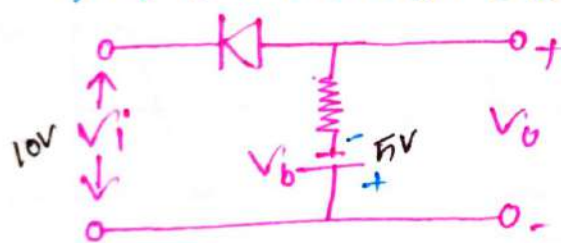
$$0 = 5V + 0V$$

$$5V - = 0V$$

During this process the wave look like this:



Biased positive Series clipper:-



Case: 1:-

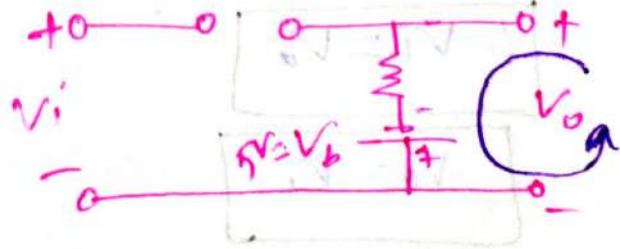
⇒ if $V_i > V_b$:-

⇒ Diode will be Act as open circuit:-

∴ Apply KVL:-

$$V_o + V_b = 0$$

$$\therefore V_o = -V_b$$



Case:2:-

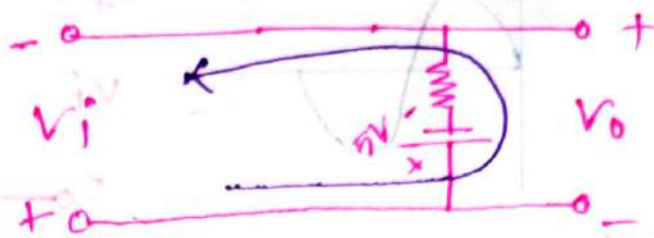
if $V_i < V_o$:-

$\Rightarrow$ Diode will be F.B & Act as short circuit.

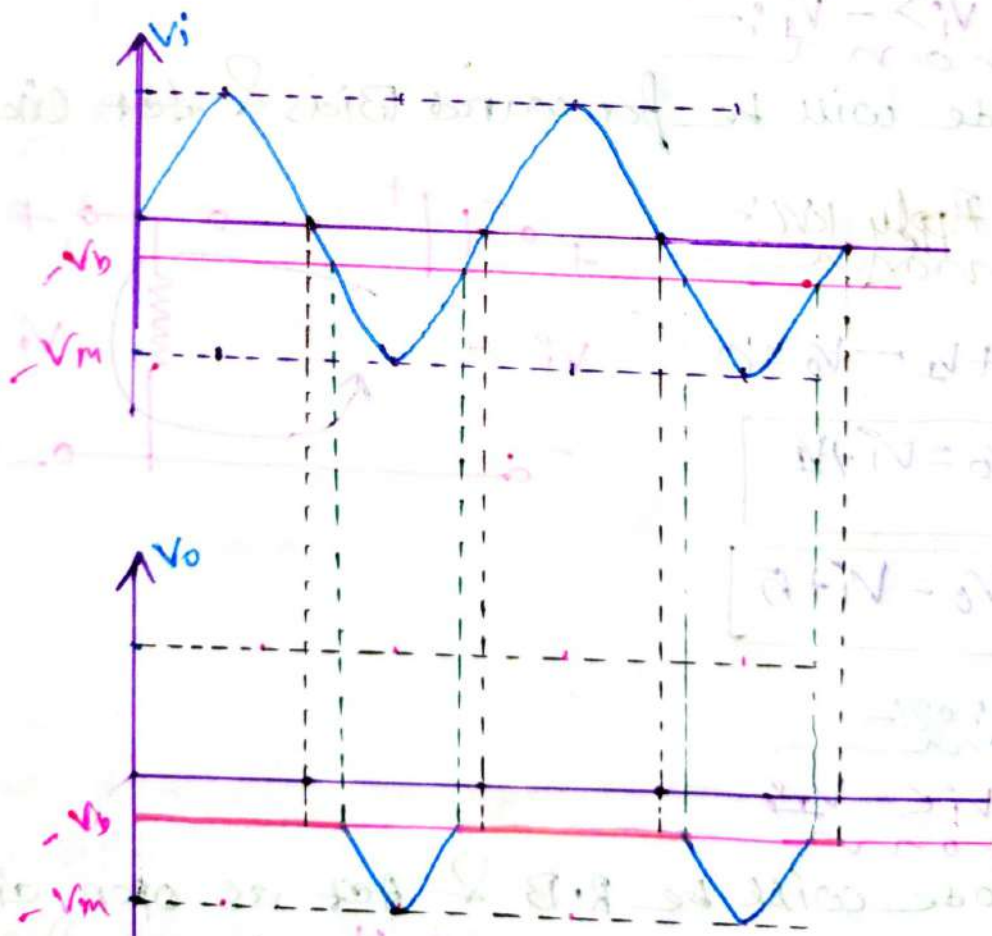
$\therefore$ Apply KVL:-

$$V_i + V_o = 0$$

$$\therefore V_o = -V_i$$

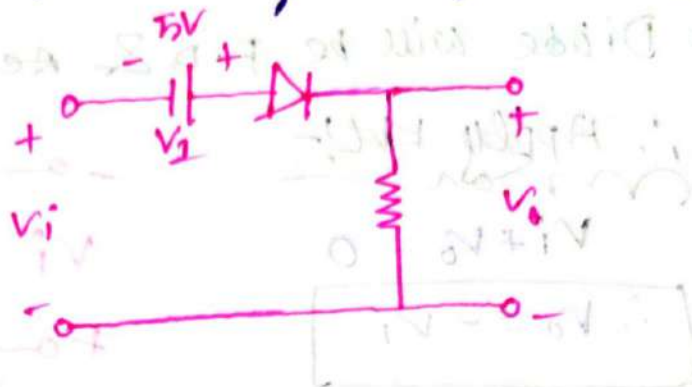
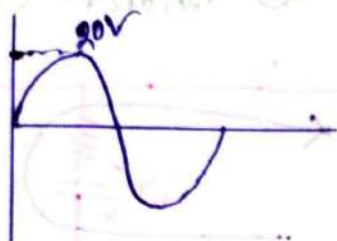


$\Rightarrow$ During this process the wave look like this:-



problem: 2.18:-

⇒ Determine the output waveform for the sinusoidal input.



Case: 01:-

if $V_i = 0$

⇒ Diode will be forward Bias.

∴ if $V_i > -V_1$:-

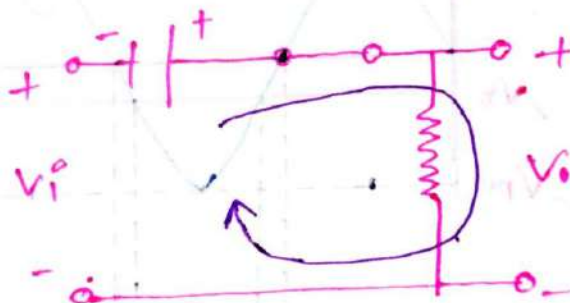
⇒ Diode will be forward Bias & Act like a short circuit

Now Apply KVL:-

$$V_i + V_1 - V_o = 0$$

$$V_o = V_i + V_1$$

$$V_o = V_i + 5$$



Case: 02:-

if $V_i < -V_1$:-

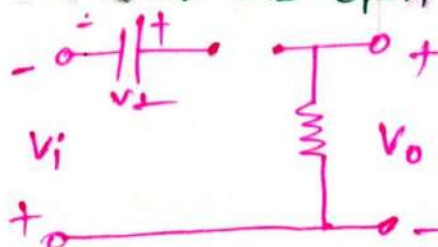
⇒ Diode will be R.B & Act as open circuit

∴ KVL:-

$$V_i + V_o - V_1 = 0$$

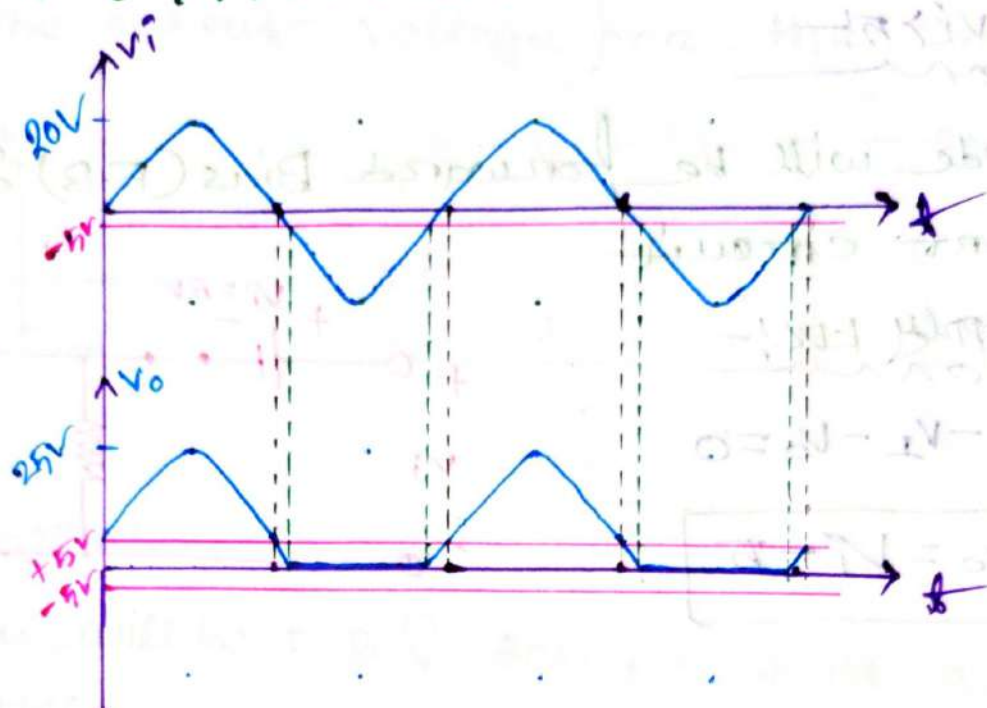
$$V_o = V_1 - V_i$$

$$V_o = 0$$



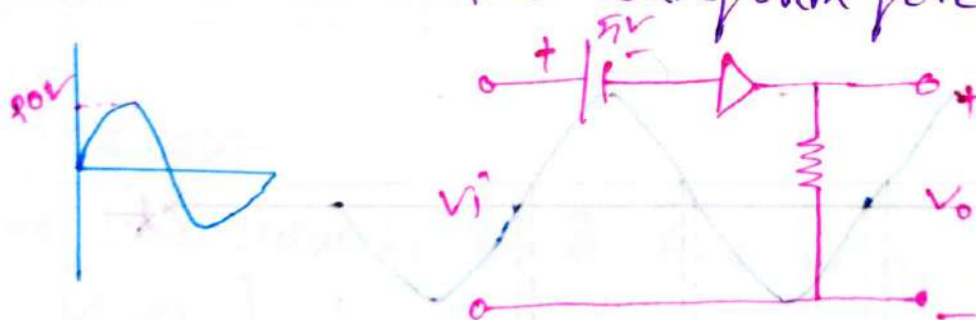
$$\left\{ \begin{array}{l} V_1 = 0 \\ V_i = 0 \end{array} \right.$$

So, the waveform for output (V_o):-



problem 2.18.1:-

⇒ Determine the output waveform for the input



Case 1:-

if $V_i = 0$;

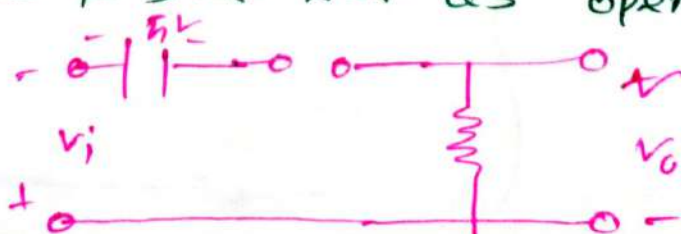
→ The Diode Act like an open circuit!

∴ if $V_i < 5V$:-

→ Diode will be R.B & Act as open circuit!

∴ KVL

$$V_o = V_i$$



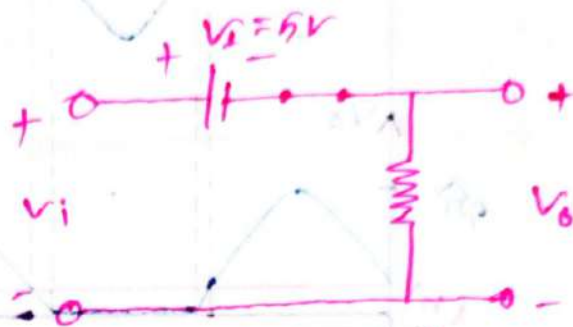
Case: 01:-
if $V_i > 5V$:-
then

→ Diode will be forward Bias (F.B) & Act as Short circuit

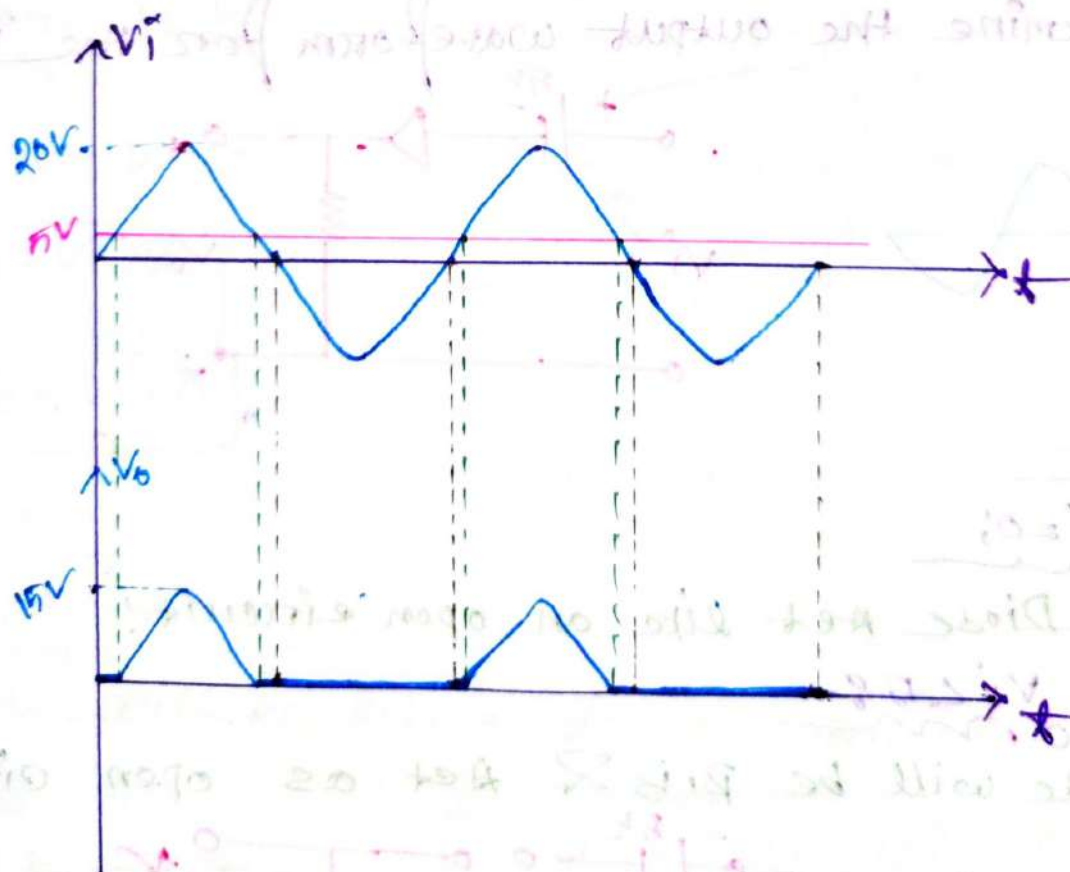
Now Apply KVL:-

$$V_i - V_1 - V_o = 0$$

$$\therefore V_o = V_i - 5$$



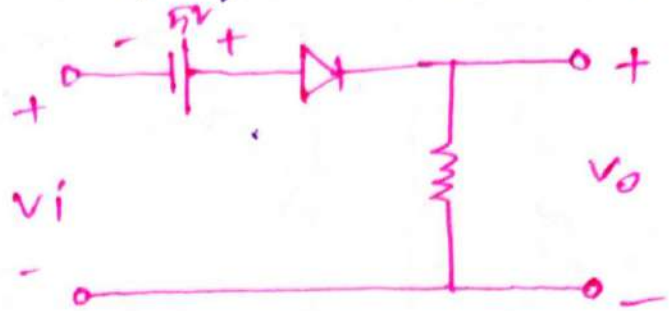
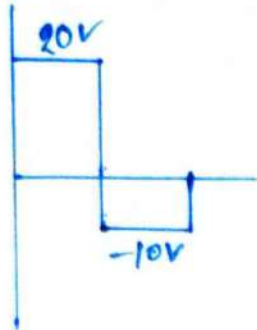
∴ So the waveform for output (V_o):-



$$V_o = V_i - 5$$

Problem 2.29:-

2) Find the output voltage for the Input.



Case 1:-

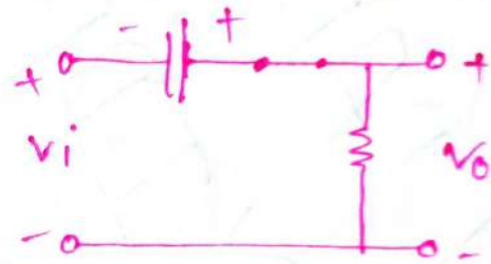
if $V_i = 20V$:-

The Diode will be F.B & Act As short circuit:-

Apply KVL:-

$$V_i + 5V - V_o = 0$$

$$\therefore V_o = V_i + 5V$$

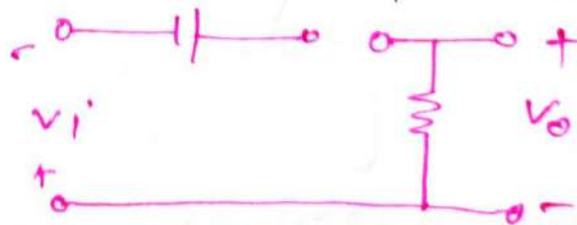


Case 2:-

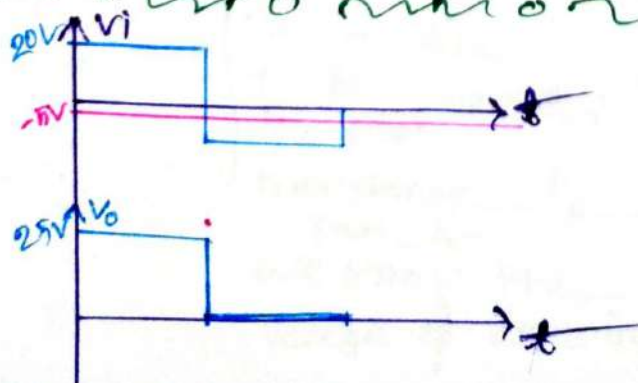
if $V_i = -10V$:-

The Diode will be R.B & Act as open circuit:-

$$\therefore V_o = 0$$



$\therefore$ The waveform of output (V_o):-



$$\therefore V_o = 20 + 5 = 25V$$

WATER FILTER

$$V_1 + V_2 - V_3 = 0$$

$$V_1 + V_2 = V_3$$

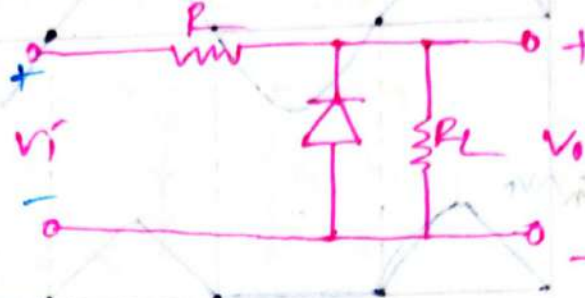
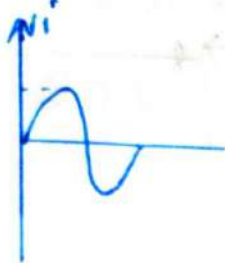
$$V_1 = 0$$

$$V_1 + V_2 = 0$$

$$V_1 = -V_2$$

Parallel Clippers:-

Negative Unbiased parallel clipper:-



Case (1):-

if $V_i > 0$:-

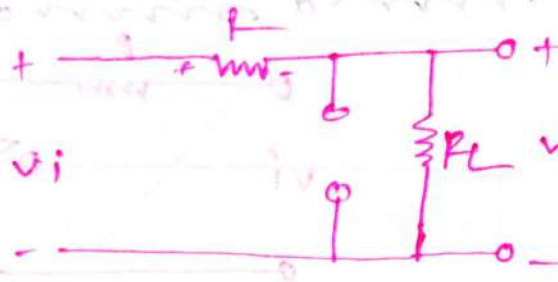
→ Diode will be R.B, act As open circuit

KVL:-

$$V_i - V_R - V_o = 0$$

$$\therefore V_o = -V_R + V_i$$

$$\Rightarrow \boxed{V_o \approx V_i}$$



here V_R is very small so we can exclude it.

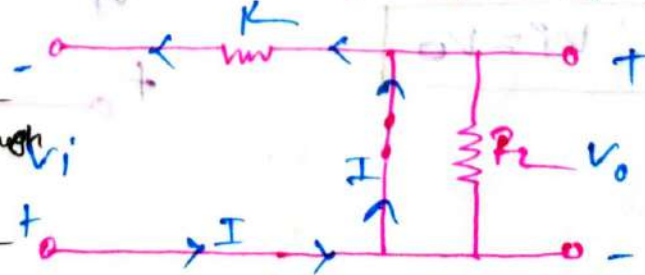
Case (2):-

if $V_i < 0$:-

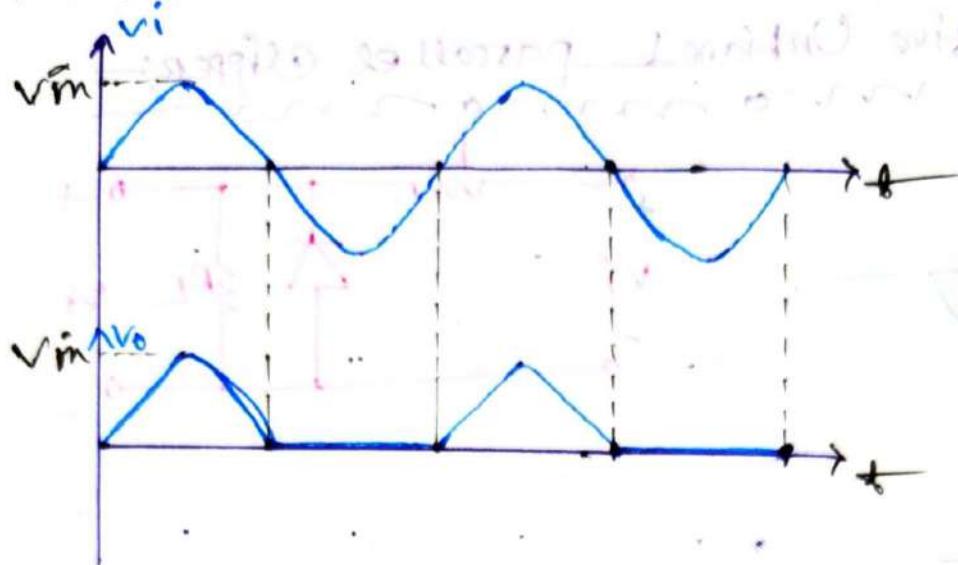
→ Diode will be F.B, Act As short circuit

$$\therefore \boxed{V_o = 0}$$

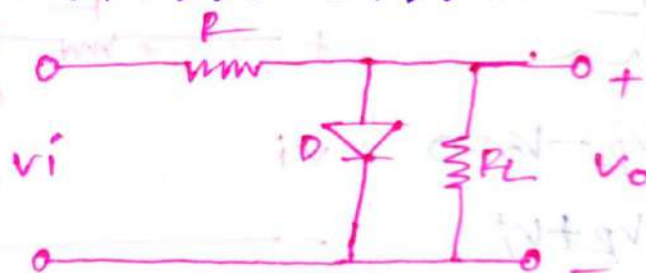
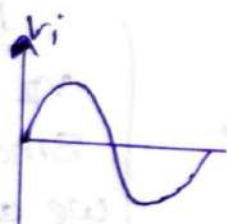
Since current like to pass through the less resistance path & we know the voltage of an wire = 0V



∴ So, the waveform for output (V_o) :-



⊗ Positive Unbiased parallel clipper :-



Case 1 :-

if $V_i < 0$:-

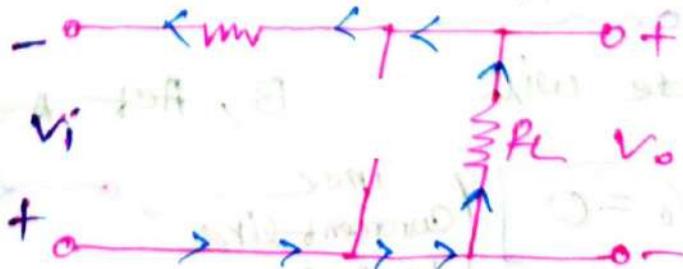
∴ Diode will be

∴ KVL :-

$$V_i - V_o = 0$$

$$\therefore V_i = V_o$$

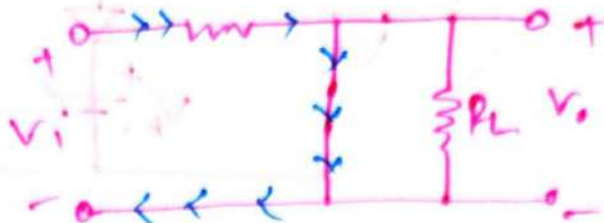
In P.B, Act as open circuit



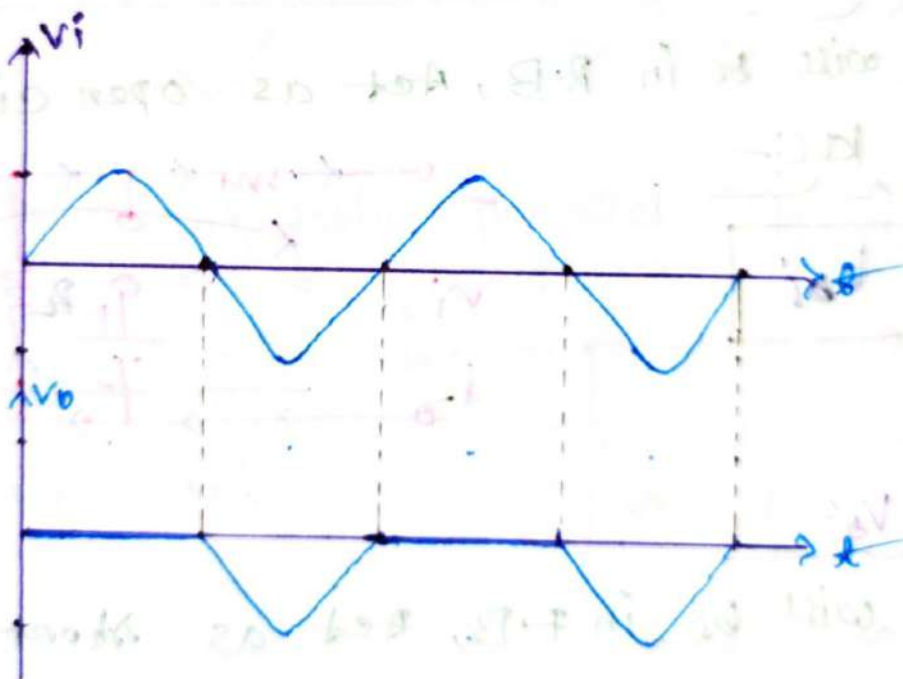
case 2:-
if $v_i > 0$:-
on

⇒ Diode will be F.B, Act as short circuit:-

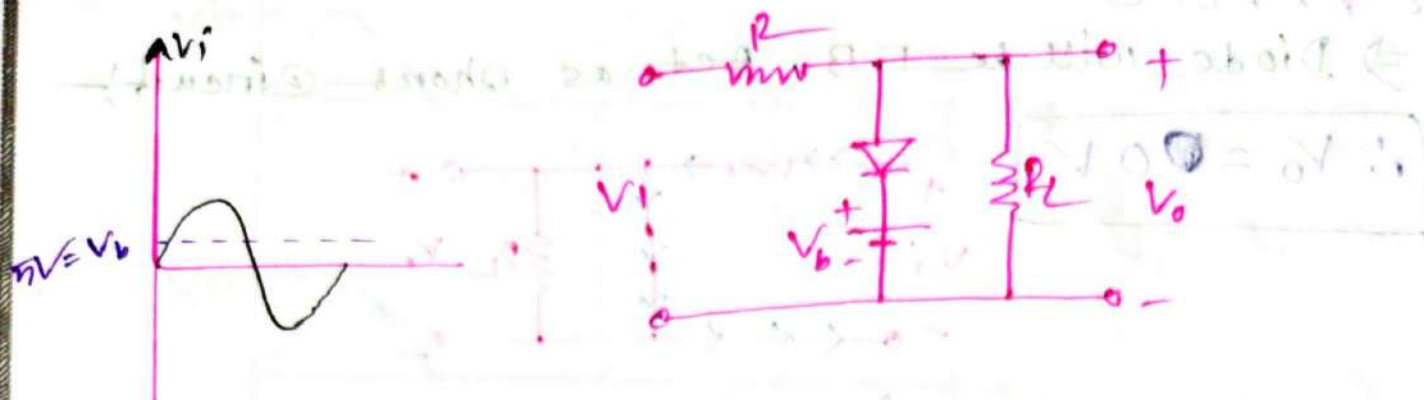
$\therefore V_o = 0V$



∴ So the waveform for output (v_o):-



Biased positive parallel clipper:-



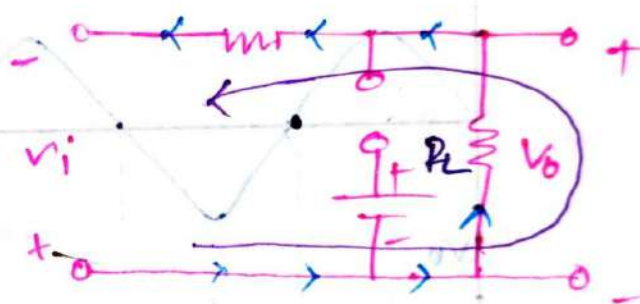
Case: 1:-

if $V_i < V_b$:-

$\Rightarrow$ Diode will be in R.B, Act as open circuit:-

$\therefore$ Apply KVC:-

$$V_o = V_i$$

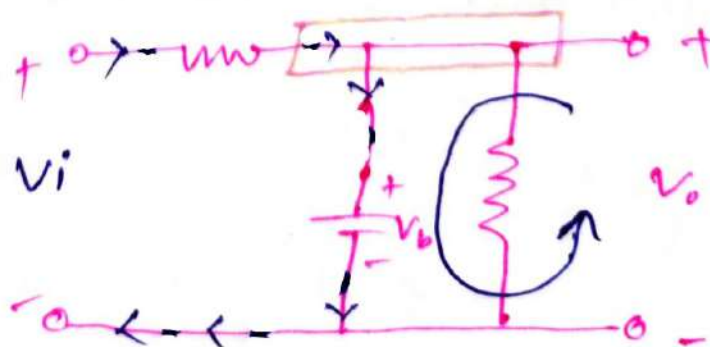


Case: 2:-

if $V_i > V_b$:-

$\Rightarrow$ Diode will be in F.B, Act as short circuit:-

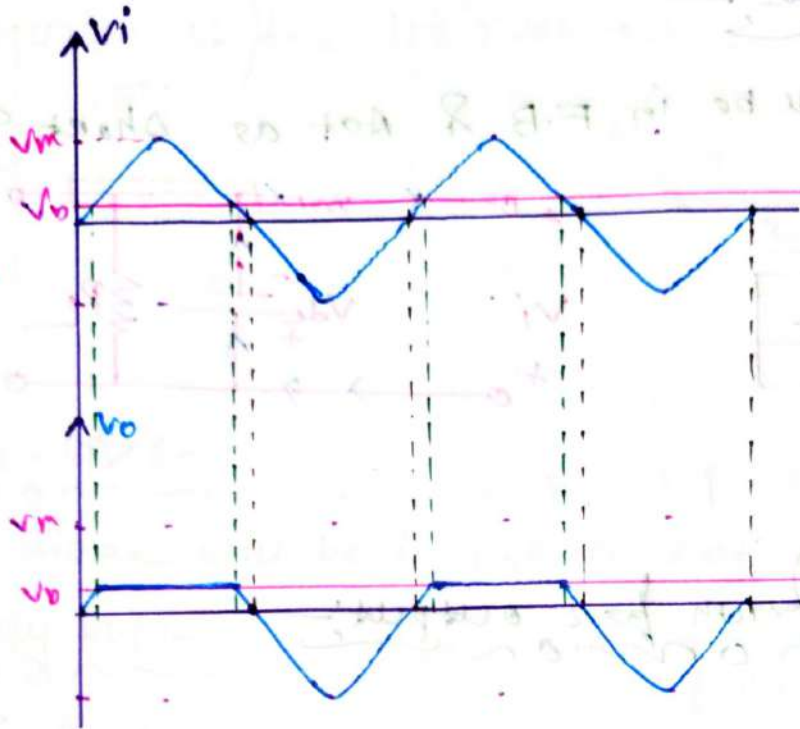
$\therefore$ Apply KVC:-



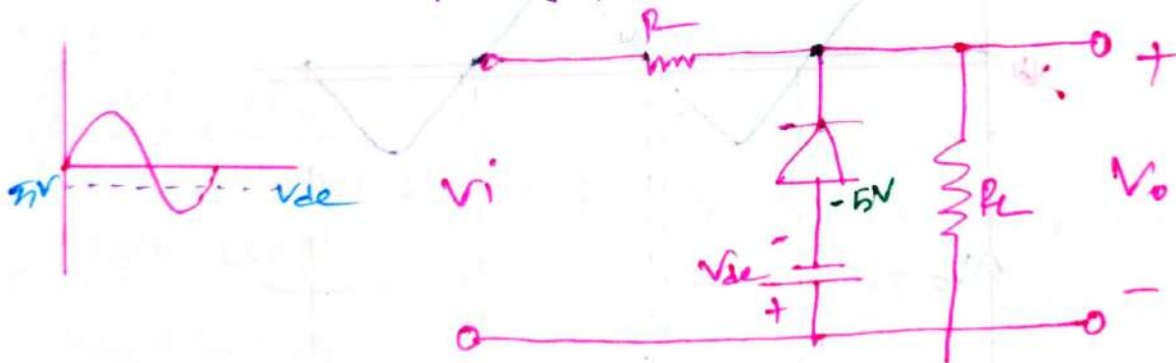
$$\therefore V_b - V_o = 0$$

$$V_o = V_b$$

∴ So, the waveform for output:-



Basic Negative parallel Clipper:-



Case 1:-

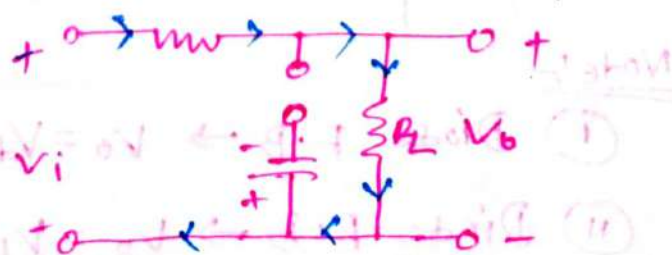
if $V_i > V_o$:-

→ The diode will be in R.O.S & acts like an open circuit

Apply KVL:-

$$V_i - V_o = 0$$

$$\therefore V_o = V_i$$



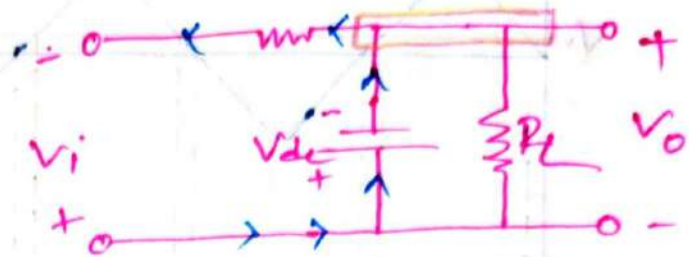
Case : 2:-
more

if $V_i < -V_{de}$:-
more

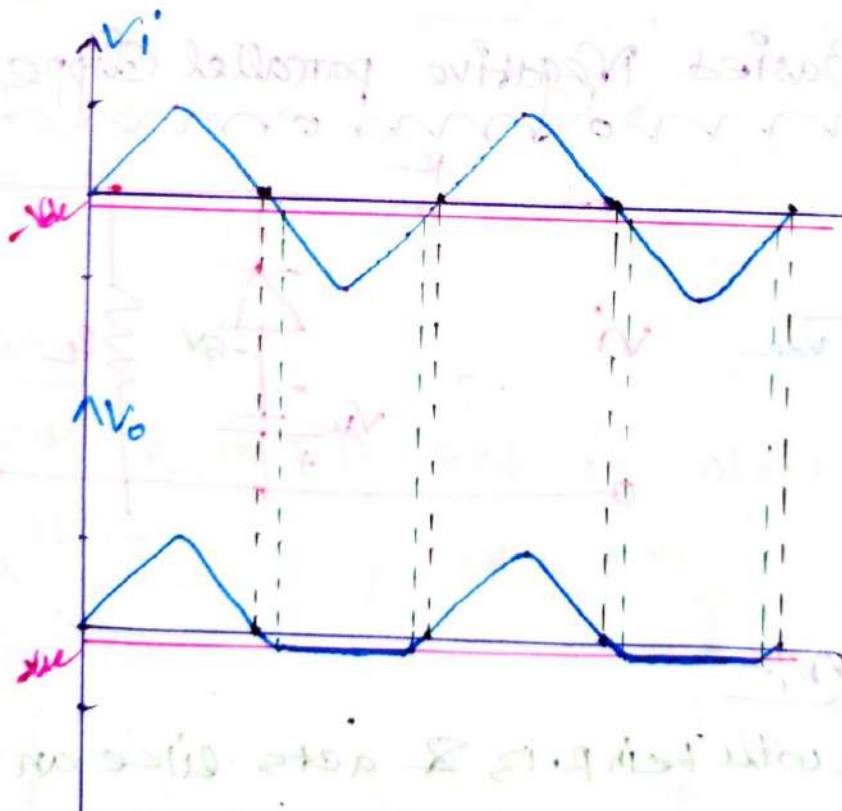
The diode will be in F.B & Act as short circuit

∴ Apply KVL:-
more

$$V_o = -V_{de}$$



∴ The waveform for output:-
more



Note:-

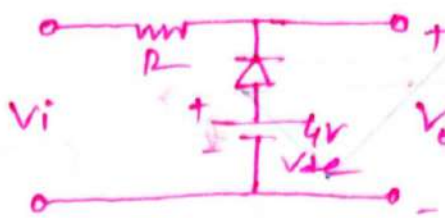
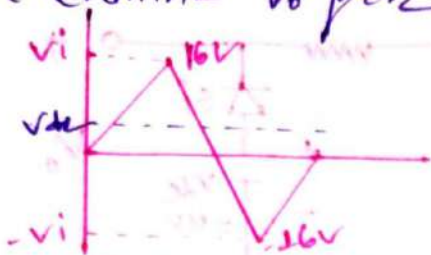
(i) Diode F.B $\rightarrow V_o = V_{de}$

(ii) Diode R.B $\rightarrow V_o = V_i$

$$V = 0V$$

problem: 2.20

⇒ Determine V_o for the network



Case 1:-

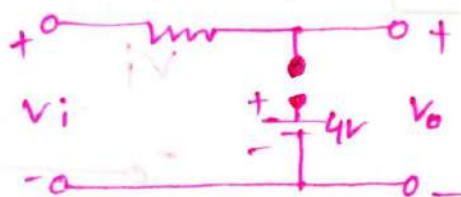
if $V_i > 4V$:-

⇒ The diode will be in F.B & Act As open circuit

∴ Apply KVL:-

$$V_i - V_o = 0$$

$$\therefore V_o = V_i$$



Case 2:-

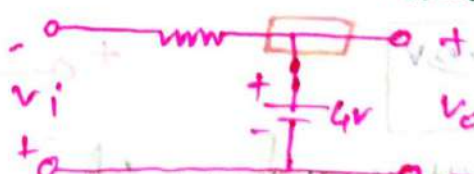
if $V_i < 4V$:-

⇒ The diode will be in F.B & act as short circuit

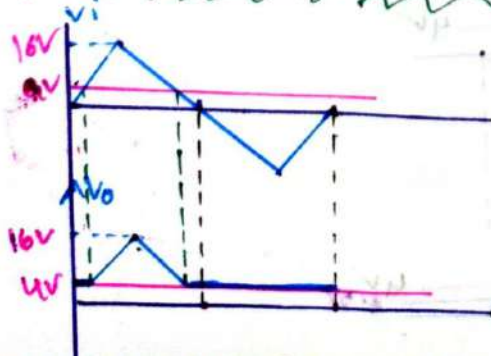
Apply KVL:-

$$V_{de} + V_o = 0$$

$$\therefore V_o = -V_{de} = -4V$$

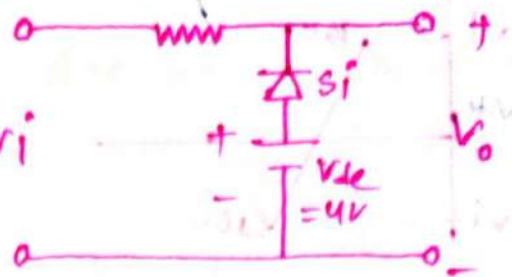
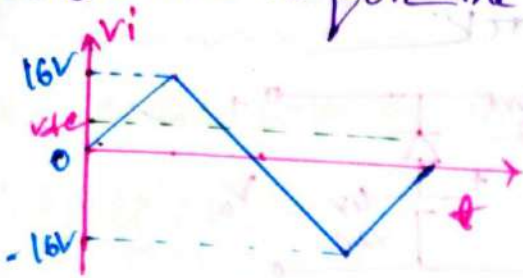


So, the wave form for output:-



Problem 2.21:-

⇒ Determine V_o for the network

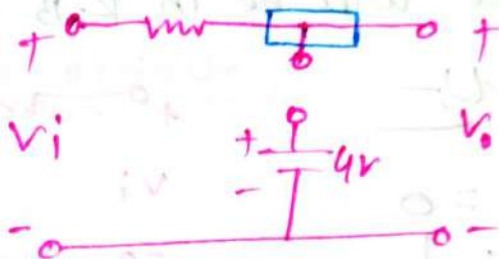


Case 1:-

if $v_i > 4V$:-

⇒ The diode is in P.B, Act as open circuit

$$\therefore V_o = v_i$$



Case 2:-

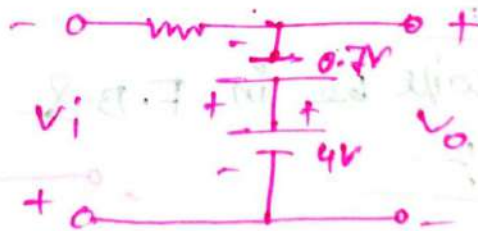
if $v_i < 4V$:-

⇒ The diode is in F.B, Act as short circuit

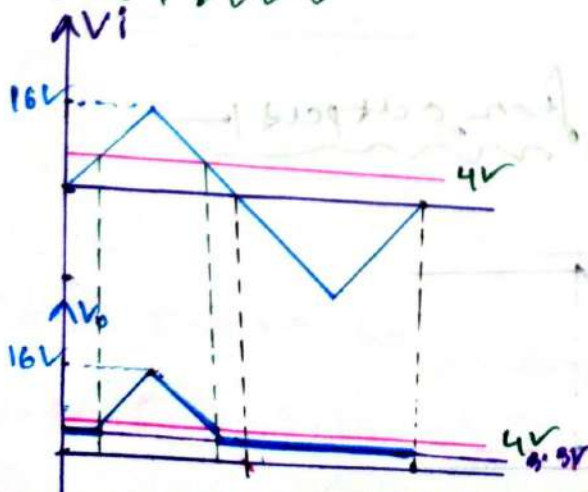
∴ Apply KVL:-

$$V_o + 0.7 - 4 = 0$$

$$\therefore V_o = 3.3V$$



∴ the output waveform:-



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TIME : _____

DATE : / /

#CLAMPERS

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Clamper :-

⇒ A circuit which adds D.C. value to an A.C. waveform without changing its shape.

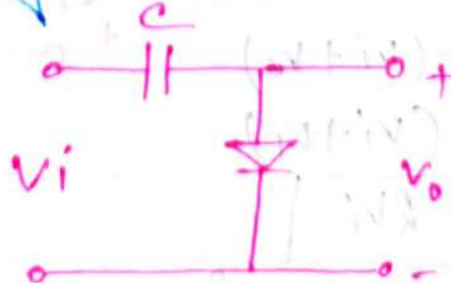
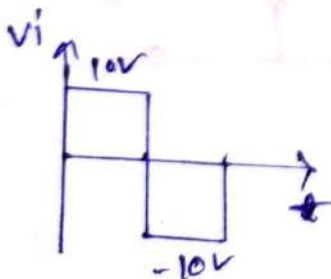
⇒ A clamper is used to re-insert D.C. voltage level into an A.C. waveform which has lost its D.C. by passing through an electronic processing circuit such as amplifier.

Classification of Clamper :-

1. Negative clamper (add -ve D.C. voltage) ↓ waveform shift

2. Positive clamper (add +ve D.C. voltage) ↑ waveform shift

Negative clamper :-



Note:-

1. Start from that cycle which makes Diode forward Bias.

Case: 1:-

1. $V_i > 0$ & the half cycle:-

⇒ Diode is in f.b, Act as short circuit & capacitor start charging

∴ Apply KVL:-

$$V_i - V_c - V_o = 0$$

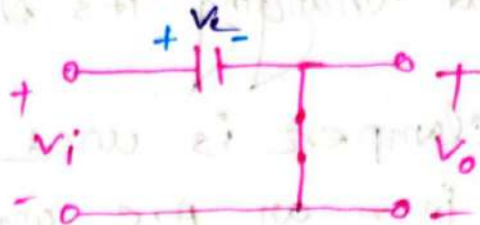
$$\therefore V_c = V_i - V_o$$

$$\therefore V_c = V_i - 0$$

$$\therefore V_c = V_i$$

$$V_o = 0$$

∴ the terminal is short & Act as an wire



Case: 2:-

2. $V_i < 0$ or -ve half cycle:-

⇒ Diode is in R.B, Act as open circuit & capacitor start discharging.

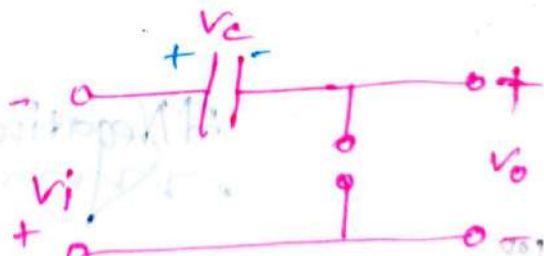
∴ Apply KVL:-

$$V_c + V_i + V_o = 0$$

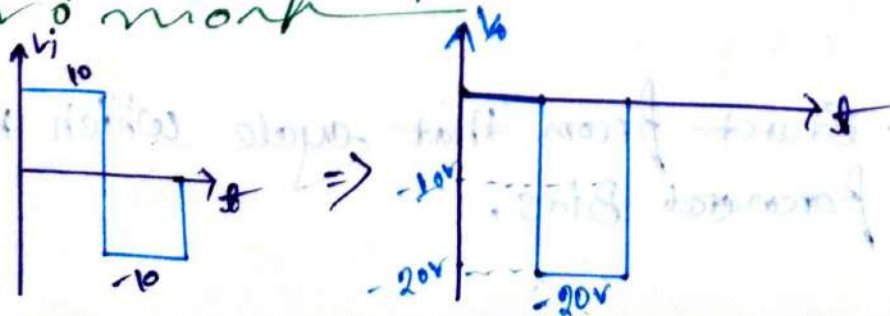
$$\therefore V_o = -(V_i + V_c)$$

$$= -(V_i + V_i)$$

$$\therefore V_o = -2V_i$$



So the output waveform:-

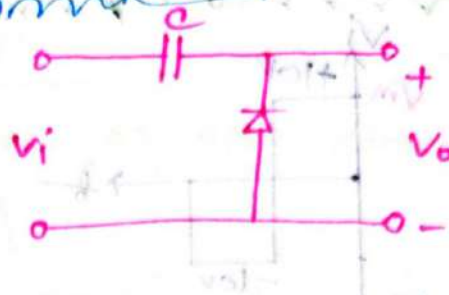
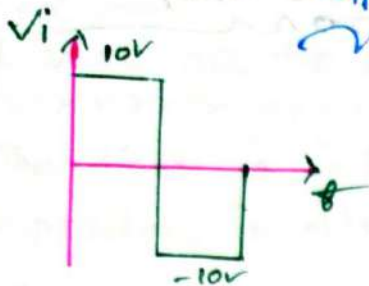


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positive clamper:-



Case 1:-
now

if $V_i < 0$ or negative half cycle:-

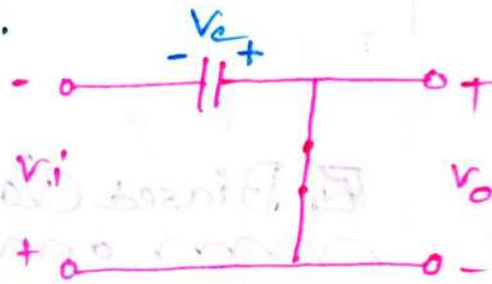
$\Rightarrow$ The Diode is in F.B & Act as short circuit also the capacitor is charging.

Now Apply KVL:-
now

$$V_o = 0V$$

$$\& V_i + V_o - V_c = 0$$

$$V_c = V_i$$



Case 2:-
now

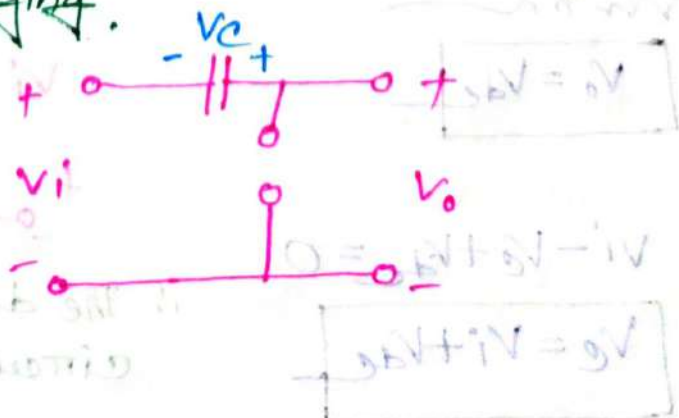
if $V_i > 0$ or positive half cycle:-

$\Rightarrow$ The Diode is in R.B & Act as open circuit also the capacitor is discharging.

Apply KVL:-
now

$$V_i + V_c = V_o$$

$$V_o = 2V_i$$



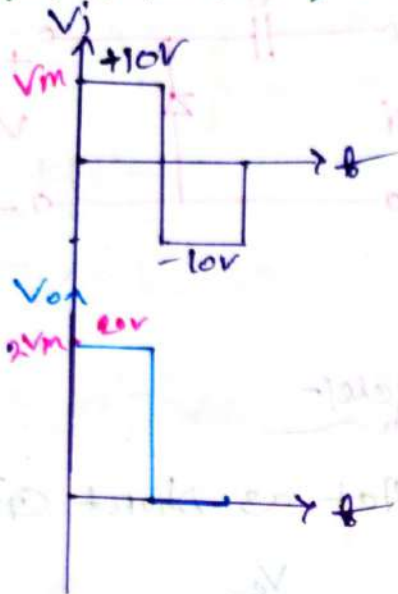
TOPIC NAME : _____

DAY : _____

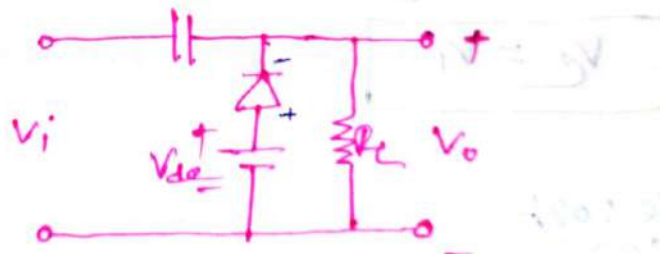
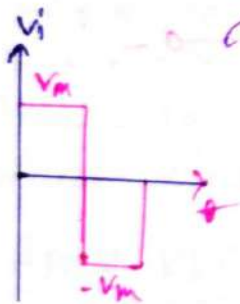
TIME : _____

DATE : / /

∴ The output waveform look like this:-



Biased Clamper:-



Case 1:-

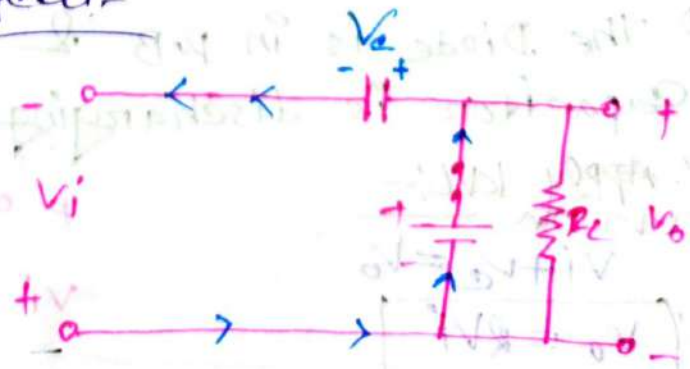
if $V_i < 0$ or -ve half cycle:-

Apply KVL:-

$$V_o = V_{dc}$$

$$V_i - V_c + V_{dc} = 0$$

$$V_c = V_i + V_{dc}$$



∴ The diode is in F.B & Act as short circuit also Capacitor is charging.

Case: 2:-
Non

if V_{i0} or the half cycle:-

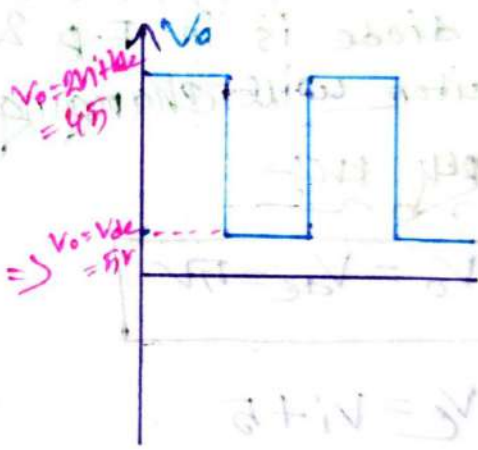
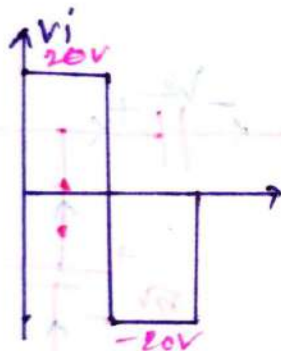
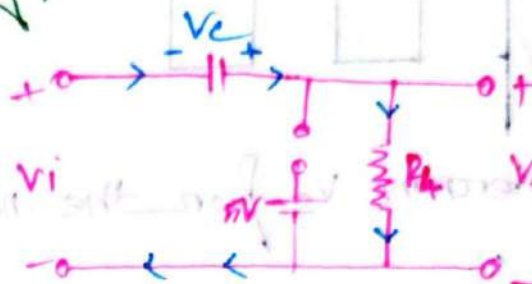
⇒ The diode is in R.B. & Act as open circuit also Now Capacitor is discharging.

Apply KVL:-

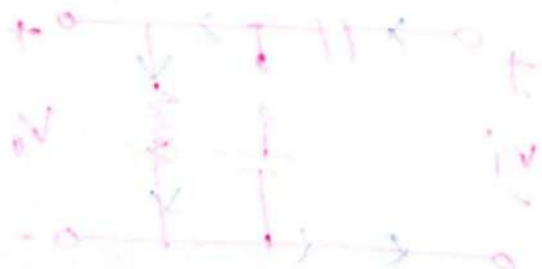
$$V_o = V_i + V_c$$

$$\therefore V_o = V_i + V_i + V_{de}$$

$$\therefore V_o = 2V_i + V_{de}$$



$$V_o = 2V_i + V_{de} = 45V$$

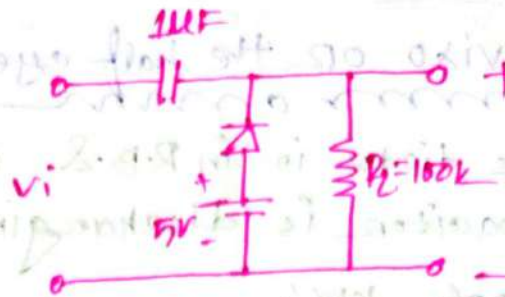
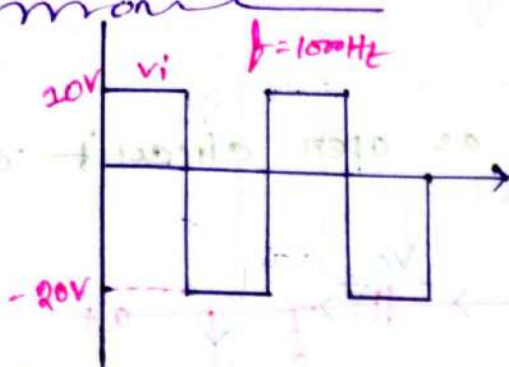


$$V_o = V_i + V_c$$

$$V_o = 2V_i + V_{de}$$

$$V_o = 2V_i + V_{de} = 45V$$

Problem: 2.22:-



⇒ Determine v_o for the network.

⇒ Case 1:-

for $-v_o$ half cycle on $v_i < 0$:-

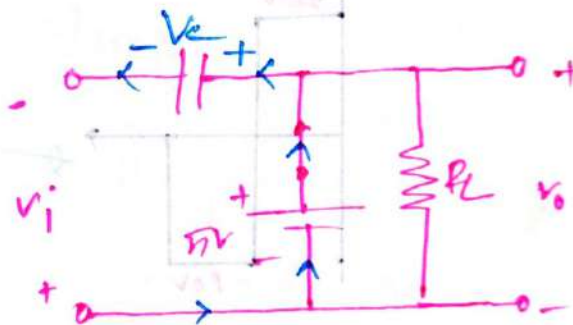
⇒ The diode is in F.B & Act as short circuit & also the Capacitor will charging

Apply KVL:-

$$v_o = v_{de} = 5V$$

$$\therefore v_c = v_i + 5$$

$$\therefore v_c = 20 + 5 = 25V$$



⇒ Case 2:-

for $+v_o$ half cycle: on $v_i > 0$

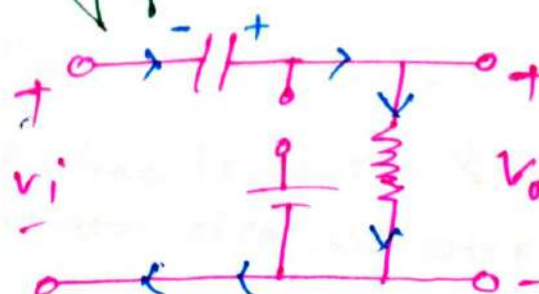
⇒ The diode is in R.D & Act as open circuit, also the capacitor will discharging.

Apply KVL:-

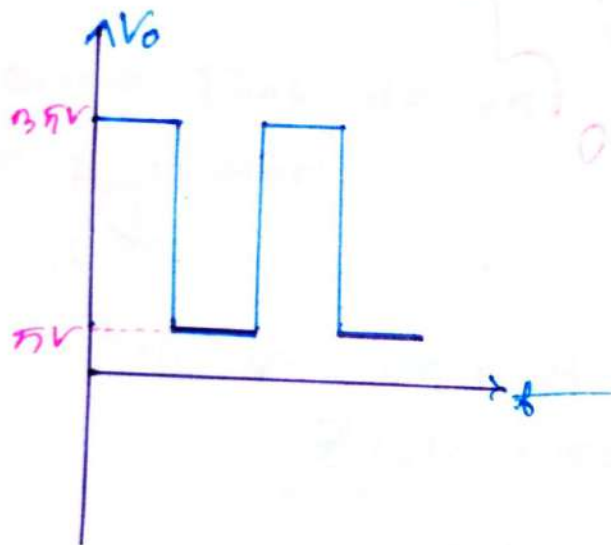
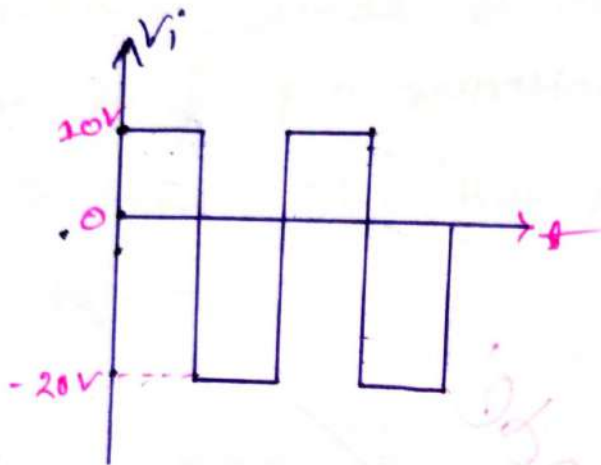
$$v_o = v_i + v_c$$

$$= 10 + 25V$$

$$v_o = 35V$$



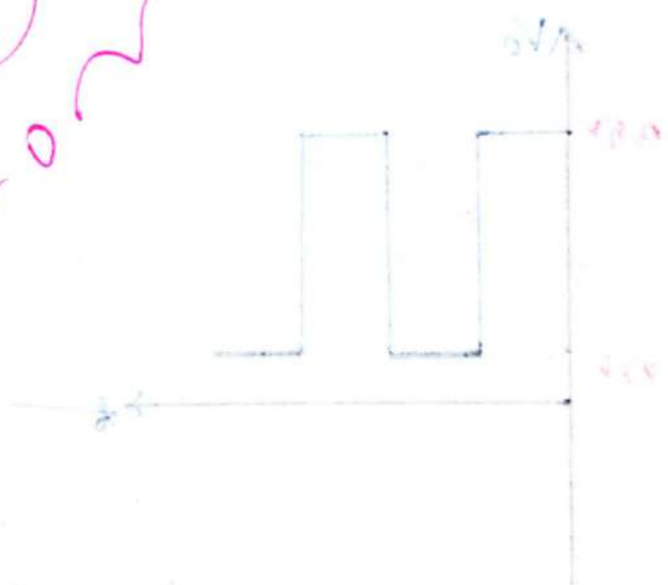
⇒ After completing the process the waveform look like this:-
mon



∴ The output swing of $30V$ matches the input swing

Ans:-

Zener Diode



∴ The output voltage of the regulator is independent of the load current.

Zener Diode:-

⇒ A Zener Diode is a special type of p-n junction Semiconductor diode. It can work in Forward Bias as that of p-n junction diode. It can also work in Reverse Bias that makes it special.

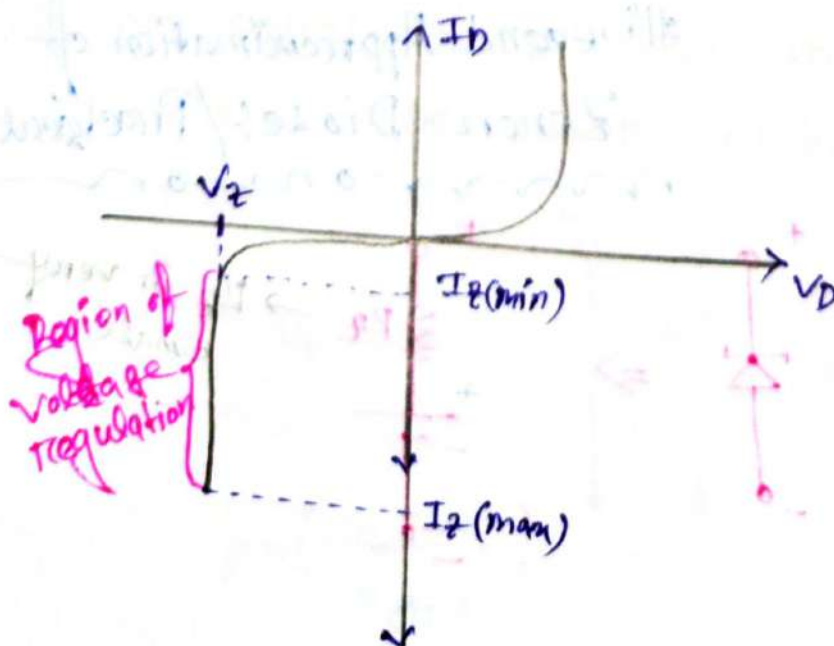
Denote by



Note:-

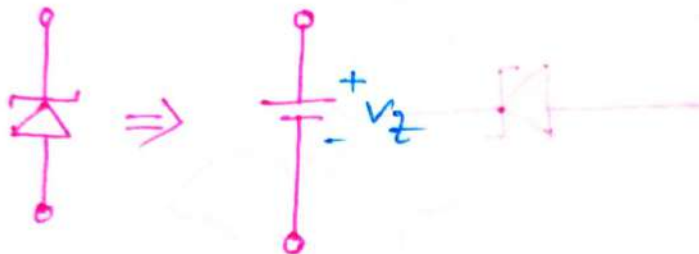
⇒ In reverse Bias the Zener diode works as a Voltage Regulator.

V-I Characteristic of Zener Diode:-



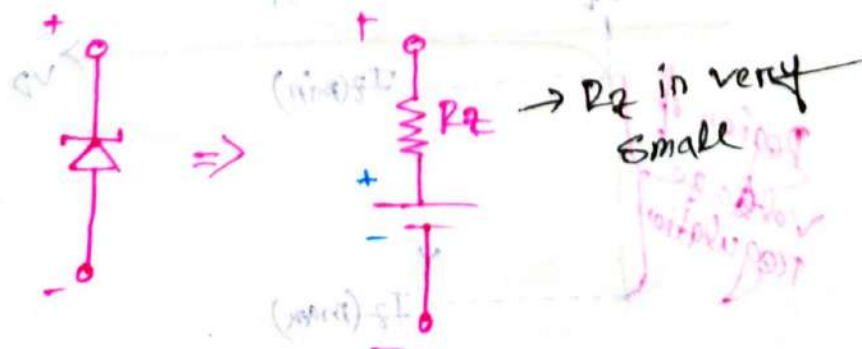
$\Rightarrow$ A voltage Regulator circuit keeps the load voltage constant irrespective change in Input voltage or load ~~resistance~~ resistance.

Ideal Approximation of a Zener diode:-



$\Rightarrow$ Zener Diode operating in breakdown region ideally act like a battery of value equal to breakdown voltage.

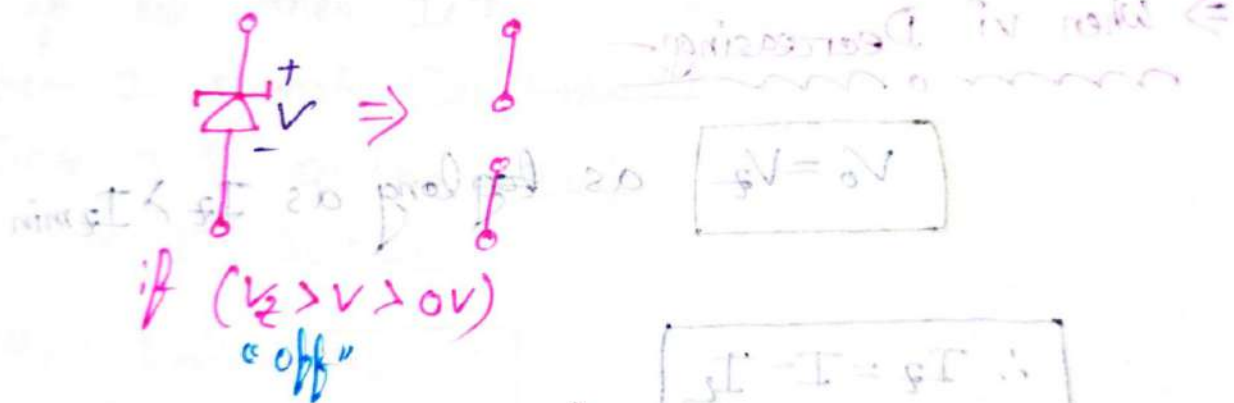
Second Approximation of Zener Diode: / Practical Zener Diode.



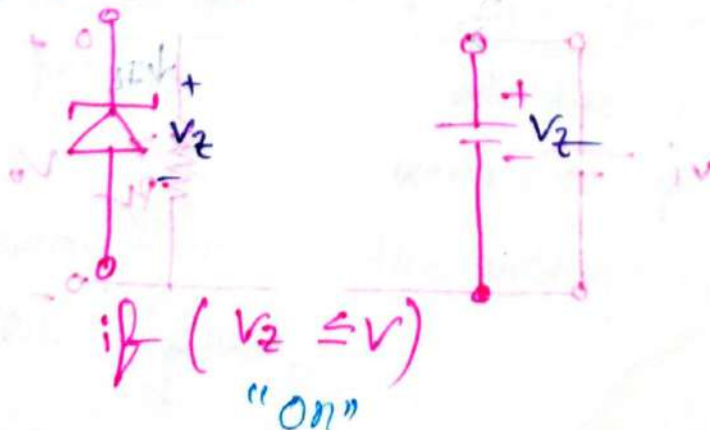
⇒ Total voltage drop across Zener diode is equal to the breakdown voltage and voltage drop across Zener Resistance R_z .

Zener Diode "off" State
& "on" state Condition

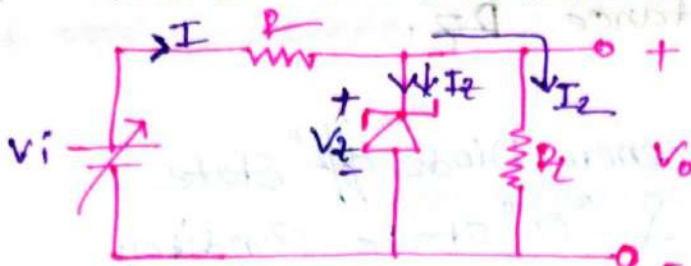
⇒ The "off" state is defined by a voltage less than V_z but greater than 0V. The Zener equivalent is the open circuit



⇒ The "on" state is defined by a voltage greater than V_z or equal to V_z . Then the Zener equivalent is act like a DC power supply



Regulating Action of Zener diode with Varying Input Voltage:-



⇒ When V_i Increasing:-

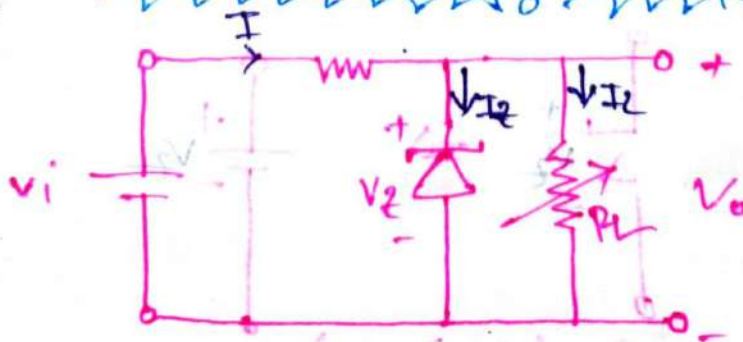
$$V_o = V_Z \text{ as long as } I_Z < I_{Zmax}$$

⇒ When V_i Decreasing:-

$$V_o = V_Z \text{ as long as } I_Z > I_{Zmin}$$

$$\therefore I_Z = I - I_L$$

Regulating Action of Zener diode with Varying Load Resistance:-



$\Rightarrow$ When R_L increasing:

$\Rightarrow$ If $R_L \uparrow$ then $I_L \downarrow$

but I remain Constant

Since $I_L \downarrow$ so that $I_Z \uparrow$

$$\text{C2, } I = I_Z + I_L$$

Therefore output voltage $V_o = V_Z$

as long as $I_Z < I_{Zmax}$

$\Rightarrow$ When R_L Decreasing:

$\Rightarrow$ If $R_L \downarrow$ then $I_L \uparrow$

but I remain Constant

Since $I_L \uparrow$ so that $I_Z \downarrow$

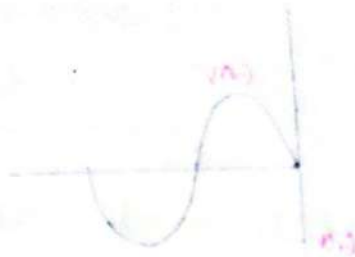
$$\text{C2 } I = I_Z + I_L$$

Output Voltage $V_o = V_Z$

As long as $I_Z > I_{Zmin}$

problem: 2.25:-

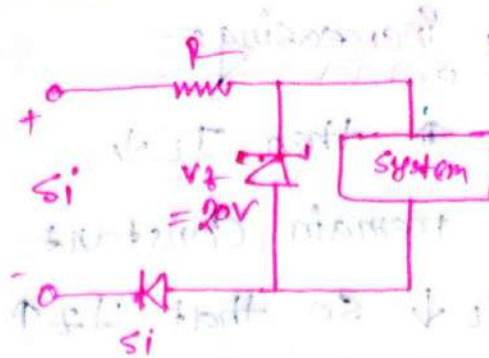
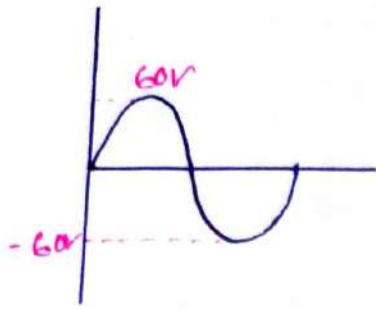
$\Rightarrow$ The network is designed to the limit voltage to 20V during the positive portion of the applied voltage and to 0V for a negative excursion of the applied voltage. plot the waveform of the voltage across the system for applied signal.



$$V_{OS} = 20V$$

$$V_O = 0V$$

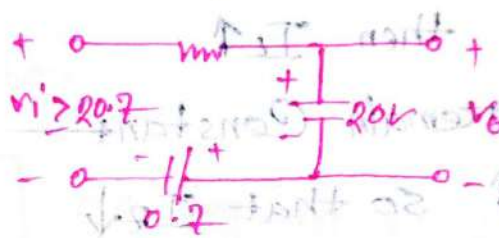




Case 1:-
if $V_i > 20.7$

The Zener diode will be "on" & act as a fixed 20V power supply

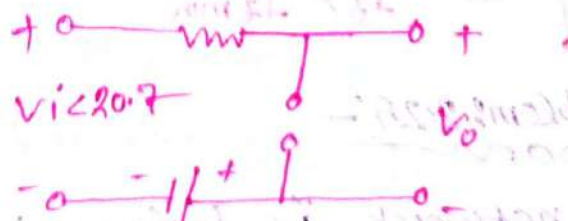
$$\therefore V_o = 20V$$



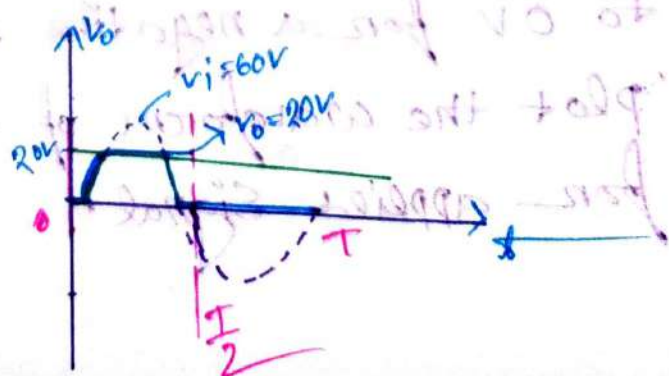
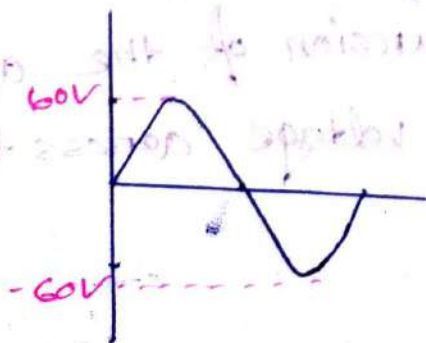
Case 2:-
if $V_i < 20.7$

The Zener diode will be "off" & act as open circuit

$$\therefore V_o = 0V$$

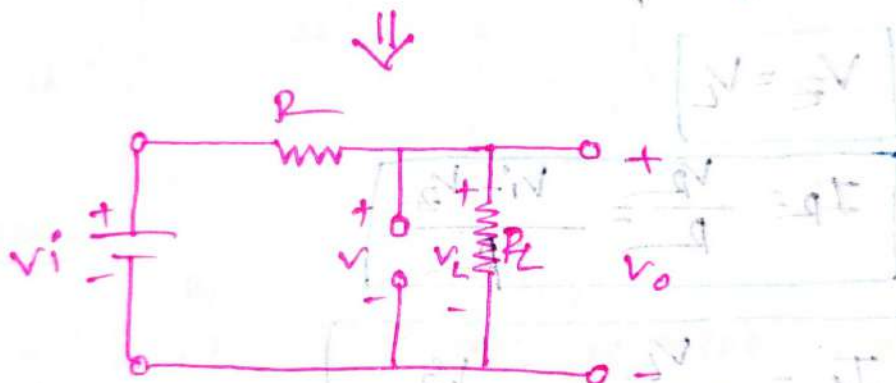
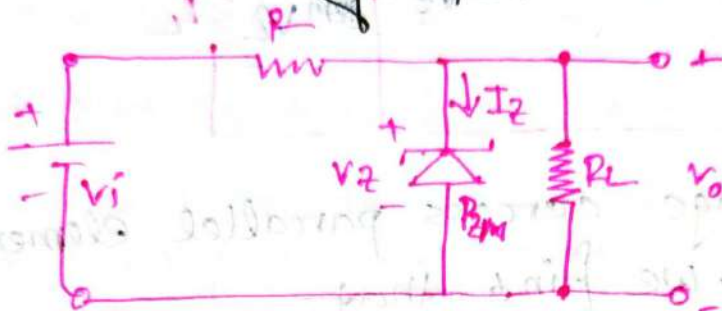


$\therefore$ The output waveform look like this:



How to Determine the State of the Zener diode:-

⇒ For Determine the Zener diode state, 1st remove it from the network and calculate the voltage across the resulting open circuit.



Apply VDR:-

$$V_Z = V = V_o = \frac{R_L}{R + R_L} \times V_i$$

$$R_T = R + R_L$$

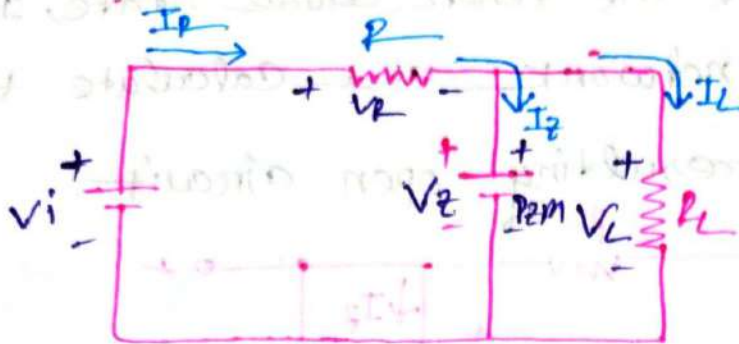
$$\therefore V = V_Z = \frac{R_L}{R + R_L} \times V_i$$

Note:-

1. if $V \geq V_Z$ the Zener diode is "on."

2. if $V < V_Z$ the Zener diode is "off"

If the diode is "on" Then
Determine V_L , I_Z , I_R , P_Z



⇒ Since voltage across parallel elements must be the same, we find that

$$V_Z = V_L$$

$$I_R = \frac{V_R}{R} = \frac{v_i - V_Z}{R}$$

$$I_L = \frac{V_L}{R_L} = \frac{V_Z}{R_L}$$

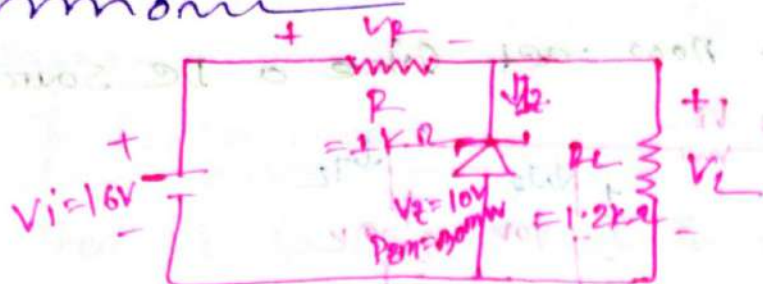
$$I_Z = I_R - I_L$$

$$P_Z = V_Z I_Z = I_Z^2 R_Z = \frac{V_Z^2}{R_Z}$$

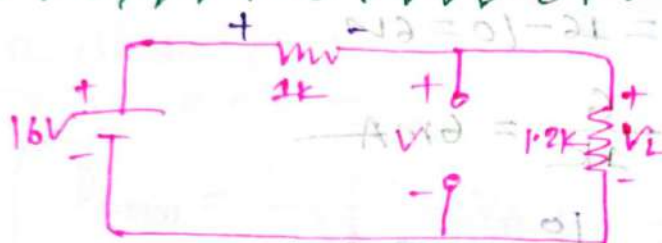
Note:-

⇒ P_Z must be less than P_{ZM}
otherwise the diode will be destroyed.

problem: 2.26

(a) Determine V_L , V_P , I_2 , & P_2 (b) Repeat part (a) with $R_L = 3k\Omega$

⇒ 1st Determine V for the given circuit:



we know-

$$V = \frac{R_L}{R_T} \times V_i = \frac{1.2k}{(1.2+1)k} \times 16V = 8.7272V$$

Since, $V \neq V_2 = 8.7272 \neq 10V$

So, the diode is "off" state

$$\therefore V_L = V = 8.7272V$$

$$V_P = V_i - V_L = 16 - 8.7272 = 7.2728V$$

$$I_2 = 0A$$

$$P_2 = V_2 I_2 = V_2 \times 0 = 0W$$

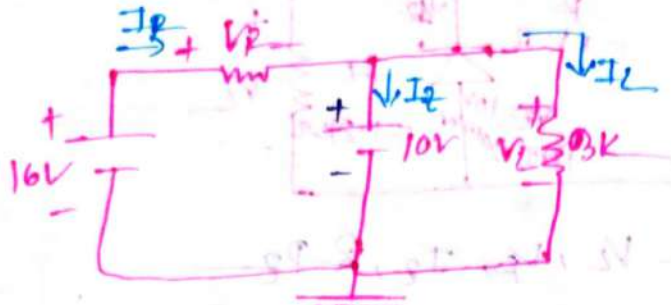
if $R_L = 3k\Omega$

so/

$$V = \frac{R_L}{R_T} \times V_i = \frac{3k}{4k} \times 16 = 12V$$

Since $V_1 > V_2 = 12 > 10V$ the diode is in "on" state.

So, the diode is now act like a DC source



$$\therefore V_L = V_Z = 10V$$

$$V_R = V_1 - V_Z = 16 - 10 = 6V$$

$$I_R = \frac{V_R}{R} = \frac{6}{1k} = 6mA$$

$$I_L = \frac{V_L}{R_L} = \frac{10}{3k} = 3.33mA$$

$$\therefore I_Z = I_R - I_L = (6 - 3.33)mA = 2.67mA$$

$$\therefore P_Z = V_Z I_Z = 10 \times 2.67mA = 26.7mW$$

$$\text{Since } P_Z < P_{ZM} = 26.7mW < 30mW$$

So, the diode is safe

if $P_Z = 30.1mW$ which is greater than P_{ZM} then the coil be "Inna Lillah"

Fixed V_i , Variable R_L

Case 1:-
mon

if R_L too small:-

Then V_L is also small & if

$$V_L < V_Z$$

so, the Zener diode will be in "off" state.

$\therefore$ To determine the min R_L that will turn the Zener diode "on":-

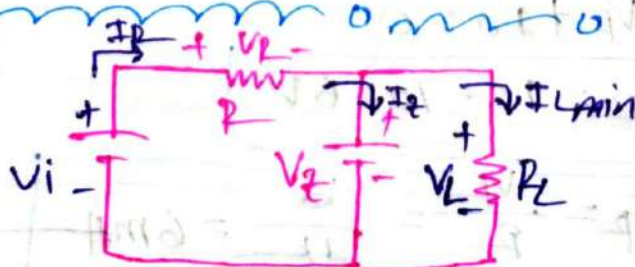
$$R_{Lmin} = \frac{R}{V_Z} \times V_Z = \frac{R}{V_i - V_Z} \times V_Z$$

$$V_L = V_Z = \frac{R_L}{R_T} \times V_i$$

$$I_{Lmax} = \frac{V_L}{R_L} = \frac{V_Z}{R_{Lmin}}$$

After the diode is on then

Determine V_R , I_R , I_Z , I_{Lmin} , R_{Lmax}



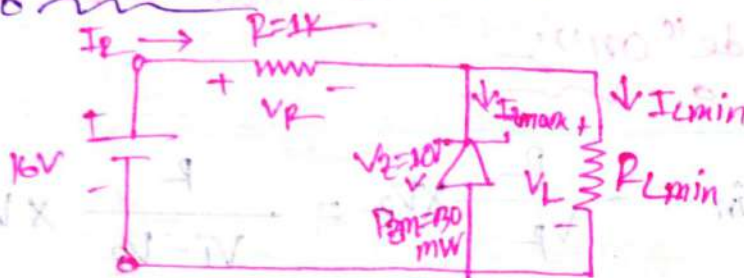
$$\therefore V_R = V_i - V_Z$$

$$I_R = \frac{V_R}{R}$$

$$I_{Lmin} = I_R - I_{Zmax}$$

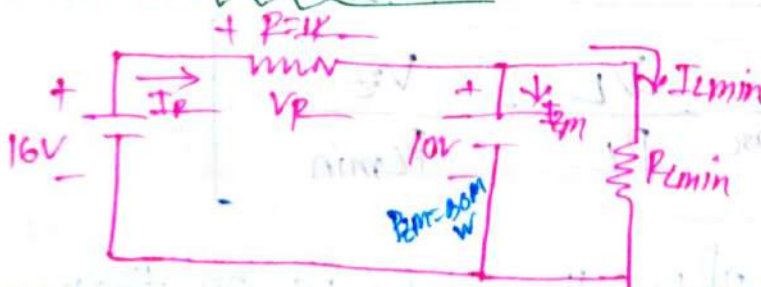
$$R_{Lmin} = \frac{V_Z}{I_{Lmin}}$$

Problem: 2.27:-



⇒ Determine V_R , I_R , I_{Zmax} , I_{Lmin} , R_{Lmin} , V_L , P_{Lmax} , I_{Lmax}

⇒ Int redraw the circuit:-



For V_R we can Apply KVL:-

$$-16 + V_R + 10 = 0$$

$$\therefore V_R = 16 - 10 = 6V$$

$$\therefore I_R = \frac{V_R}{R} = \frac{6}{2k} = 6mA$$

We know:-

$$\Rightarrow P_{Z_m} = V_Z I_{Z_m}$$

$$\therefore I_{Z_m} = \frac{P_{Z_m}}{V_Z} = \frac{30 \text{ mW}}{10} = 3 \text{ mA}$$

$$\therefore I_{Z_m} = 3 \text{ mA}$$

$$\Rightarrow I_{L_{\min}} = I_L - I_{Z_m}$$

$$\therefore I_{L_{\min}} = 6 \text{ mA} - 3 \text{ mA}$$

$$I_{L_{\min}} = 3 \text{ mA}$$

$$\Rightarrow R_{L_{\max}} = \frac{V_Z}{I_{L_{\min}}}$$

$$R_{L_{\max}} = \frac{10}{3 \text{ mA}} = 3.33 \text{ k}\Omega$$

$$\Rightarrow V_L = V_Z = 10 \text{ V}$$

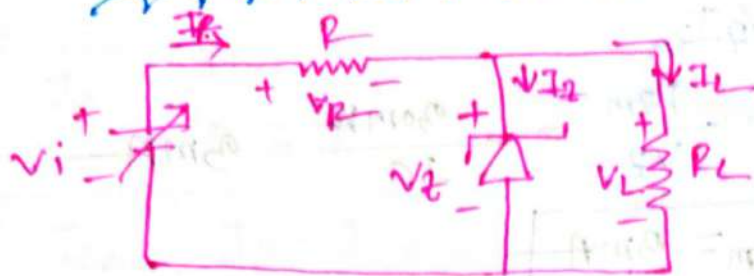
$$\Rightarrow R_{L_{\min}} = \frac{R}{\frac{V_Z}{V_P} \times V_Z}$$

$$\therefore R_{L_{\min}} = \frac{1 \text{ k}}{6} \times 10 = 1.667 \text{ k}\Omega$$

$$\Rightarrow I_{L_{\max}} = \frac{V_L}{R_{L_{\min}}}$$

$$\therefore I_{L_{\max}} = \frac{10 \text{ V}}{1.667 \text{ k}} = 5.99 \text{ mA}$$

Fixed: R_L , Variable: V_i



To turn the diode "on" the minimum turn on voltage

$$V_{i\min} = \frac{R_T}{R_L} \times V_Z$$

$$V_L = \frac{R_L}{R_T} \times V_i$$

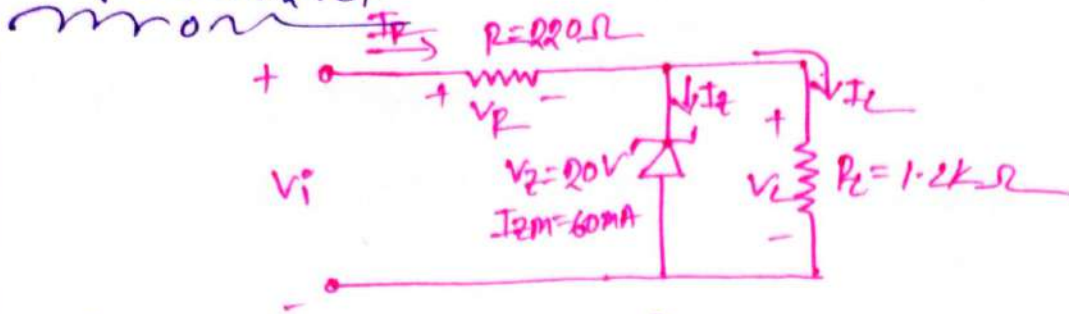
$$I_{P\max} = I_{Zm} + I_L$$

$$V_{i\max} = V_{P\max} + V_Z$$

OR

$$V_{i\max} = I_{P\max} * R + V_Z$$

problems 2.28/-



⇒ Determine the range of V_i that will maintain the Zener diode in the "on" state.

⇒ In given:-

$$R = 220 \Omega$$

$$V_z = 20V$$

$$I_{zm} = 60mA$$

$$R_L = 1.2k \Omega$$

we know:-

$$\Rightarrow V_{imin} = \frac{R_T}{R_L} \times V_z = \frac{R_L + R}{R_L} \times V_z = \frac{220 + 1.2k}{1.2k} \times 20$$

$$V_{imin} = 23.667V$$

$$\Rightarrow V_L = V_z = 20V$$

$$\Rightarrow I_L = \frac{V_L}{R_L} = \frac{20}{1.2k} = 16.667mA$$

$$\Rightarrow I_{Rmax} = I_{zm} + I_L = 16.667mA + 60mA = 76.667mA$$

$$\Rightarrow V_{Rmax} = I_{Rmax} \times R = 76.667mA \times 220\Omega = 16.866V$$

$$\Rightarrow V_{imax} = V_{Rmax} + V_z = 16.886 + 20 = 36.87V$$

Ans:-

Bipolar Junction Transistors (BJT)

Ch 03

Introduction:-

⇒ BJT consists of three terminals:-

1. collector & C → Moderately Doped
2. Base & B → Lightly Doped
3. emitter & E → Heavily Doped

There are two types of BJT:-

1. p-n-p type
2. n-p-n type.

Transistor:-

⇒ Transistor is a 3 layer Semiconductor device.

It consists:-

- ⇒ Two n- and one p-type layers = n-p-n transistor
- ⇒ Two p- and one n-type layers = p-n-p transistor

Note:-

⇒ A single p-n junction has two different types of Bias:-

1. Forward Bias
2. Reverse Bias

⇒ Thus, a two-p-n junction device has four types of Bias.

position of terminals & Symbol of BJT

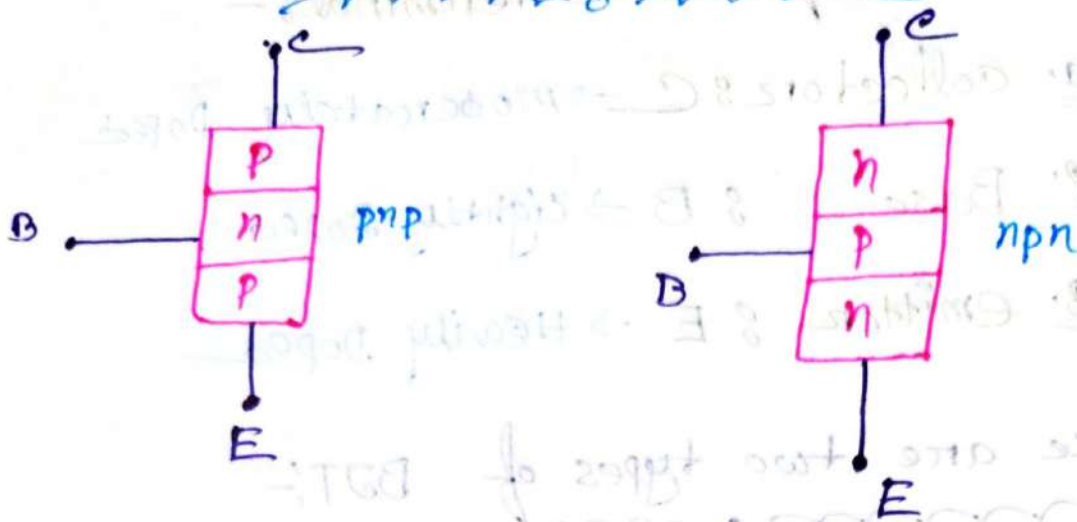


Fig 1: position of terminal

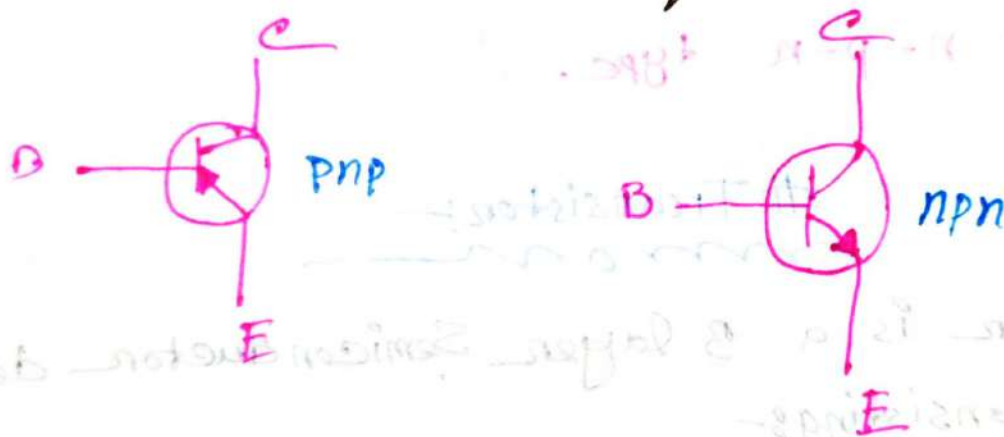


Fig 2: Type of BJT

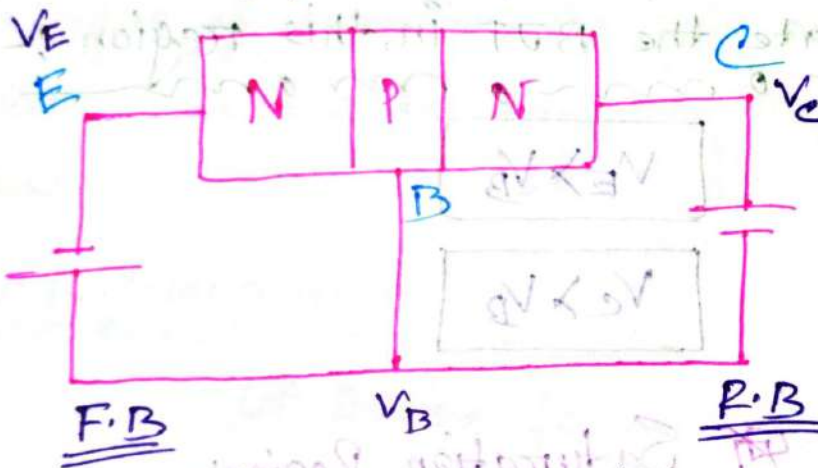
2. Forward Bias
2. Reverse Bias

Different Regions of operation

⇒ For N-P-N

1. Active - Region → Amplifier
 2. Cut-off Region
 3. Saturation Region
- } → Switch

Active Region:-



For Forward Bias:-

$$V_B > V_E$$

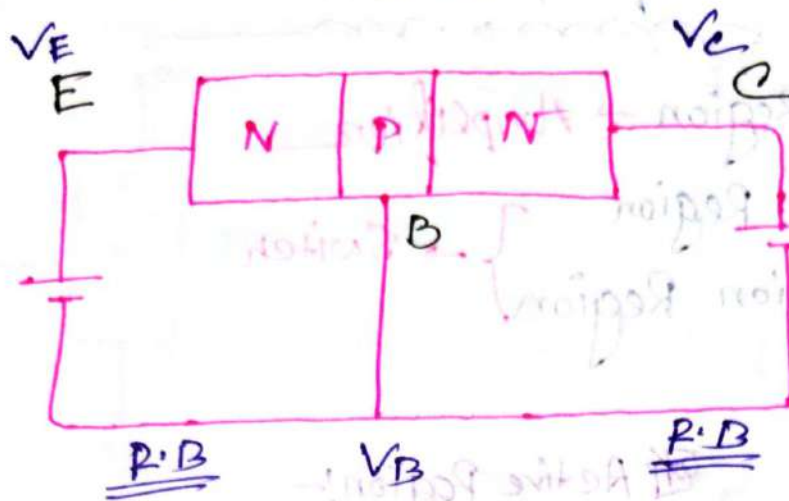
For Reverse Bias:-

$$V_B < V_C$$

So that, to operate the BJT in Active Region

$$V_E < V_B < V_C$$

Cut-off Region:-

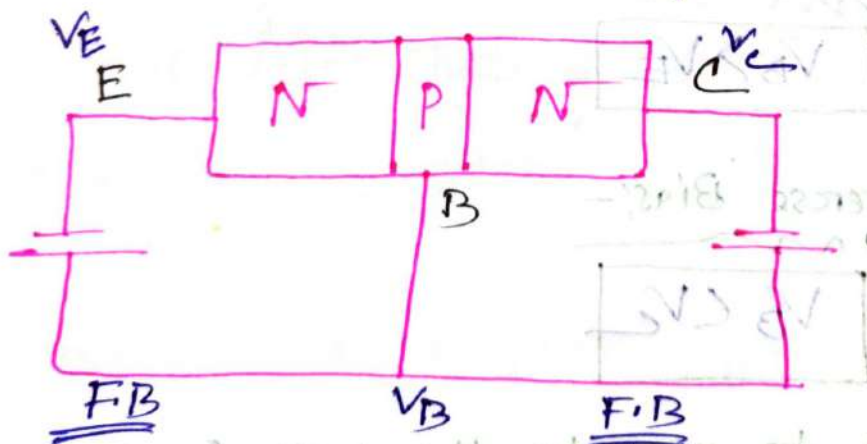


∴ to operate the BJT in this region:-

$$V_E > V_B$$

$$V_C > V_B$$

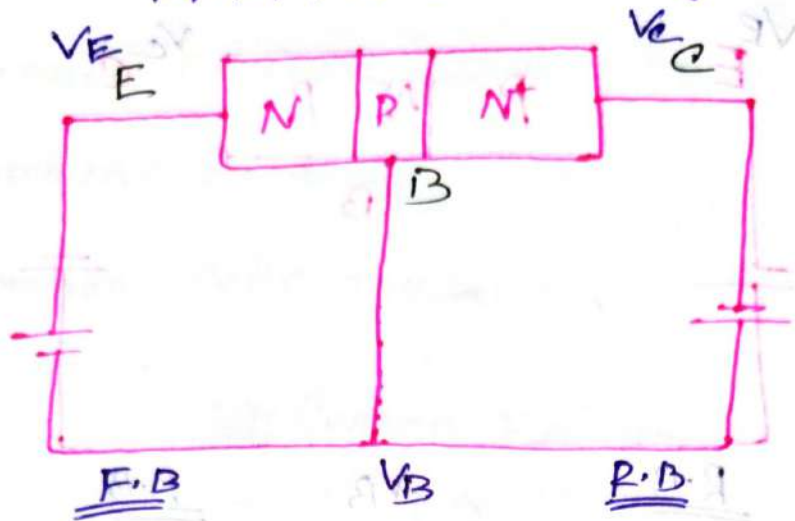
Saturation Region:-



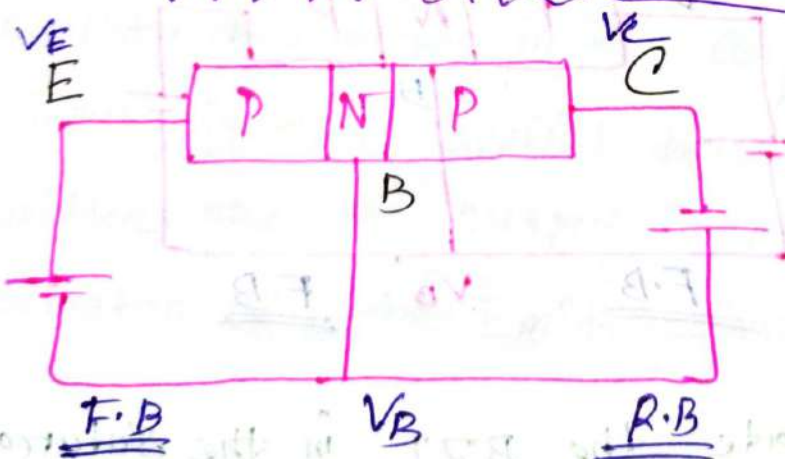
∴ to operate the BJT in this region:-

$$V_E < V_B$$

$$V_C < V_B$$

Reverse Active Region

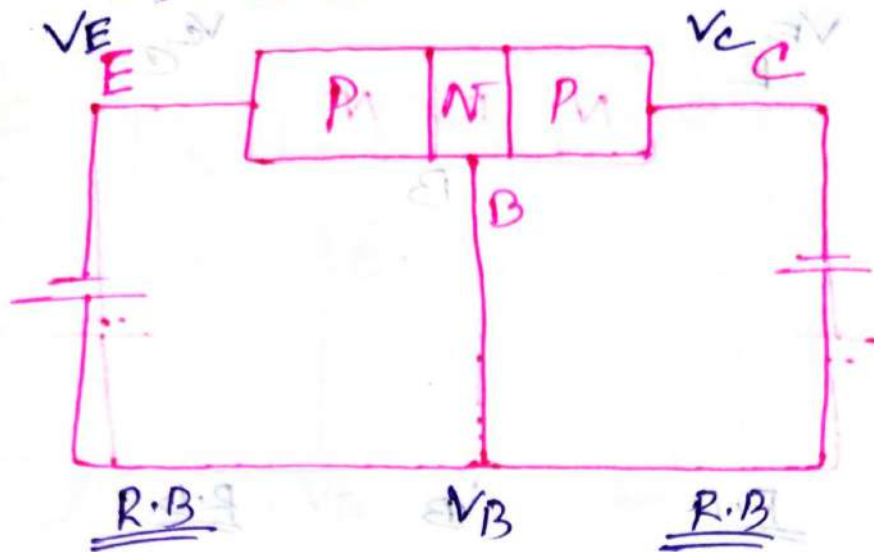
⇒ In this region of operation, the gain provided by the BJT is very less, and due to that, this region of operation is avoided.

⇒ P-N-P Transistor:-Active Region:-

So, In case of PNP transistor, to operate it in the active region,

$$V_E > V_B > V_C$$

Cut-off region:-

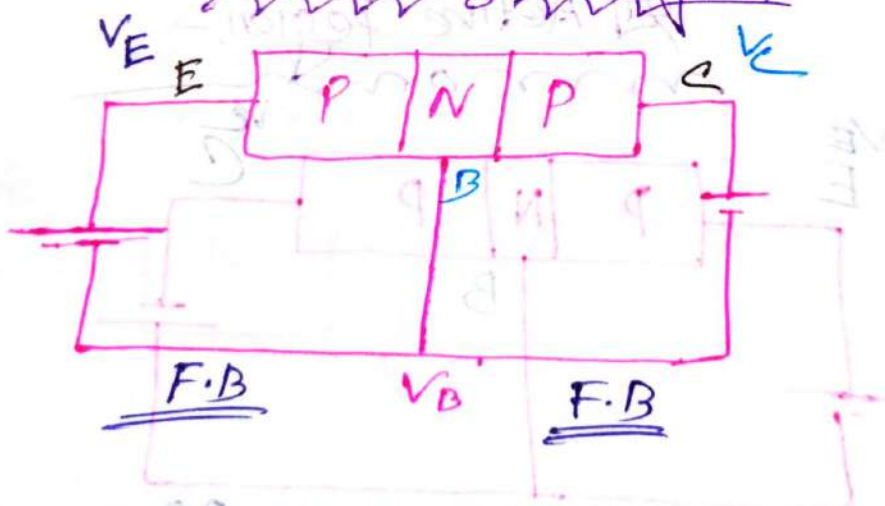


∴ To operate the BJT in the cut-off region:-

$$V_B > V_E$$

$$V_B > V_C$$

Saturation Region:-



∴ To operate the BJT in the Saturation Region

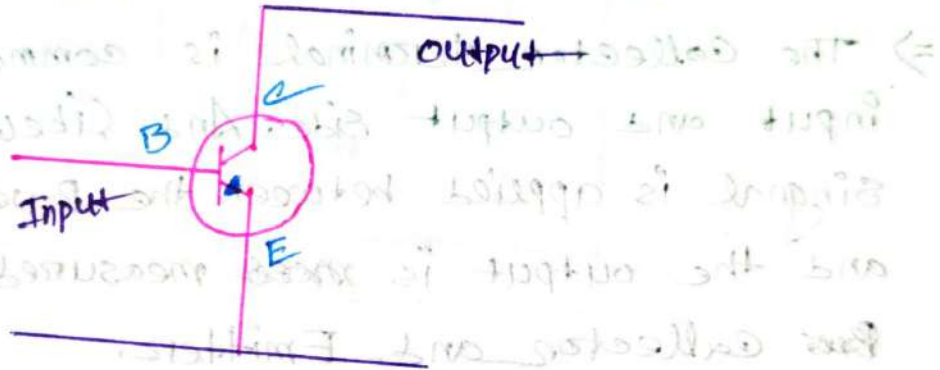
$$V_E > V_B$$

$$V_C > V_B$$

#BJT Configuration:-

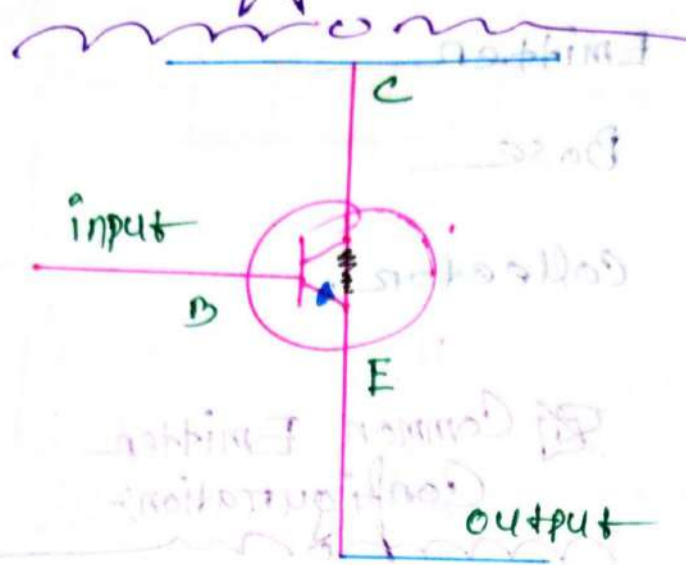
1. Common Emitter
2. Common Base
3. Common Collector

Common Emitter Configuration:-



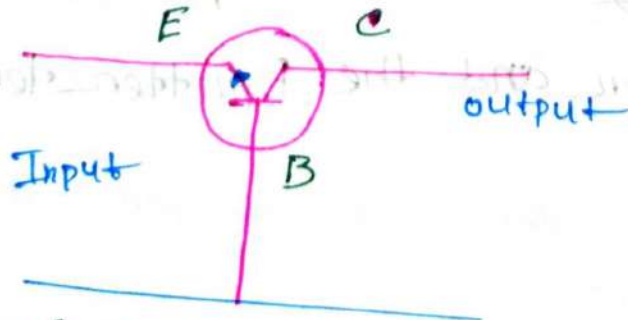
- ⇒ The Emitter terminal is common between the input and output. That means in this configuration.
- ⇒ The input signal is applied between the Base and the Emitter, and the output is measured between the collector and the Emitter terminal.

Common Collector Configuration:-



⇒ The collector terminal is common between the input and output side. And likewise in the input signal is applied between the base and collector and the output is measured between the collector and emitter.

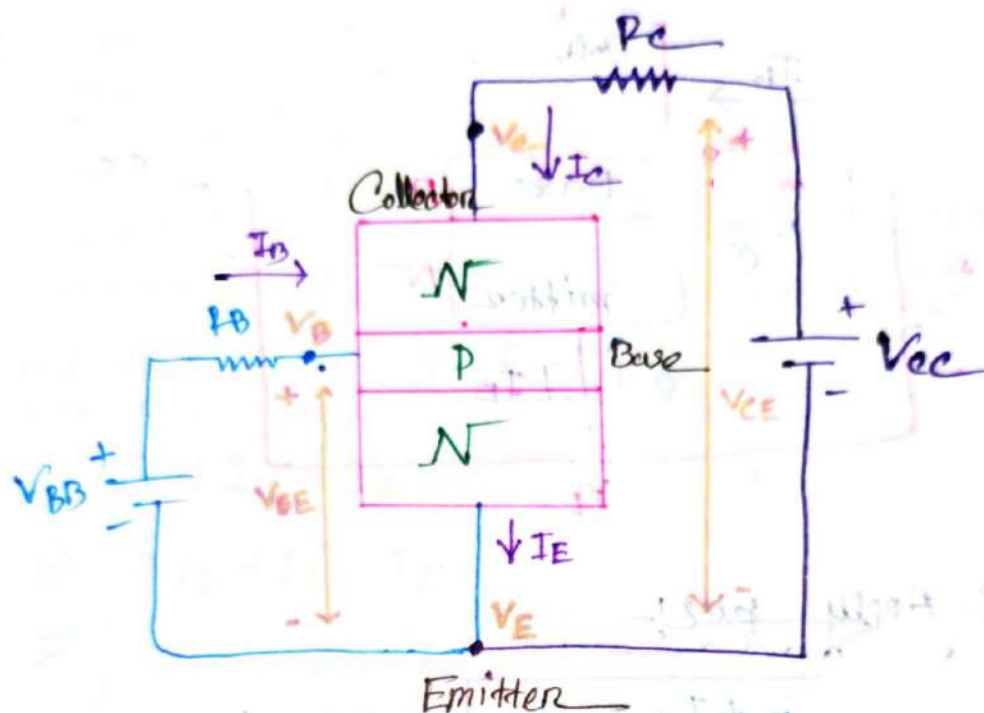
Common-Base Configurations:-



⇒ The Base terminal is common between the input and output side.

#Working of BJT

In the Active region:-



here,

V_{BE} = Source voltage for Base

V_{CE} = Source voltage for Collector

V_{BE} = Voltage Difference between the Base & the Emitter

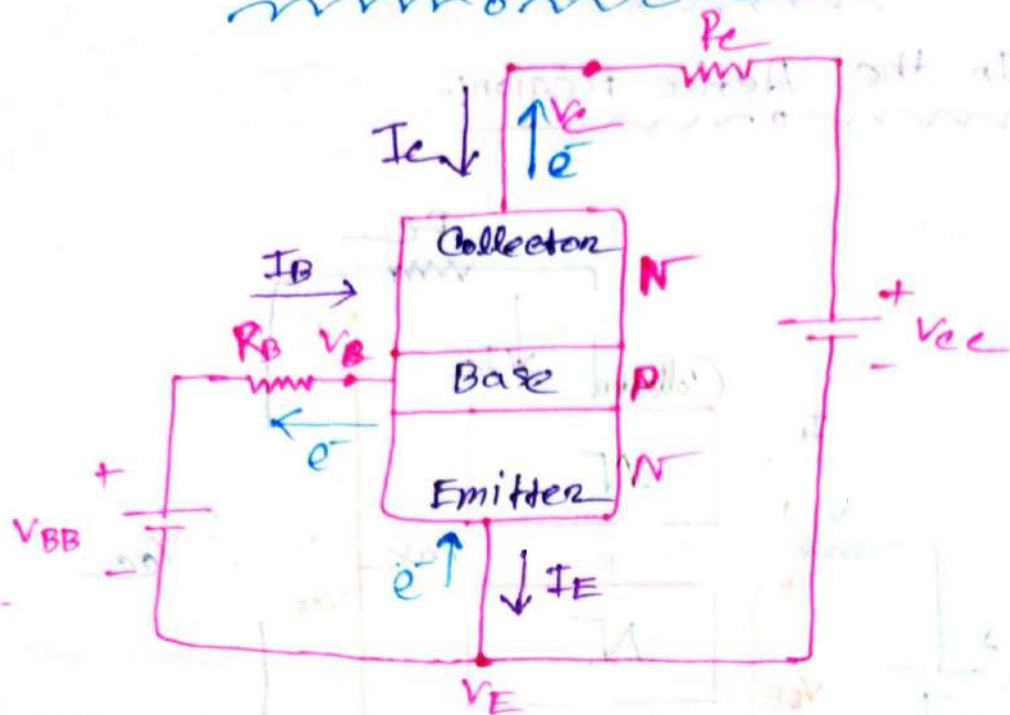
$$\therefore V_{BE} = V_B - V_E$$

$$V_{EB} = V_E - V_B = -V_{BE}$$

V_{CE} = Voltage difference between the collector & the Emitter

$$\therefore V_{CE} = V_C - V_E$$

Current In BJT



Now Apply KCL:-

$$I_B + I_C = I_E \quad \text{--- (1)}$$

$$I_E \cong I_C$$

$$\Rightarrow I_C = \alpha I_E$$

$$\Rightarrow I_E = \frac{I_C}{\alpha}$$

[Since very small amount of e^- are able to go V_B direction]

α = fraction of the Emitter Current is following through the collector terminal.

Now put the I_E value in (1) eq:-

$$I_B + \alpha I_E = I_E$$

$$\therefore I_B = (1 - \alpha) I_E \quad \text{--- (11)}$$

Now put the value of I_E in eq (11) :-

$$I_B = (1 - \alpha) \times \frac{I_C}{\alpha}$$

$$\therefore I_B = \frac{1-\alpha}{\alpha} \times I_C$$

$$\Rightarrow I_C = \frac{\alpha}{1-\alpha} \times I_B$$

Let:-

$$\beta = \frac{\alpha}{1-\alpha}$$

$$\therefore I_C = \beta I_B$$

β = Current gain of the BJT/
The amplification factor

Now put the I_C in eq (1):

$$I_C + I_B = I_E$$

$$\Rightarrow \beta I_B + I_B = I_E$$

$$\Rightarrow I_E = I_B (\beta + 1)$$

$\Rightarrow$ BJT also known as the current control device.

The total current

$$I_{ET} = I_C + I_{CO}$$

I_{CO} = leakage current

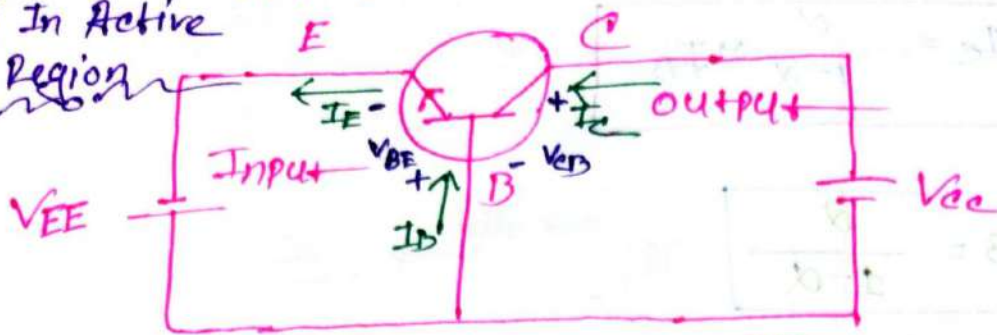
$$\therefore \alpha_{dc} = \frac{I_C}{I_E}$$

$$\beta_{dc} = \frac{I_C}{I_B}$$

$$\alpha = \frac{\beta}{1+\beta}$$

Common Base Configuration:-

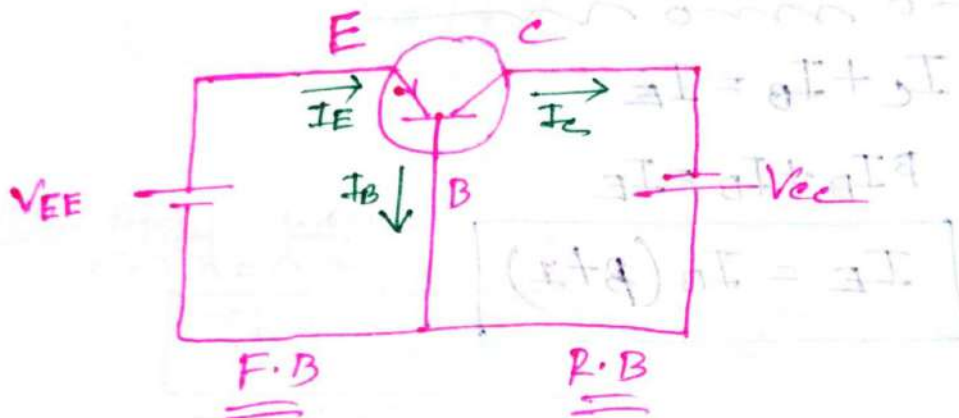
In Active Region



F.B

R.B

Fig 1: n-p-n Transistor



F.B

R.B

Fig 2: p-n-p Transistor

$$\therefore \boxed{I_E = I_B + I_C}$$

Now:- In this configurations, the behavior of the device can be describe by two characteristics,

1. Input Characteristics
2. Output characteristics

Note:- these characteristics are similar to the V-I characteristics of Diode.

Q For Fig 1 / n-p-n transistor:-

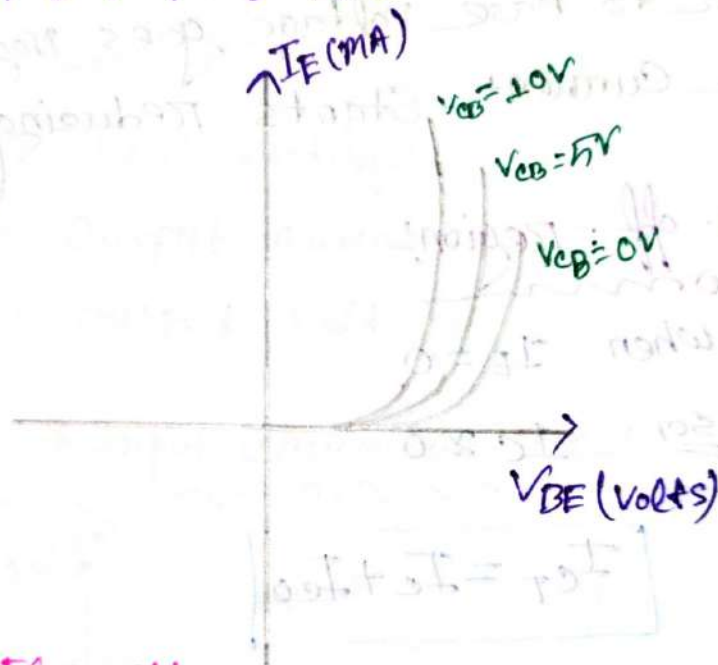
Input Element:-

1' I_E = Input Current

2' V_{BE} = Input Voltage.

* Input Characteristics:-

$$\therefore R_{in} = \frac{\Delta V_{BE}}{\Delta I_E}$$

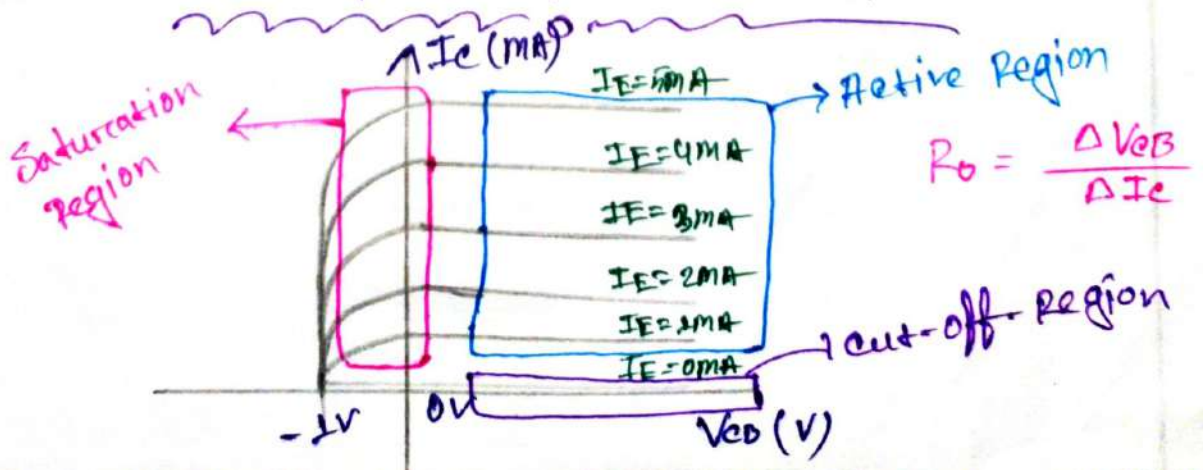


Output Element:-

1' I_C = Output Current

2' V_{CB} = output Voltage

* Output Characteristics:-



For Active - Region:-

$$I_C = \alpha I_E$$

$$\therefore \frac{I_C}{I_E} = \alpha$$

For Saturation Region:-

$\Rightarrow$ If we reduce the value of V_{CB} , that means the Collector to base voltage goes Negative, then the Collector Current Starts reducing

For cut-off - region:-

when $I_E = 0$

$$\underline{\underline{\text{So } I_C \approx 0}}$$

$$\therefore \boxed{I_{CT} = I_C + I_{CO}}$$

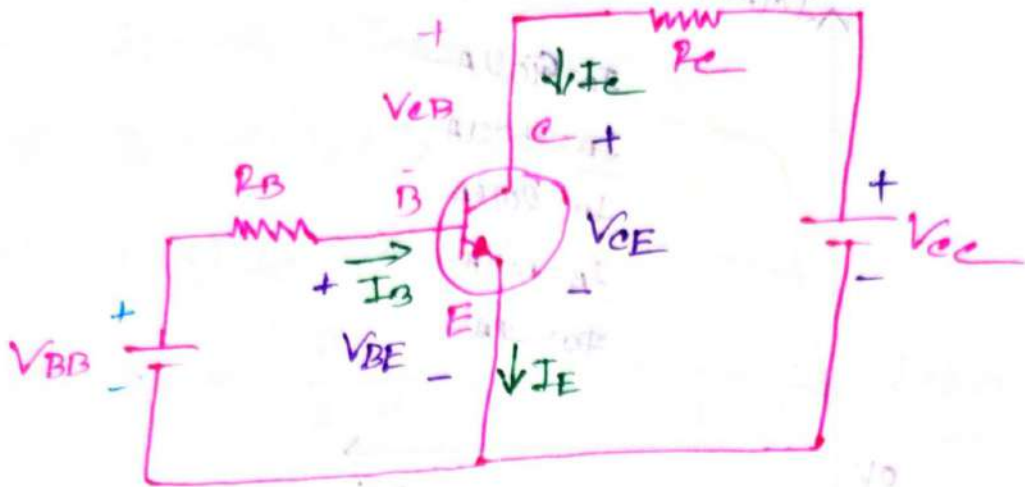
I_{CT} = Total Collector Current

I_C = Collector Current

I_{CO} = Reverse Saturation Current.



Common Emitter Configuration:-



here,

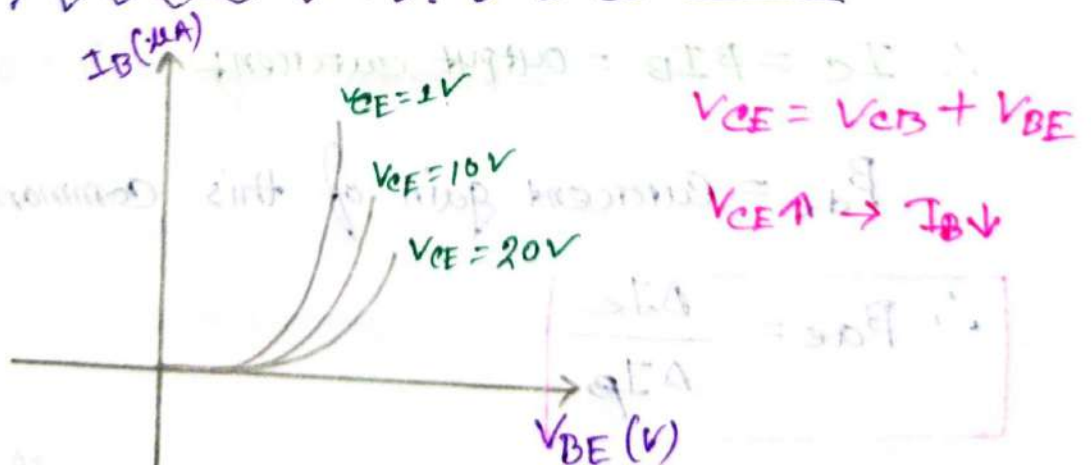
I_B = Input current

V_{BE} = Input Voltage.

I_E = Output current

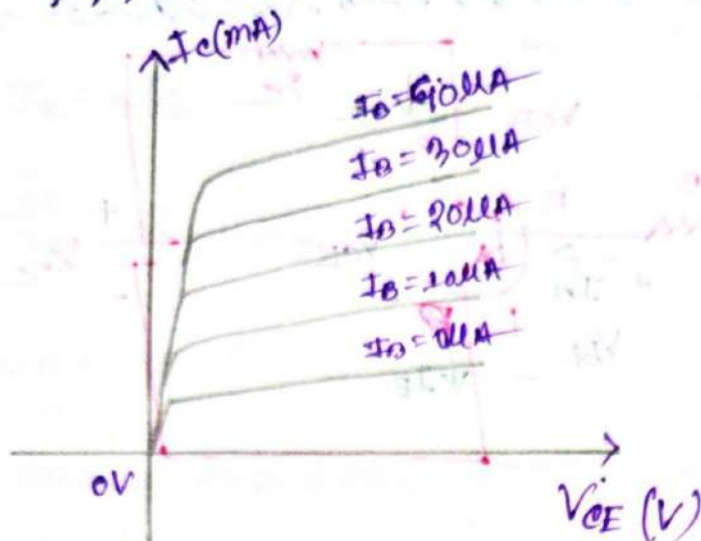
V_{CE} = Output Voltage.

* Input Characteristics:-



$$r_i = \frac{\Delta V_{BE}}{\Delta I_B}$$

* Output - Characteristics



For Active region:-

$$V_{CE} \uparrow \rightarrow V_{CB} \uparrow$$

$$V_{CE} = V_{CB} + V_{BE}$$

$$\therefore V_{CE} \uparrow \rightarrow I_c \uparrow$$

If $I_B = \text{Input current}$

$$\therefore I_c = \beta I_B = \text{output current}$$

β_{ac} = Current gain of this common-Emitter Configuration.

$$\therefore \beta_{ac} = \frac{\Delta I_c}{\Delta I_B}$$

For Saturation Region:-

$$V_{CB} \uparrow \rightarrow V_{BE} \uparrow$$

For cut-off region:

$$I_c = \alpha I_E + I_{CBO}$$

$$\Rightarrow I_c = \alpha (I_c + I_B) + I_{CBO}$$

$$\Rightarrow (1 - \alpha) I_c = \alpha I_B + I_{CBO}$$

$$\Rightarrow I_c = \frac{\alpha}{1 - \alpha} I_B + \frac{1}{1 - \alpha} I_{CBO}$$

$$\therefore I_c = \beta I_B + \frac{1}{1 - \alpha} I_{CBO}$$

if: $I_B = 0$

$$\text{Then } I_c = \frac{1}{1 - \alpha} I_{CBO}$$

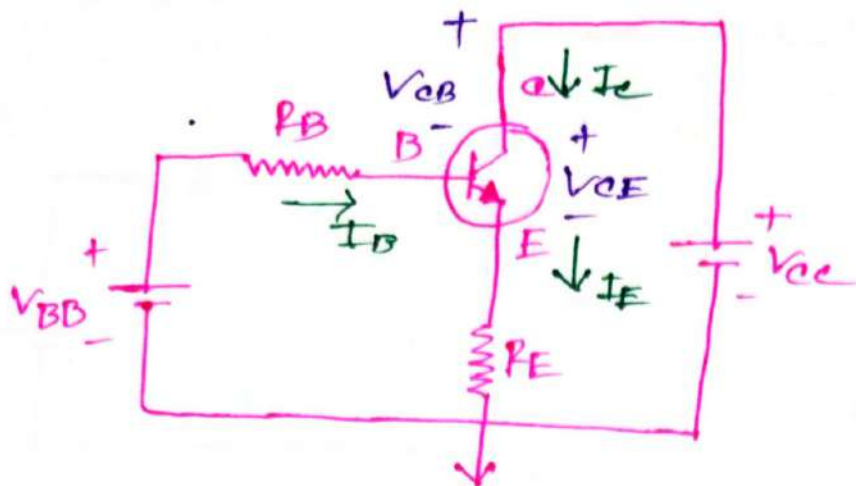
$$\therefore I_c = (\beta + 1) I_{CBO}$$

$$\therefore I_{CEO} = (\beta + 1) I_{CBO}$$

I_{CEO} = leakage current

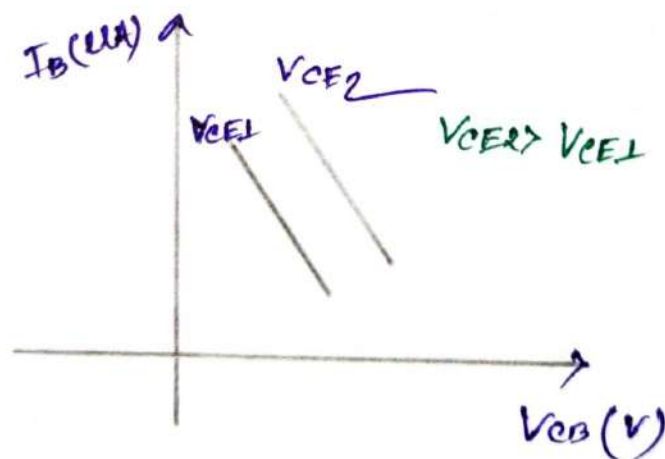
$$\left. \begin{aligned} \beta &= \frac{\alpha}{1 - \alpha} \\ \therefore \beta + 1 &= \frac{1}{1 - \alpha} \end{aligned} \right\}$$

#Common Collector Configuration:



here, I_B = Input current
 V_{BE} = Input voltage
 $I_E \Rightarrow$ Output current
 V_{CE} = Output voltage.

* Input Characteristics



$$V_{CE} = V_{BE} + V_{BE}$$

$$r = \frac{I_E}{I_B}$$

Relationship between α, β, r

We know

$$I_E = I_e + I_B$$

$$\therefore r = \frac{I_e + I_B}{I_B} = \frac{I_e}{I_B} + 1 \quad \text{--- (1)}$$

We also know:-

$$I_e \approx \beta I_B$$

Now, put I_e in eq (1) =)

$$\therefore r = \frac{\beta I_B}{I_B} + 1 \approx \beta + 1$$

$$\therefore r = \beta + 1$$

$$\Rightarrow r = \frac{\alpha}{1-\alpha} + 1$$

$$[\because \beta = \frac{\alpha}{1-\alpha}]$$

$$\therefore r = \frac{1}{1-\alpha}$$

Now multiply Both side with α :-

$$\alpha r = \frac{\alpha}{1-\alpha}$$

$$\therefore \alpha \cdot r = \beta$$

$$\underline{\underline{\text{So, } \boxed{\beta = \alpha r}}}$$

Biasing Dec
chooy

Biasing:-

- ⇒ Apply DC voltages to establish a fixed level of current and voltage.
- ⇒ The applied DC establishes an operating point (Q-point) that define the region for the signal amplification.

A BJT to be biased in active operating region:-

1. BE Junction = Forward Biased.

2. BC Junction = Reverse Biased.

BJT BIAS CONFIGURATION:-

1. Fixed-Bias Configuration.

2. Emitter Bias Configuration.

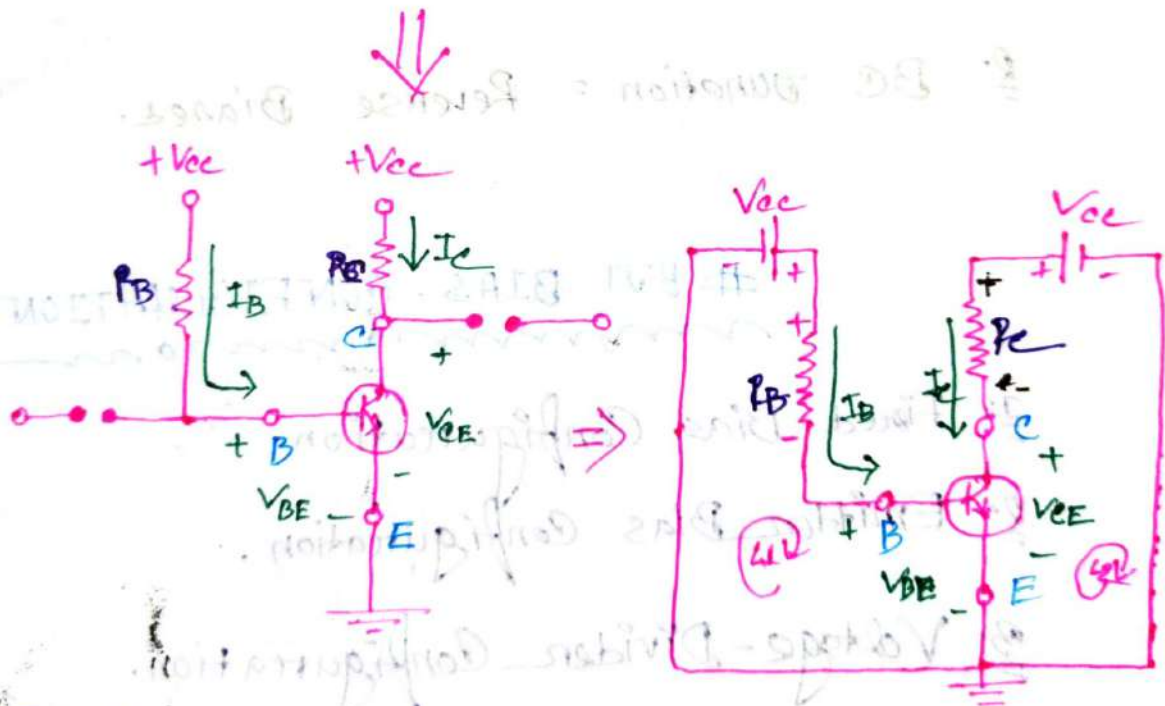
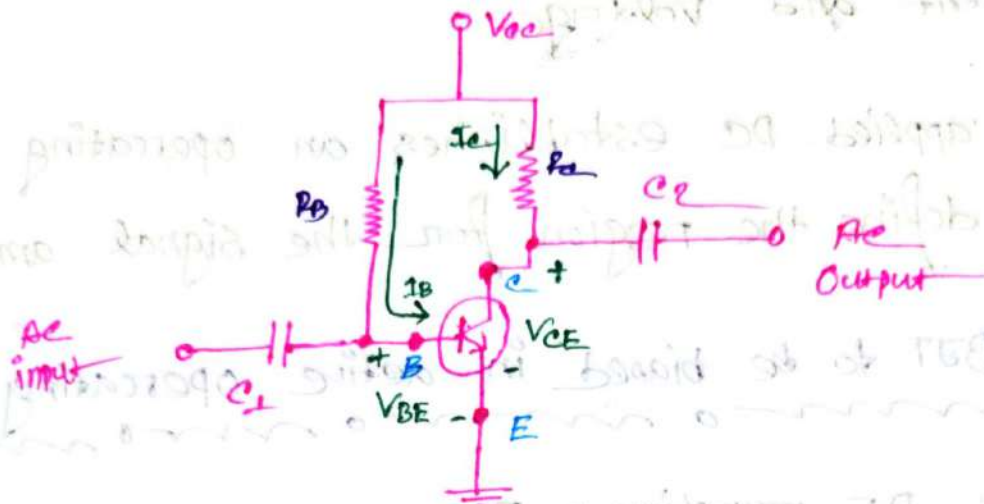
3. Voltage-Divider Configuration.

4. Collector Feedback Configuration.



Fixed-Bias Configuration:-

Base-Bias Configuration:-



⇒ Now For Output current I_C and output voltage V_{CE} Apply KVL:-
In the Input Loop (L1):-

$$V_{CC} - I_B R_B - V_{BE} = 0$$

$$\therefore I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

We know:-

$$\beta = \frac{I_c}{I_B}$$

$$\therefore I_c = \beta(I_B) \\ = \beta \left(\frac{V_{cc} - V_{BE}}{R_B} \right)$$

$$\therefore \boxed{I_c = \beta \times \frac{V_{cc} - V_{BE}}{R_B}}$$

Where:-

$$V_{BE} = 0.7 \rightarrow \text{Si} / 0.3 \rightarrow \text{Ge}$$

V_{cc} = Biasing potential.

R_B = Base Resistance.

Now, For V_{CE} :-

=> Apply KVL In the output Loop (22):-

$$V_{CE} + I_c R_c - V_{cc} = 0$$

$$\therefore \boxed{V_{CE} = V_{cc} - I_c R_c}$$

Put the value of I_c :-

$$V_{CE} = V_{cc} - \beta \times R_c \times \frac{V_{cc} - V_{BE}}{R_B}$$

In this case:- $V_E = 0V$! Since it's connected to ground:-

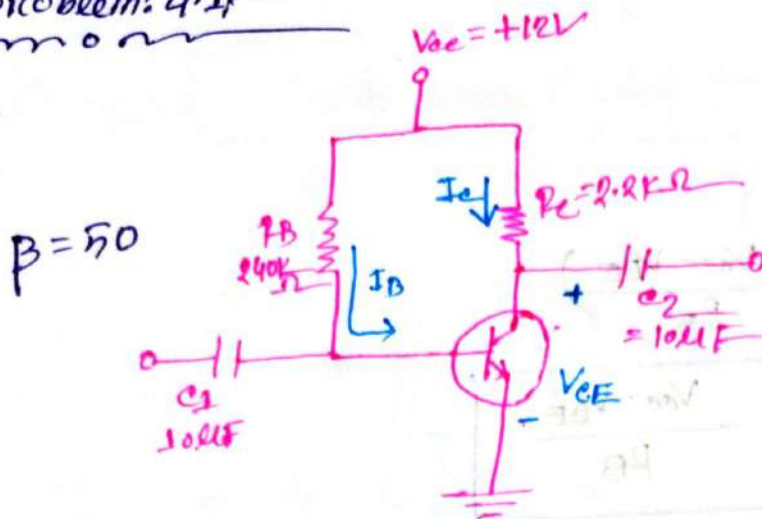
$$\therefore V_{DE} = V_B - V_E = V_B - 0 = V_B$$

$$\boxed{V_{DE} = V_B}$$

$$\therefore V_{CE} = V_c - V_E = V_c - 0 = V_c$$

$$\boxed{V_{CE} = V_c}$$

problem: 4.4



⇒ Determine:-

- I_{BQ} and I_{CQ}
- V_{CEQ}
- V_B and V_E
- V_{BE}

⇒ we draw the circuit:-

Is given:-

$$R_B = 240k\Omega$$

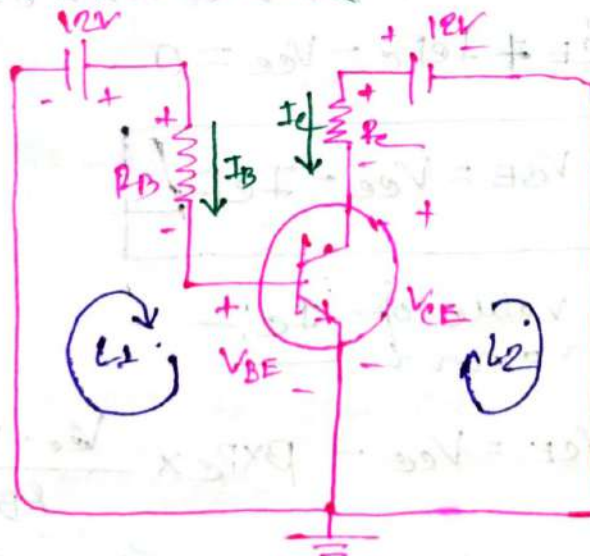
$$R_C = 2.2k\Omega$$

$$V_{CC} = 12V$$

$$\beta = 50$$

$$V_{BE} = 0.7V$$

①



For I_{BQ} :-

Apply KVL in Loop 1 (L1):-

$$12V - I_{BQ}R_B - V_{BE} = 0$$

$$\therefore I_{BQ} = \frac{12 - 0.7}{240k} = 47.08\mu A$$

For I_{CQ} :-

we know:-

$$\beta = \frac{I_C}{I_B}$$

$$\therefore I_C = \beta \times I_B$$

$$= 50 \times 47.08 \mu A$$

$$= 2.35 mA$$

ii

For V_{CEQ} :-

Apply KVL in Loop 2 (L2):-

$$V_{CE} + I_{CQ} R_C - 12 = 0$$

$$\therefore V_{CE} = 12 - I_{CQ} R_C$$

$$= 12 - 2.35 mA \times 2.2 k$$

$$= 6.83 V$$

iii

we know:-

$$V_{BE} = V_B - V_E$$

$$\therefore V_B = V_{BE} + V_E = 0.7 V + 0 = 0.7 V$$

$$V_{CE} = V_C - V_E$$

$$\therefore V_C = V_{CE} + V_E = 6.83 + 0 = 6.83 V$$

iv

we know:-

$$V_{BE} = V_B - V_E$$

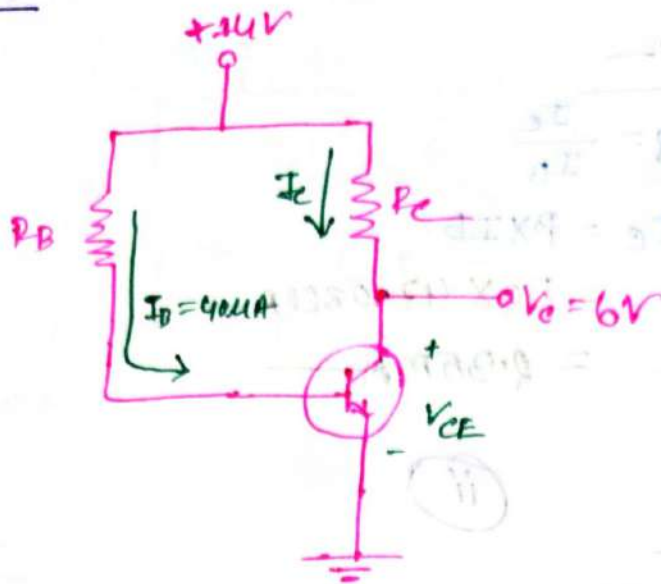
$$= 0.7 - 6.83$$

$$= -6.13 V$$

Ans:-

problem's

$$\beta = 80$$



⇒ Determine

(i) I_C

(ii) R_C

(iii) R_B

(iv) V_{CE}

⇒ 1st redraw the circuit:-

In given:-

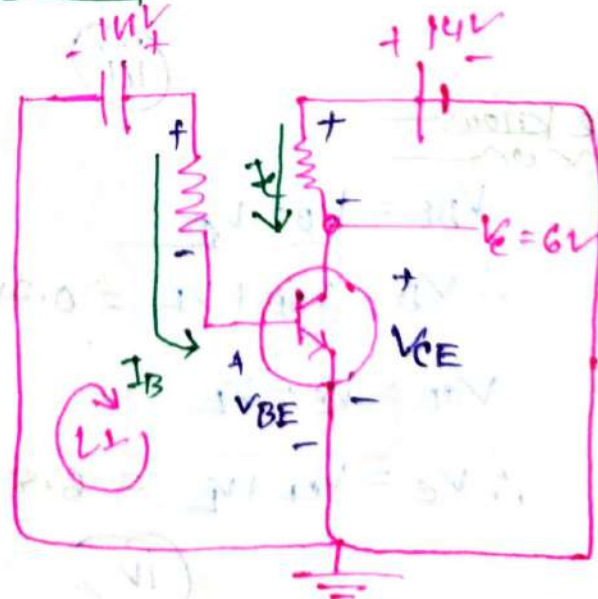
$$I_B = 40\mu A$$

$$V_{CC} = 14V$$

$$\beta = 80$$

$$V_C = 6V$$

$$V_{BE} = 0.7V$$



We know:-

$$\beta = \frac{I_C}{I_B}$$

$$\therefore I_C = I_B \times \beta = 40\mu A \times 80 = 3.2mA$$

(iv)

We know:-

$$V_{CE} = V_C - V_E$$

$$\therefore V_{CE} = 6 - 0 = 6V$$

(ii)

1st Find the Voltage Difference Across the R_E

$$\begin{aligned}\therefore V_{RE} &= V_{EE} - V_E \\ &= 14 - 6 \\ &= 8V\end{aligned}$$

We know:-

$$R_E = \frac{V_{RE}}{I_E}$$

$$= \frac{8V}{0.2mA}$$

$$= 2.5K\Omega$$

(iii)

For R_B :-1st Find V_{RB} :- $\therefore$ Apply KVL in Loop (1):-

$$14V - V_{RB} - V_{BE} = 0$$

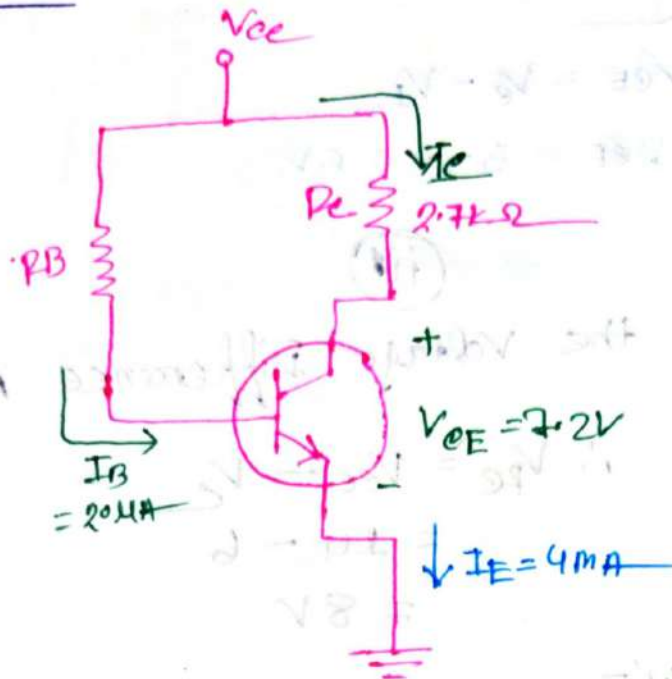
$$\begin{aligned}\therefore V_{RB} &= 14 - 0.7 \\ &= 13.3V\end{aligned}$$

We know:-

$$I_B = \frac{V_{RB}}{R_B}$$

$$\therefore R_B = \frac{V_{RB}}{I_B} = \frac{13.3}{40\mu} = 332.5K\Omega$$

problems 2:-



⇒ Determine:-

(a) I_E

(b) V_{CE}

(c) β

(d) R_B

⇒ 1st red draw the circuit:-

Is given:-

$$R_E = 2.7k\Omega$$

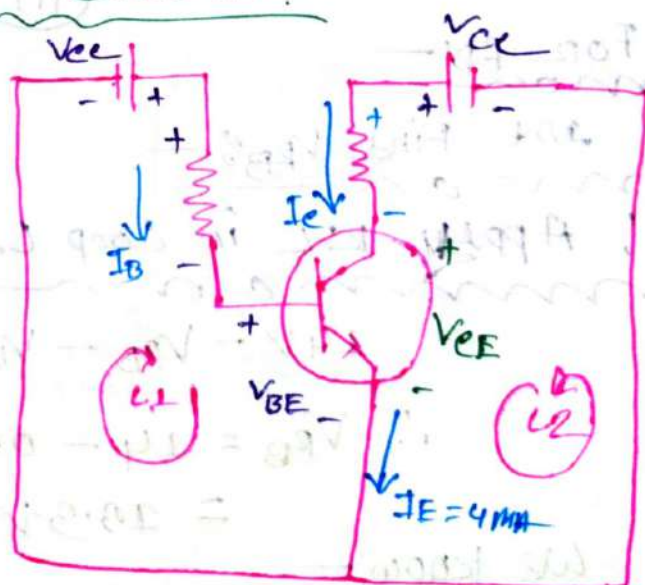
$$V_{CE} = 7.2V$$

$$I_E = 4mA$$

$$V_{BE} = 0.7V$$

$$I_B = 20\mu A$$

(1)



For I_E :-

we know:-

$$I_E = I_B + I_C$$

$$\therefore I_e = I_E - I_D$$

$$= 4 \text{ mA} - 20 \mu\text{A}$$

$$= 3.98 \text{ mA}$$

(ii)

For V_{ee} :Apply KVL in Loop (L2):

$$V_{ee} - I_e R_e - V_{EE} = 0$$

$$\therefore V_{ee} = I_e R_e + V_{EE}$$

$$= 3.98 \text{ mA} \times 2.7 \text{ k}\Omega + 7.2 \text{ V}$$

$$= 17.046 \text{ V}$$

(iii)

we know:

$$\beta = \frac{I_e}{I_B} = \frac{3.98 \text{ mA}}{20 \mu\text{A}} = 199$$

(iv)

For R_B :Apply KVL in Loop (L1):

$$V_{ee} - I_B R_B - V_{BE} = 0$$

$$\therefore R_B = \frac{V_{ee} - V_{BE}}{I_B}$$

$$= \frac{17.046 - 0.7}{20 \mu\text{A}}$$

$$= 862.3 \text{ k}\Omega$$

Transistor Saturation:-

⇒ The transistor is said to be saturated when the collector current I_c is larger than or equal to the maximum collector current.

$$\therefore I_{cmax} \leq I_c$$

In this condition the collector current I_c representing the collector current in saturation condition.

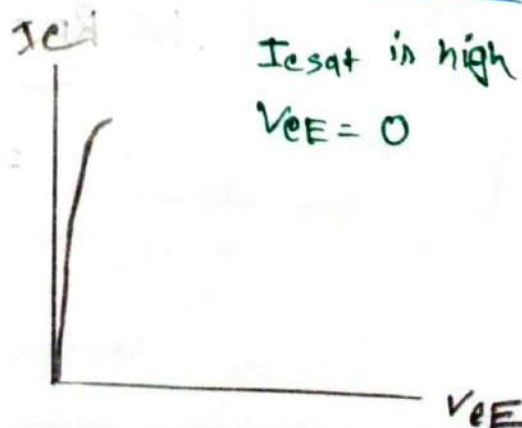
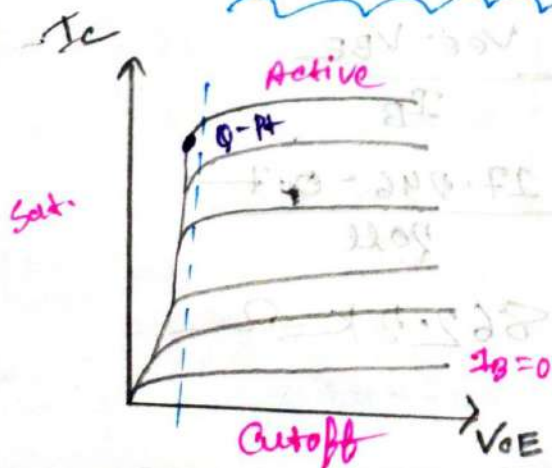
$$\therefore I_{cmax} \leq I_{csat}$$

☑ In Saturation Region:-

1. CBJ (Collector Base junction) → Forward Bias

2. EBJ (Emitter Base junction) → Forward Bias

Analysis the Output ch: for Common Emitter Tr:-



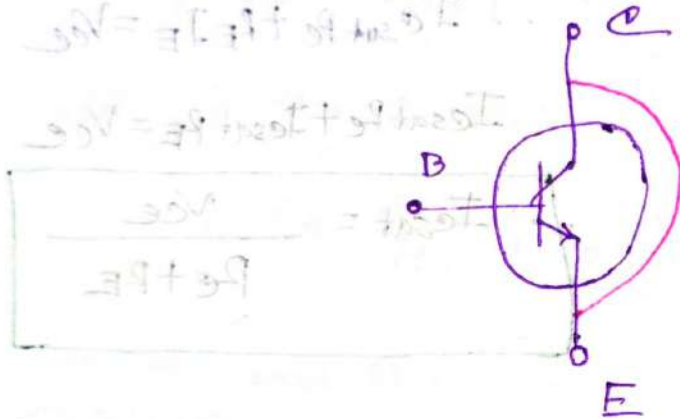
For R_{EE} in Sat Condition:-

$$I_{E\text{sat}} \times R_{EE} = V_{EE}$$

$$\therefore R_{EE} = \frac{0V}{I_{E\text{sat}}} = 0 \Omega$$

This means there is no resistance between the Collector and the Emitter terminals.

$\Rightarrow$ So we can short circuit the two terminal.

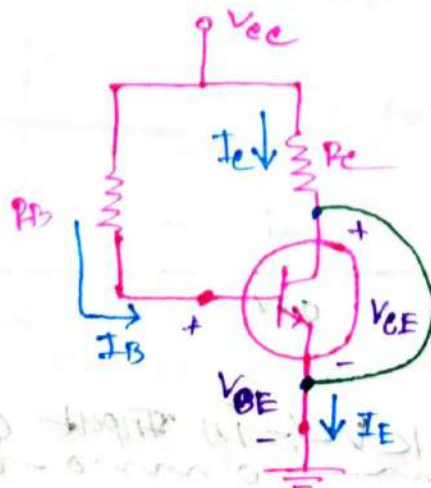


☑ Saturation Condition for the Fixed-Bias Configuration:-

$$\therefore V_{CE} = V_{CC} - I_E R_E$$

If $I_E = 0$:-

$$V_{CE} = V_{CC}$$



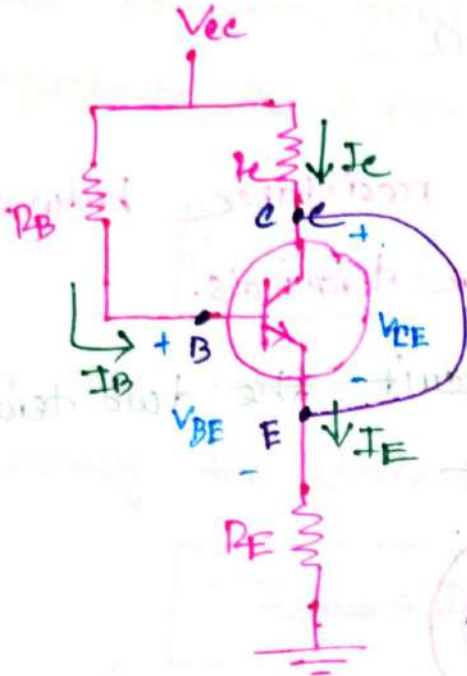
$$I_{E\text{sat}} = I_{E\text{max}}$$

Apply KVL in output loop:-

$$V_{CC} - I_{E\text{sat}} R_E = 0$$

$$\therefore I_{E\text{sat}} = I_{E\text{max}} = \frac{V_{CC}}{R_E}$$

Saturation Condition for Emitter Bired Configuration:-



$$I_{sat} = I_{emax}$$

∴ Apply KVL in output loop:-

$$V_{cc} - I_{sat} R_C - I_E R_E = 0$$

$$∴ I_{sat} R_C + R_E I_E = V_{cc}$$

$$I_{sat} R_C + I_{sat} R_E = V_{cc}$$

$$I_{sat} = \frac{V_{cc}}{R_C + R_E}$$

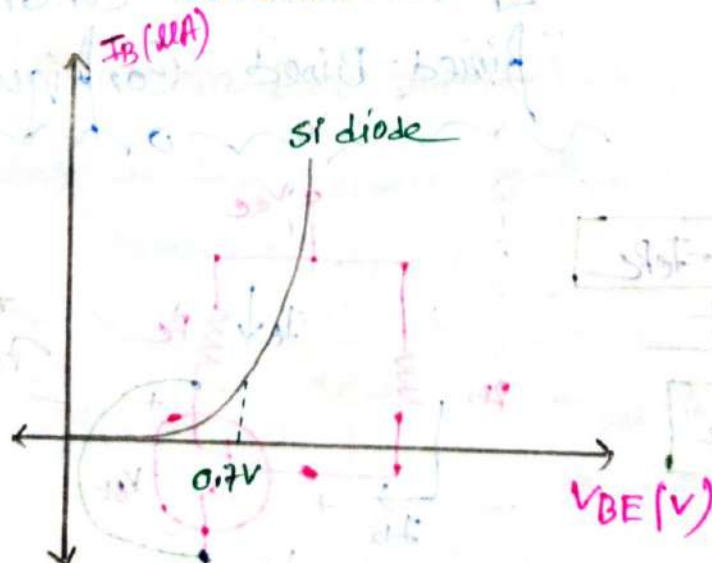
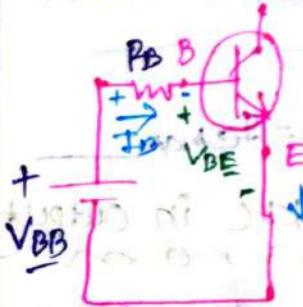
we know

$$I_E \approx I_C$$

$$∴ I_{sat} = I_E$$

Load-Line Analysis:-

For Input:-



Apply KVL in Input loop:-

$$V_{BB} - I_B R_B - V_{BE} = 0$$

For Draw the load line we need 2 points:-

$$P_1 = (0, \quad), P_2 = (\quad, 0)$$

At $V_{BE} = 0$:-

$$I_B = -\frac{V_{BB}}{R_B}$$

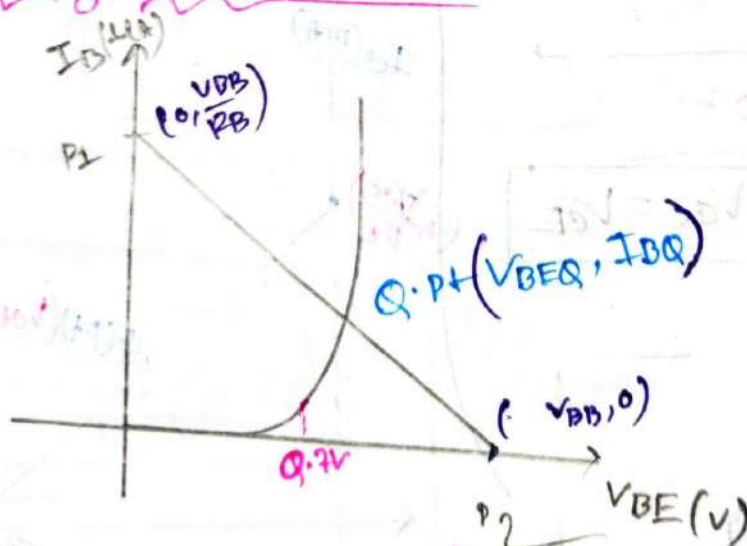
At $I_B = 0$:-

$$V_{BE} = V_{BB}$$

So, points 2 are:-

$$P_1 = (0, \frac{V_{BB}}{R_B}), P_2 = (V_{BB}, 0)$$

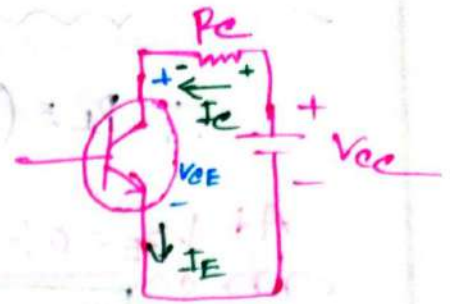
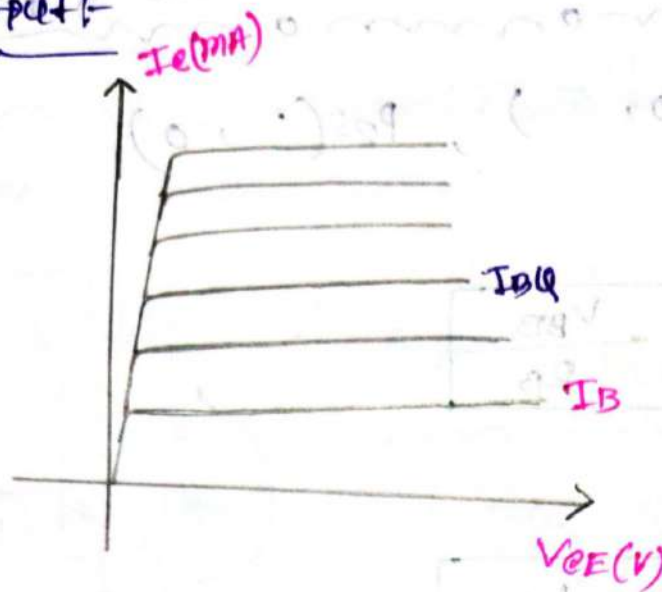
So, the load line is:-



Notes:-

- 1. If we increase R_B the slope will decrease
- 2. If we decrease R_B the slope will increase.

For Output:-



Apply KVL in output loop:-

$$V_{CC} - I_C R_C - V_{CE} = 0$$

For Draw the load line we need 2 points:-

At $V_{CE} = 0$

$$I_C = \frac{V_{CC}}{R_C}$$

At $I_C = 0$

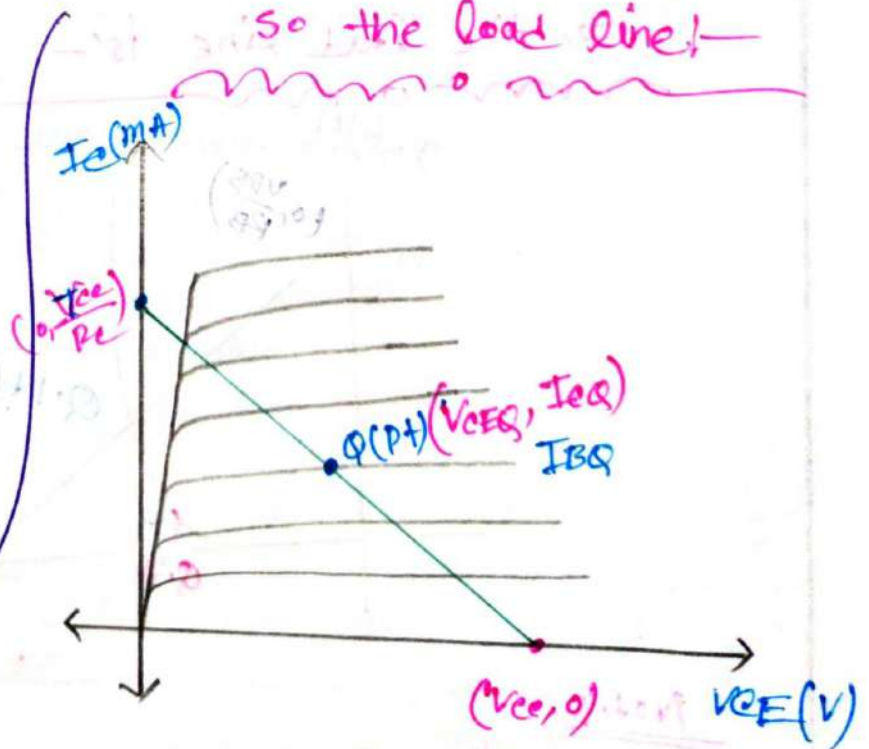
$$V_{CC} = V_{CE}$$

So the load line:-

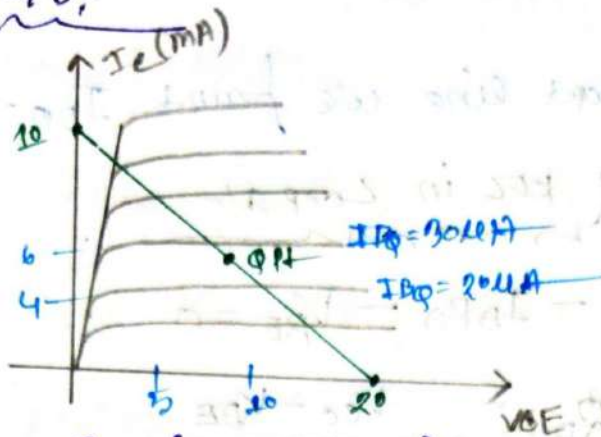
∴ So, points 2 are:-

$$P_1 \left(0, \frac{V_{CC}}{R_C} \right)$$

$$P_2 (V_{CC}, 0)$$

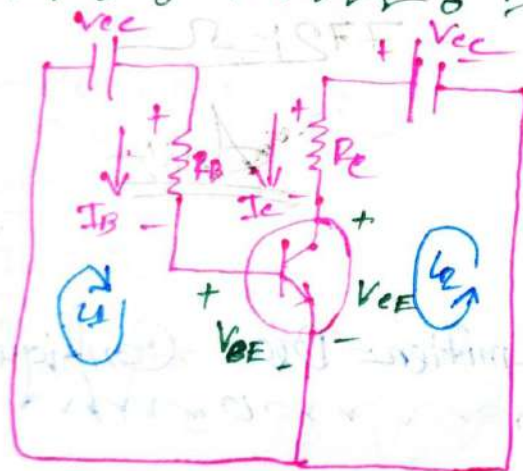


problems 4.3:-



⇒ Determine Q point. also, find V_{CE} , R_E , R_B for a fixed-bias configuration.

⇒ Draw the fixed-bias configuration circuit:-



For Q point:-

$$V_{CEQ} = 8V$$

$$I_{CQ} = 5mA$$

$$\therefore Q = (8, 5mA)$$

At $I_C = 0$:-

$$V_{CE} = V_{CE} = 20V$$

For R_E :-

At $V_{CE} = 0$:-

$$I_C R_E = V_{CE}$$

$$R_E = \frac{V_{CE}}{I_C}$$

$$= \frac{20}{10mA}$$

$$R_E = 2k\Omega$$

$$\left. \begin{array}{l} I_C = 10mA \\ V_{CE} = 20V \end{array} \right\}$$

For V_{CE} :-

Apply KVL in loop 2:-

$$V_{CE} - V_C - V_{CE} = 0$$

$$\Rightarrow V_{CE} - I_C R_E - V_{CE} = 0$$

For R_B :-

⇒ From the load line we found $I_{BQ} = 25 \mu A$

Now, Apply KVL in Loop 1:-

$$V_{CC} - I_B R_B - V_{BE} = 0$$

$$\therefore R_B = \frac{V_{CC} - V_{BE}}{I_B}$$

$$= \frac{(20 - 0.7V)}{25 \mu A}$$

$$= 772 k\Omega$$

here,

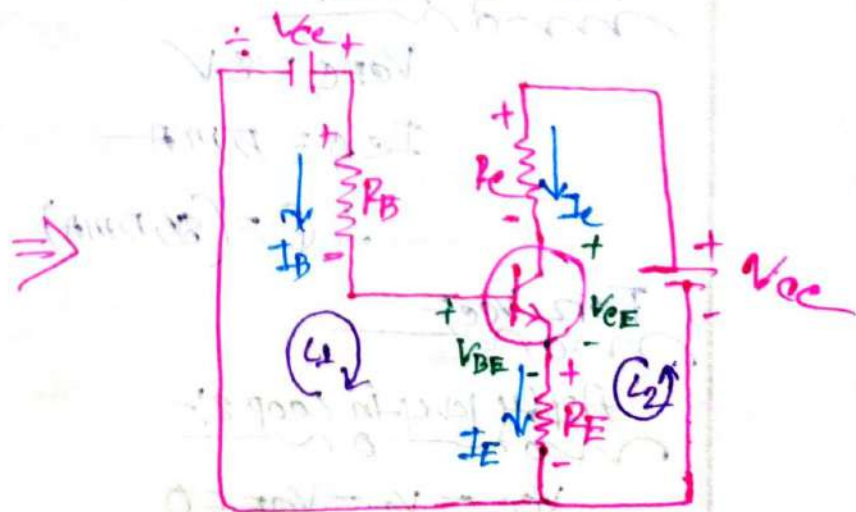
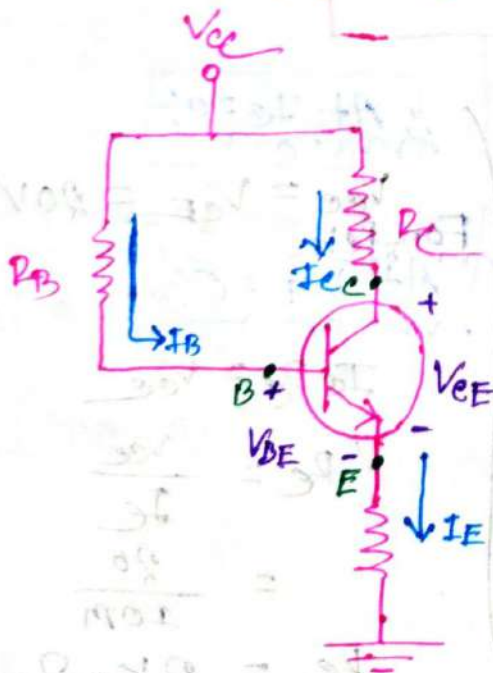
$$V_{BE} = 0.7V$$

$$I_B = I_{BQ} = 25 \mu A$$

$$V_{CC} = 20V$$

Ans:-

Emitter-Bias Configuration:-



Now For Input current I_B Apply KVL in the Input Loop (L1):-

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$\Rightarrow V_{CC} - I_B R_B - V_{BE} - (\beta + 1) I_B R_E = 0$$

$$\Rightarrow I_B \{ (\beta + 1) R_E + R_B \} = V_{CC} - V_{BE}$$

$$\therefore I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E}$$

We know:-

$$I_E = I_C + I_B$$

$$= \beta I_B + I_B$$

$$I_E = I_B (\beta + 1)$$

For Output Current I_C :-

We know:-

$$I_C = \beta I_B$$

$$I_C = \beta \times \left\{ \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E} \right\}$$

Now For V_{CE} :-

$\Rightarrow$ Apply KVL in the output Loop (L2):-

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$\Rightarrow V_{CC} - I_C R_C - V_{CE} - I_C R_E = 0$$

$$\Rightarrow V_{CC} - I_C (R_C + R_E) - V_{CE} = 0$$

$$\therefore V_{CE} = V_{CC} - I_C (R_C + R_E)$$

We know:-

$$I_E \approx I_C$$

Now:-

$$V_E = I_{E R_E} - 0 = I_{E R_E}$$

For V_E :-we know:-

$$V_{CE} = V_C - V_E$$

$$\therefore V_C = V_{CE} + V_E$$

$$= V_{CC} - I_{C R_E} - I_{E R_E} + I_{E R_E}$$

$$V_C = V_{CC} - I_{C R_E}$$

Since
 $I_E \approx I_C$

For V_B :-we know:-

$$V_{BE} = V_B - V_E$$

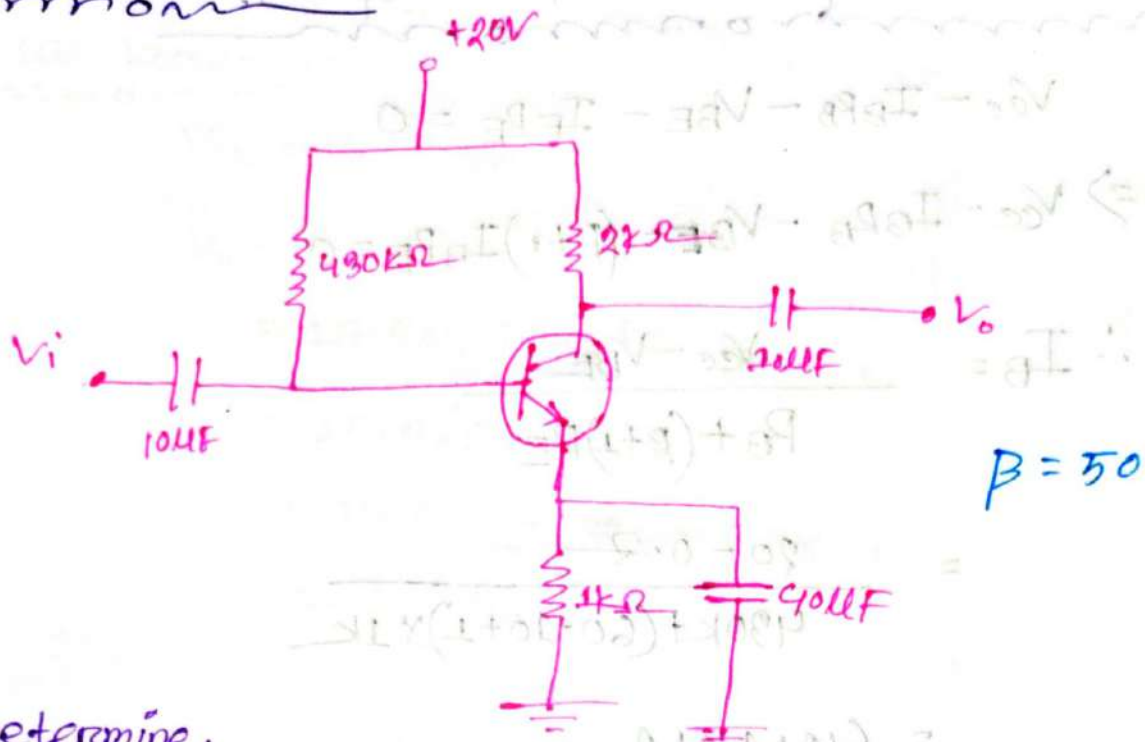
$$\therefore V_B = V_{BE} + V_E$$

$$\therefore V_B = V_{BE} + I_{E R_E}$$

$$\text{or } V_B = V_{BE} + I_{C R_E}$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

problem 4.4:-



⇒ Determine,

- (a) I_B , (b) I_C , (c) V_{CE} , (d) V_C , (e) V_E , (f) V_B
(g) V_{BE}

⇒ 1st redraw the circuit:-

Is given:-

$$R_B = 430k\Omega$$

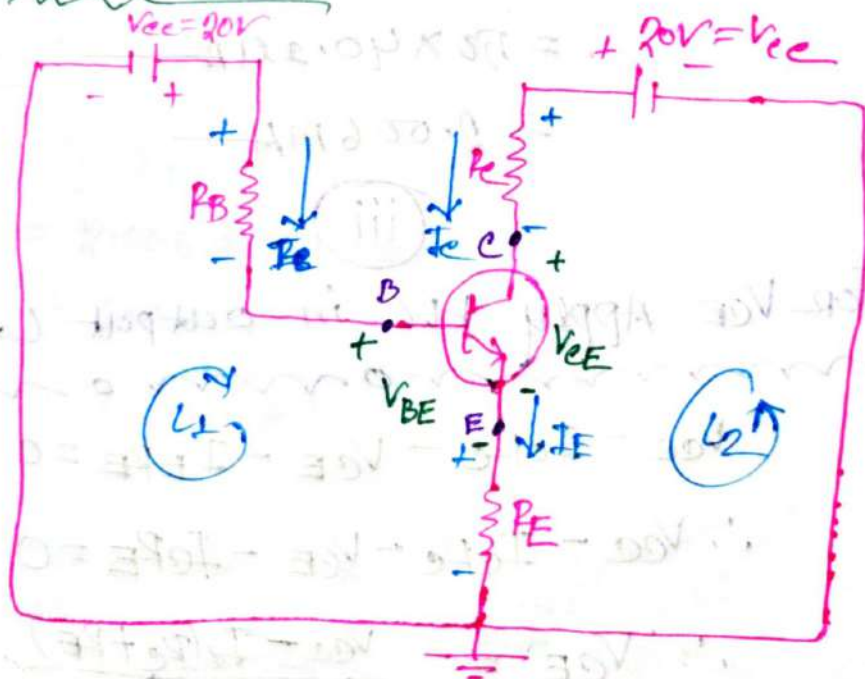
$$R_C = 2k\Omega$$

$$R_E = 1k\Omega$$

$$V_{CC} = 20V$$

$$\beta = 50$$

$$V_{BE} = 0.7V$$



(4)

For I_B Apply KVL in input loop (L1):-

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$\Rightarrow V_{CC} - I_B R_B - V_{BE} - (\beta + 1) I_B R_E = 0$$

$$\begin{aligned} \therefore I_B &= \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E} \\ &= \frac{20 - 0.7}{490k + (60 - 10 + 1) \times 1k} \\ &= 40.1 \mu A \end{aligned}$$

(ii)

we know:-

$$I_C = \beta I_B$$

$$= 50 \times 40.1 \mu A$$

$$= 2.006 \text{ mA}$$

(iii)

For V_{CE} Apply KVL in output loop (L2):-

$$V_{CC} - I_E R_E - V_{CE} - I_E R_E = 0$$

$$\therefore V_{CC} - I_E R_E - V_{CE} - I_E R_E = 0$$

$$\therefore V_{CE} = \frac{V_{CC} - I_E (R_E + R_E)}{1}$$

$$= 20 - 2.006 \text{ m} (2k + 1k) = 13.982 \text{ V}$$

(iv)

we know:-

$$V_{CE} = V_C - V_E$$

$$\therefore V_C = V_{CE} + V_E$$

$$= 15.982 + I_{E R_E}$$

$$= 15.982 + I_{E R_E}$$

$$= 15.982 + 2.006 \text{ mA} \times 1 \text{ k}$$

$$= 17.98 \text{ V}$$

OR

$$V_C = V_{CC} - I_{C R_C}$$

$$= 20 - 2.006 \text{ mA} \times 2 \text{ k}$$

$$= 15.98 \text{ V}$$

(v)

we know:-

$$V_E = I_{E R_E}$$

$$= 2.006 \text{ mA} \times 1 \text{ k}$$

$$= 2.006 \text{ V}$$

(vi)

we know:-

$$V_{BE} = V_B - V_E$$

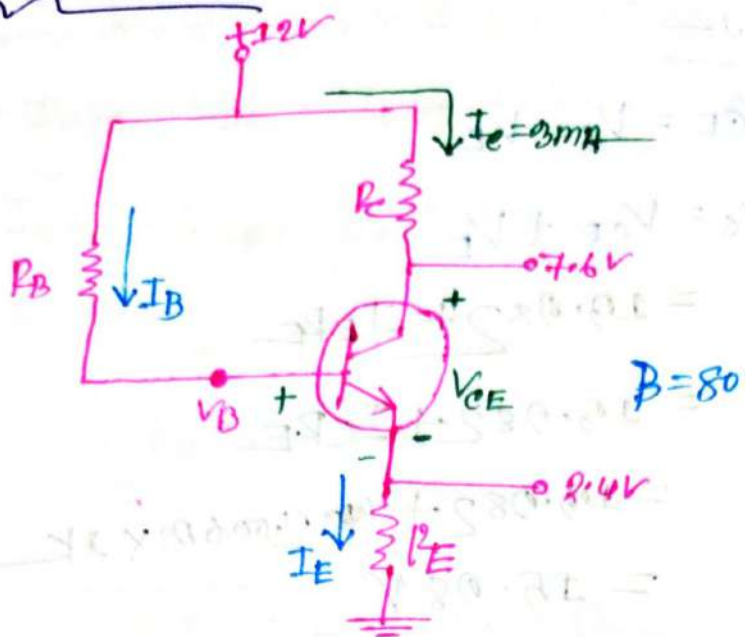
$$\therefore V_B = V_{BE} + V_E = 0.7 + 2.006 = 2.706 \text{ V}$$

(vii)

we know:-

$$V_{BC} = V_B - V_C = 2.706 - 15.98 = -13.27 \text{ V}$$

problem 2:-



⇒ For the emitter-bias circuit Determine.

- ① R_C ② R_E ③ R_B ④ V_{CE} ⑤ V_B

⇒ 1st redraw the circuit:-

⇒ Is given:-

$$V_{CC} = 12V$$

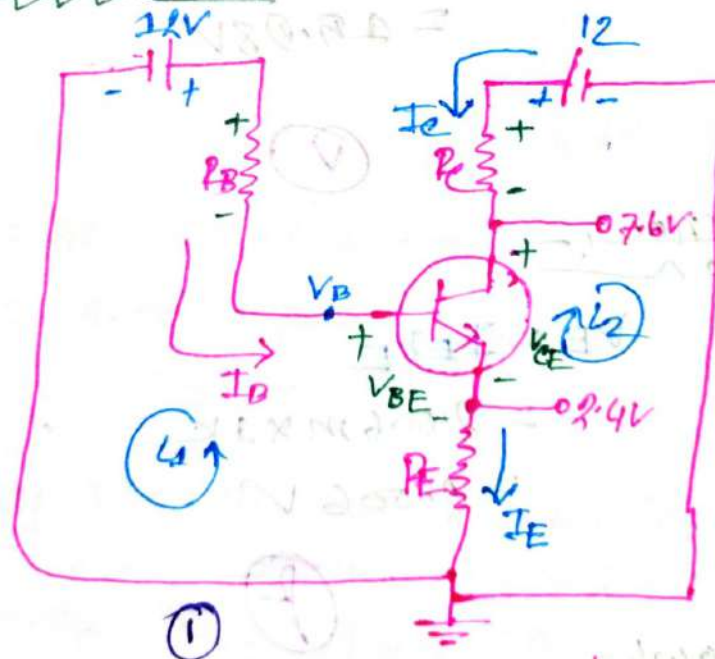
$$I_C = 3mA$$

$$\beta = 80$$

$$V_C = 7.6V$$

$$V_E = 2.4V$$

$$V_{BE} = 0.7V$$



For R_E :-

The voltage difference across the resistor R_E :-

$$V_{RE} = 12 - 7.6 = 4.4V$$

We know:-

$$R_E = \frac{V_{RE}}{I_E} = \frac{4.4}{3mA} = 1.467K\Omega$$

(ii)

we know:-
mon

$$V_E = I_E R_E$$

$$\therefore R_E = \frac{V_E}{I_E}$$

$$= \frac{2.4}{3m}$$

$$= 0.8k\Omega$$

$$\left(\begin{array}{l} \text{here,} \\ I_E \approx I_E \end{array} \right)$$

(iii)

we know:-
mon

$$\beta = \frac{I_E}{I_B}$$

$$\therefore I_B = \frac{I_E}{\beta} = \frac{3m}{80} = 37.5\mu A$$

(iv)

$\Rightarrow$ For V_{CE} Apply KVL in output loop (22)
mon

$$12 - I_E R_E - V_{CE} - I_E R_E = 0$$

$$\therefore V_{CE} = 12 - I_E (R_E + R_E)$$

$$= 12 - 3m(1.467k + 0.8k)$$

$$= 5.100V$$

(v)

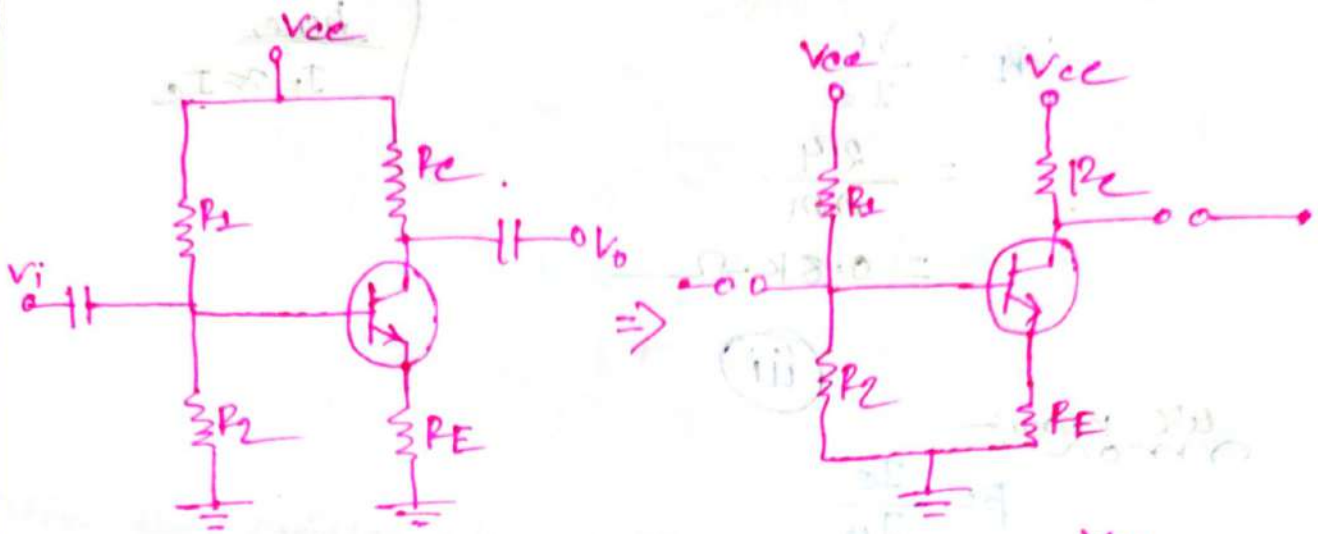
we know:-
mon

$$V_{BE} = V_B - V_E$$

$$\therefore V_B = V_{BE} + V_E = 0.7 + 2.4 = 3.1V$$

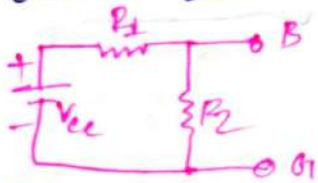
Anse

Voltage-Divider Bias Configuration:

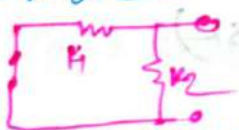


Now:

Calculate R_{th} and E_{th} across the terminals B-G



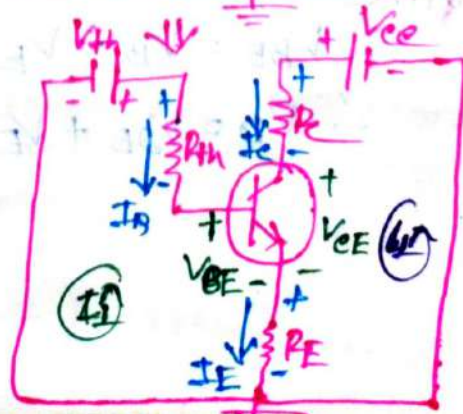
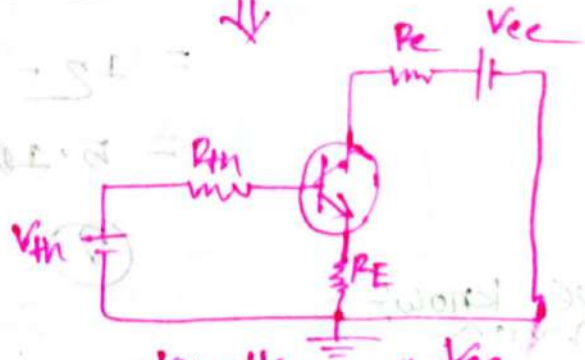
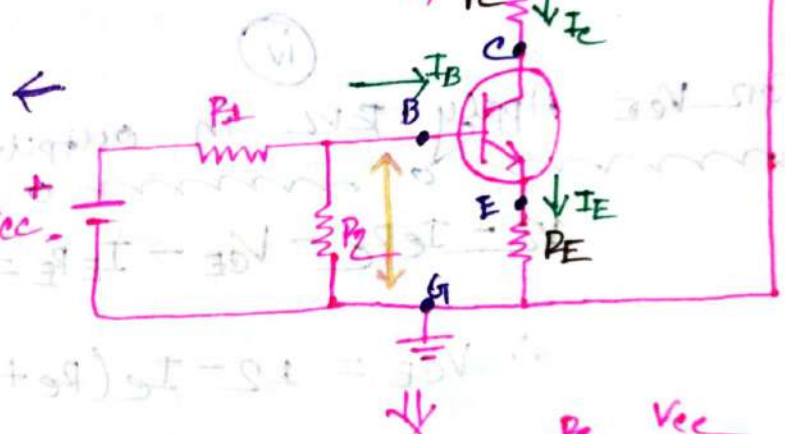
For R_{th} :-



$$R_{th} = \left(\frac{1}{R_1} + \frac{1}{R_2} \right)^{-1}$$

For E_{th} :-

$$V_{B2} = E_{th} = \frac{R_2}{R_1 + R_2} \times V_{cc}$$



For Current I_B Apply KVL in Loop 1:-

$$V_{th} - I_B R_{th} + V_{BE} - I_E R_E = 0$$

$$\Rightarrow V_{th} - I_B R_{th} - V_{BE} - I_B (\beta + 1) R_E = 0$$

$$\Rightarrow V_{th} - I_B \{ R_{th} + (\beta + 1) R_E \} - V_{BE} = 0$$

$$\Rightarrow I_B = \frac{V_{th} - V_{BE}}{R_{th} + (\beta + 1) R_E}$$

$$\begin{aligned} I_E &= I_C + I_B \\ &= \beta I_B + I_B \\ &= I_B (\beta + 1) \end{aligned}$$

For Output Current I_E :-

we know:-

$$I_E = \beta I_B$$

$$\therefore I_E = \beta \times \frac{V_{th} - V_{BE}}{R_{th} + (\beta + 1) R_E}$$

For V_{CE} :-

Apply KVL in output loop (L2):-

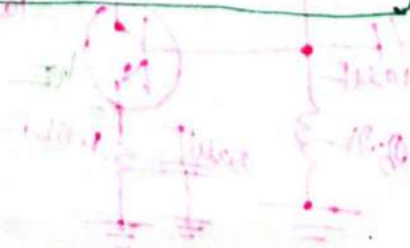
$$+V_{CC} - I_E R_E - V_{CE} - R_E I_E$$

$$\Rightarrow V_{CC} - I_E R_E - V_{CE} - I_E R_E$$

$$\Rightarrow V_{CE} = V_{CC} - I_E (R_E + R_E)$$

here,

$$I_E \approx I_C$$



Approximate Analysis:-

1st Check the below condition:-

$$R_E \geq 10 R_2$$

if Satisfied

Then:-

$$V_B = \frac{R_2}{R_1 + R_2} \times V_{CC}$$

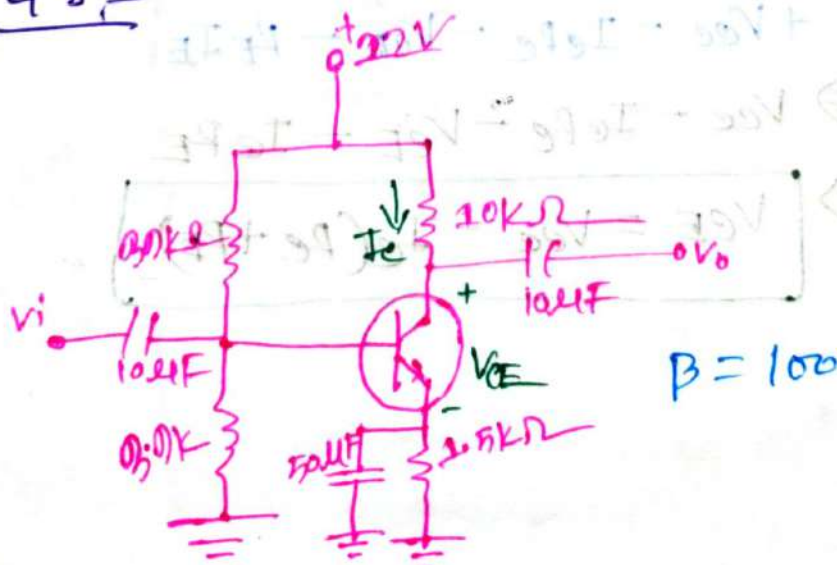
$$V_E = V_B - V_{BE}$$

$$I_E = \frac{V_E}{R_E}$$

$$I_C \approx I_E$$

$$V_{CEQ} = V_{CC} - I_C (R_C + R_E)$$

problems 4.8:-

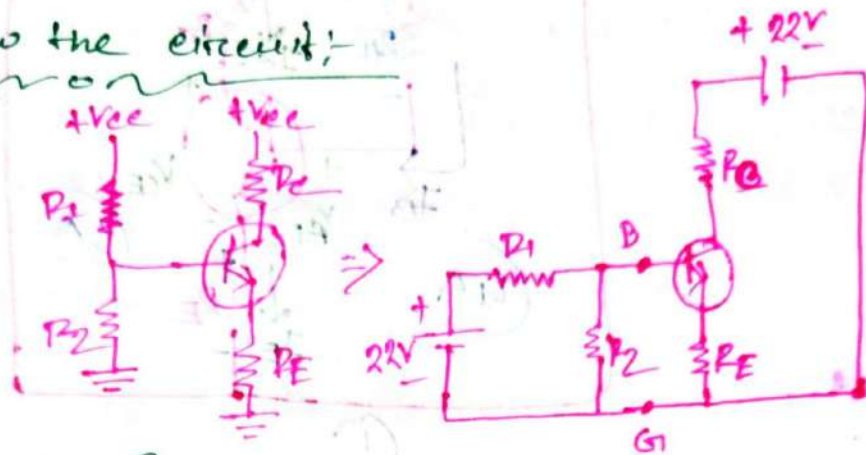


⇒ Determine

(i) V_{CE}

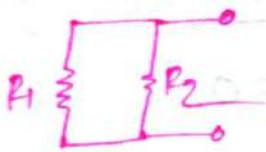
(ii) I_E

⇒ Int. re-draw the circuit:-



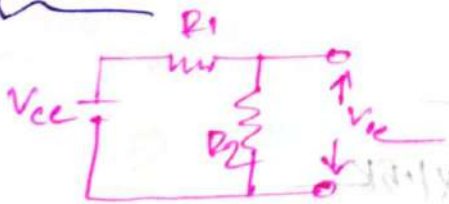
Now Find the E_{th} & R_{th} across the terminal B and G

For R_{th} :-



$$\therefore R_{th} = \left(\frac{1}{R_1} + \frac{1}{R_2} \right)^{-1} = \left(\frac{1}{30k} + \frac{1}{3.9k} \right)^{-1} = 3.54k\Omega$$

For E_{th}/V_{th} :-



$$\therefore V_{oc} = V_{R_2} = V_{th} = \frac{R_2}{R_1 + R_2} \times V_{CC}$$

$$= \frac{3.9k}{30k + 3.9k} \times 22V$$

$$= 2V$$

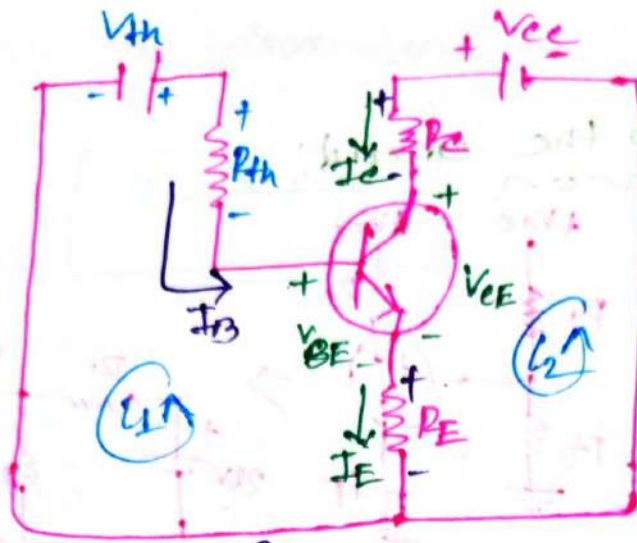
where

$$R_2 = 3.9k\Omega$$

$$R_1 = 30k\Omega$$

$$V_{CC} = 22V$$

Now Rearrange the new circuit!:-



Is given:-

$$V_{th} = 2V$$

$$R_{th} = 3.54k\Omega$$

$$R_E = 1.5k\Omega$$

$$R_C = 10k\Omega$$

$$\beta = 100$$

$$V_{cc} = +22V$$

$$V_{BE} = 0.7V$$

①

For Input current I_B :-

Apply KVL in input Loop L1:-

$$V_{th} - I_B R_{th} - V_{BE} - I_E R_E = 0$$

$$\Rightarrow V_{th} - I_B R_{th} - V_{BE} - (\beta + 1) I_B R_E = 0$$

$$I_B = \frac{V_{th} - V_{BE}}{R_{th} + (\beta + 1) R_E}$$

$$= \frac{2 - 0.7}{3.54k + (101) \times 1.5k}$$

$$= 8.38\mu A$$

here,

$$I_E = I_C + I_B$$

$$= \beta I_B + I_B$$

$$= (\beta + 1) I_B$$

For I_C :-

we know:-

$$I_C = \beta I_B$$

$$= 100 \times 8.38\mu A$$

$$= 0.84mA$$

(11)

For V_{CE} :- $\Rightarrow$ Apply KVL in output loop (L2):-

$$V_{CC} - I_E R_E - V_{CE} - I_E R_E$$

$$\Rightarrow V_{CC} - I_E R_E - V_{CE} - I_E R_E$$

$$\Rightarrow V_{CE} = V_{CC} - I_E (R_E + R_E)$$

$$= 22 - 0.84 \text{ mA} (10 \text{ k} + 1.5 \text{ k})$$

$$= 12.34 \text{ V}$$

Since
 $I_E \approx I_C$

problem: 4.9:-

$\Rightarrow$ Repeat the problem 4.8 using the approximate technique and compare the solution I_{EQ} and V_{CEQ} .

 $\Rightarrow$ 1st check the condition:-

$$\Rightarrow \beta R_E \geq 10 R_2$$

$$\Rightarrow 100 \times 1.5 \text{ k} \geq 10 \times 3.0 \text{ k}$$

$$\Rightarrow 150 \text{ k} \geq 30 \text{ k}$$

(Satisfied)

we know:-

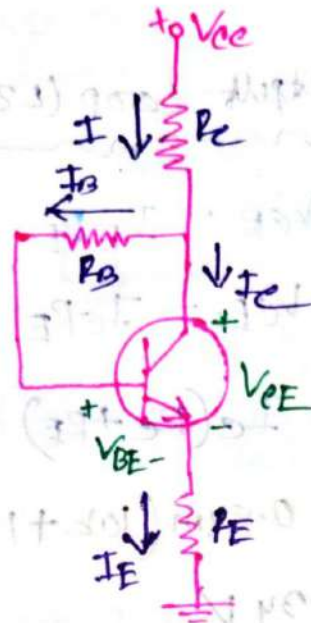
$$V_B = \frac{R_2}{R_1 + R_2} \times V_{CC} = \frac{3.0 \text{ k}}{30 \text{ k} + 3.0 \text{ k}} \times 22 = 2 \text{ V}$$

$$V_E = V_B - V_{BE} = 2 - 0.7 = 1.3 \text{ V}$$

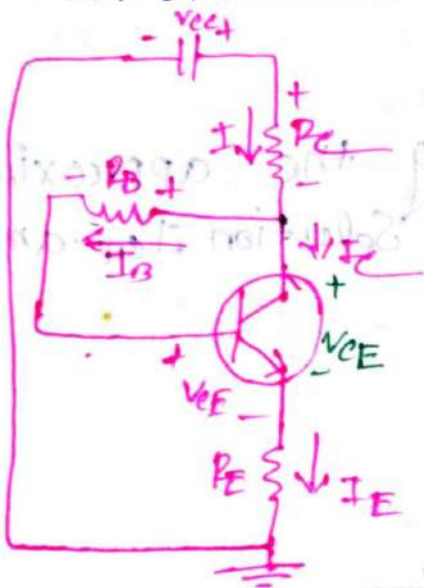
$$I_E = I_{EQ} = \frac{V_E}{R_E} = \frac{1.3}{1.5 \text{ k}} = 0.867 \text{ mA}$$

$$V_{CEQ} = V_{CC} - I_{EQ} (R_E + R_E) = 22 - 0.867 \text{ mA} (10 \text{ k} + 1.5 \text{ k}) = 12.03 \text{ V}$$

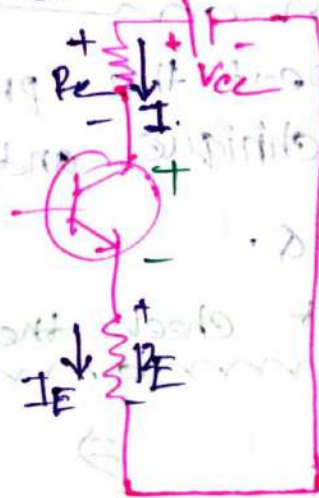
Collector Feedback Configuration:-



For Input:-



For Output:-



Now,

For current I_B :-

Apply KVL in Input Loop:-

$$V_{CC} - I_C R_C - I_B R_B - V_{BE} - I_E R_E = 0$$

$$\Rightarrow V_{CC} - (I_C + I_B) R_C - I_B R_B - V_{BE} - (I_C + I_B) R_E = 0$$

$$\Rightarrow V_{CC} - I_C R_C - I_B R_C - I_B R_B - V_{BE} - (I_C + I_B) R_E = 0$$

here

$$I_C = I_E + I_B$$

$$I_E = I_C + I_B$$

$$= \beta I_B + I_B$$

$$= (\beta + 1) I_B$$

$$\Rightarrow V_{ce} - \beta I_B R_c - I_B R_c - I_B R_B - V_{BE} - (\beta + 1) I_B R_E = 0$$

$$\Rightarrow \beta I_B R_c + I_B R_c + I_B R_B + (\beta + 1) I_B R_E = V_{ce} - V_{BE}$$

$$\Rightarrow I_B \{ (\beta + 1) R_c + R_B + (\beta + 1) R_E \} = V_{ce} - V_{BE}$$

$$\therefore I_B = \frac{V_{ce} - V_{BE}}{R_B + (\beta + 1) R_c + (\beta + 1) R_E}$$

$$\therefore I_B = \frac{V_{ce} - V_{BE}}{R_B + (R_c + R_E)(\beta + 1)}$$

For Output Current I_c :-

We know:-

$$I_c = \beta I_B$$

$$\therefore I_c = \beta \times \frac{V_{ce} - V_{BE}}{R_B + (R_c + R_E)(\beta + 1)}$$

Now for V_{ce} :-

$\Rightarrow$ Apply KVL in output loop:-

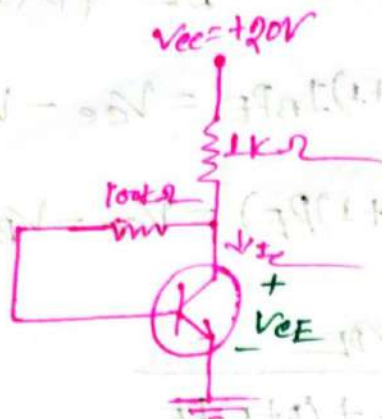
$$V_{ce} - I_c R_c - V_{CE} - I_c R_E = 0$$

$$\Rightarrow V_{ce} - I_c R_c - V_{CE} - I_c R_E = 0$$

$$\Rightarrow V_{CE} = V_{ce} - I_c (R_c + R_E)$$

Since
mon
 $I_E \approx I_c$

#problems 1:-



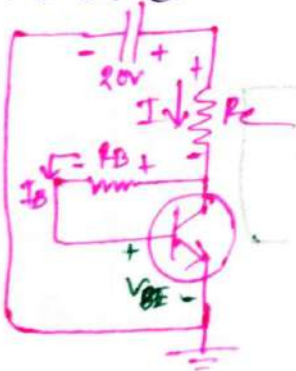
⇒ Determine the operating point where $\beta = 100$

⇒ Operating points means:-

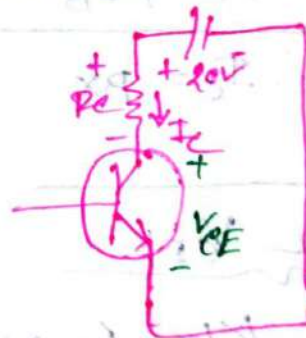
$$Q-Pt \equiv (I_c, V_{CE})$$

Now:-
Arrange the circuit:-

For:- Input



For Output



For I_B :-

⇒ Apply KVL in Input Circuit:-

$$20 - I_C R_C - I_B R_B - V_{BE} = 0$$

$$\Rightarrow 20 - I_B (\beta + 1) R_C - I_B R_B - 0.7 = 0$$

$$\Rightarrow I_B = \frac{20 - 0.7}{R_B + (\beta + 1) R_C}$$

$$= \frac{19.3}{100k + 201 \times 1k} = 26.02 \mu A$$

here:-

$$\begin{aligned} I &= I_C + I_B \\ &= \beta I_B + I_B \\ &= I_B (\beta + 1) \end{aligned}$$

I_s given:-

$$\begin{aligned} R_B &= 100k\Omega \\ \beta &= 100 \\ R_C &= 1k\Omega \\ V_{BE} &= 0.7 \end{aligned}$$

∴ For I_e :-
mon

we know:-
mon

$$I_e = \beta I_B$$

$$= 100 \times 0.011 \text{ A}$$

$$= 0.6 \text{ mA}$$

Now:- For V_{CE} :-
mon

⇒ Apply KVL in output Circuit:-
mon

$$V_{ee} - I_e R_e - V_{CE} = 0$$

$$\Rightarrow V_{CE} = V_{ee} - I_e R_e$$

$$= 20 - 0.6 \text{ mA} \times 1 \text{ k}$$

$$= 10.4 \text{ V}$$

So, the Operating points are

$$Q \equiv (0.6 \text{ mA}, 10.4 \text{ V})$$

Ans:-



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